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FINAL CLASSIFICATION GRADE:	

Department of Chemical & Environmental Engineering

ASSIGNMENT TITLE	CHEE2050 Assignment (Spring 2026)

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LEARNING OUTCOMES	
A2.1.3 Be able to apply the principles to the analysis of complex systems within a structured approach to safety, health and sustainability	
A2.2.1 Understand the principles of material and energy balances	
A2.4.2 Understand the principles on which processing equipment operates, and be able to apply methods to determine equipment size and performance of common items such as reactors and columns.	
A2.6.0.1 Understand the inherent nature of safety and loss prevention, and the principal hazard sources in chemical and related processes – including flammability, explosivity and toxicity (including biological hazards).	
A2.6.02 Understand the principles of risk assessment and of safety management, and be able to apply techniques for the assessment and abatement of process and product hazards.	
A2.6.0.3 Understand methods of identifying process hazards (e.g. HAZOP), and of assessing environmental impact.	
A4.2.3 Be able to deploy chemical engineering knowledge using rigorous calculation and results analysis to arrive at and verify the realism of the chosen design	
A5.2.1 Have developed a wide range of problem-solving skills	
A5.2.3 Recognise the importance of working effectively with others and have acquired a range of experience in achieving this.	
ASSESSMENT CRITERIA	
Assessed in the Group Design Project (Drawings, spreadsheets, HAZID workbook, report, ppt)	
Assessed in the Group Design Project through peer assessment	
ALL Assessment Criteria met (Y / N) :	**Y**

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Marking criteria: refer to marking rubrics published in Moodle.

Students' declaration: ***We certify that this assignment is our own work.*** ***NOTE: Plagiarism is a whole group responsibility***	**Date: 26/04/2026**
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Declaration of work contribution

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ASSESSOR FEEDBACK – to improve your grade in future assignments you will need to:

FINAL ASSIGNMENT GRADE:	**1ST 2:1 2:2 3RD REFERRAL**
ASSESSOR SIGNATURE:	**DATE:**

STUDENT FEEDBACK and COMMENTS: (as required) **Note to the student(s):** ***You must now make an appointment with the academic assessor, if you wish to discuss this assignment further***

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1 Executive Summary

PMB Biodiesel Co. Ltd. currently produces crude glycerol as a by-product of biodiesel manufacture and supplies it to an adjacent subsidiary facility for cosmetic-product production. This project proposes the installation of a new crude glycerol cooling package to reduce the crude glycerol temperature before it enters the downstream reactor. The required duty is to cool 6000 kg/h of crude glycerol from 110°C to 55°C, with no phase change, while maintaining a constant product flowrate and outlet temperature. The project is expected to improve downstream operation and deliver an annual benefit of approximately RMB 2.5 million.

A concept-selection study was carried out using Kepner–Tregoe screening. The selected concept is an open recirculating cooling-water system using plant/tap water as the make-up source, a cooling tower with bypass temperature control, and an AES floating-head shell-and-tube heat exchanger. This arrangement was selected because it provides a good balance of temperature control, maintainability, water-use reduction, and project deliverability. Cooling water is routed through the tube side of the heat exchanger to simplify cleaning of water-side scale, while crude glycerol is routed through the shell side. The floating-head arrangement provides better allowance for thermal expansion and improved access for inspection and maintenance compared with a fixed tube-sheet design.

The main process equipment includes crude glycerol storage and buffer tanks, crude glycerol transfer pumps, the shell-and-tube heat exchanger E-101, a cooling tower, cooling-water storage and mixing tanks, a sand-bed filter, cation and anion exchange units, and a wastewater neutralisation / settling system. The cooling system is designed as a recirculating loop rather than a once-through system. This reduces fresh-water demand and allows the cooling-water inlet temperature to the exchanger to be stabilised by adjusting the split between the cooling tower and bypass line. Supporting calculations show a heat duty of approximately 262 kW and an estimated cooling-water flowrate of about 32.5 m^3/h for the preliminary design basis.

The FEED-stage equipment specification demonstrates that the proposed exchanger and associated utility equipment are technically feasible. The E-101 heat-transfer design retains a positive duty margin under sensitivity checks, and the calculated shell-side and tube-side pressure drops are within the adopted design limits. Supporting design calculations are provided for the heat exchanger, cooling tower, water-treatment equipment, pump and control valve. Datasheets, equipment lists, valve lists and line lists have been prepared to define the main process and utility items and to support further mechanical and vendor design.

The proposed plant layout separates the core crude glycerol cooling module from the larger cooling-water and treatment equipment. The 30 m^2 designated area is reserved mainly for the crude glycerol pumps and heat exchanger, while the cooling tower, storage tanks and water-treatment equipment are arranged outside this core area. This layout improves access, ventilation and equipment separation, and helps to reduce thermal interference around the heat exchanger. Access routes are included for inspection, maintenance and heavy-vehicle movement.

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Process safety has been assessed using HAZID and HAZOP methods. The main hazards identified include the flammability and toxicity of methanol in the crude glycerol, high-temperature process streams, rotating equipment, pressure-related hazards, chemical handling during ion-exchange regeneration, and wastewater discharge risks. The HAZOP study divided the process into 12 nodes covering the crude glycerol line, heat exchanger, cooling-water system, ion-exchange system and wastewater treatment system. Key safeguards include duty/standby equipment, pressure and level protection, control-system monitoring, alarms, emergency shutdown functions, and defined start-up and shutdown sequences. The main outstanding safety issue is the practical provision of explosion-proof or blast-protection measures where site space is limited; this should be resolved during detailed design.

The control system is based on DCS/PLC supervision with field-level control loops for tank levels, crude glycerol flow, cooling-water flow, cooling-tower operation, exchanger outlet temperature, filtration monitoring and ion-exchange switching. Cascade temperature control is proposed for the heat exchanger and cooling tower because both systems are affected by time lag and weather-related disturbances. Independent safety protection is provided through pressure, level and emergency shutdown functions to prevent overpressure, pump dry-running, overflow and unsafe discharge.

The environmental assessment uses 1 tonne of cooled crude glycerol as the functional unit. The package processes approximately 48,000 tonnes of crude glycerol per year, assuming 8000 operating hours per year. The selected recirculating cooling-water design gives an estimated fresh-water consumption of 4216 m^3/y, or 0.0878 m^3/t of cooled crude glycerol, and a wastewater generation rate of 1040 m^3/y, or 0.0217 m^3/t. There is no direct steam, gas or oil use within the cooling-package boundary. Electricity consumption is estimated at 141,520 kWh/y, corresponding to 2.95 kWh/t, and the associated carbon footprint is approximately 70.4 tCO_2/y, or 1.47 $kgCO_2$/t. The main environmental control requirements are therefore wastewater verification, chemical-discharge control and thermal-discharge control rather than direct combustion emissions.

The project schedule runs from February 2026 to the end of December 2026. The critical path passes through material and energy balance development, heat-exchanger thermal and mechanical design, FEED review, vendor award, equipment fabrication, delivery, installation, pressure testing, hot commissioning, performance testing and final handover. The schedule is consistent with the project requirement for completion by the end of 2026, provided that vendor procurement and unresolved safety/layout actions are closed on time.

The economic analysis treats the project as a plant improvement rather than a new stand-alone production facility. The annual benefit is taken as RMB 2,500,000, with an estimated annual operating cost of RMB 358,757. The resulting annual net operating cash flow is approximately RMB 2.14 million. The total capital investment is estimated at RMB 4.72 million, giving a simple payback period of about 2.2 years and a DCF IRR of approximately 45%. On the basis of these base-case results, the project is economically attractive, provided that detailed design confirms the capital estimate, wastewater-treatment basis, control-system requirements and outstanding safety actions.

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Overall, the proposed crude glycerol cooling package satisfies the main technical requirements of the project brief. It provides the required 6000 kg/h crude glycerol output at 55°C, uses a maintainable shell-and-tube exchanger design, reduces water consumption through a recirculating cooling-water system, and includes the main control, safety, environmental and economic considerations required for FEED-stage approval. The project is therefore recommended to proceed to detailed design, vendor confirmation and final cost validation.

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3 Design Proposal

3.1 Introduction

PMB Biodiesel Co. Ltd. operates a biodiesel production plant in which crude glycerol is generated as a by-product. This crude glycerol is currently supplied to the company's subsidiary cosmetic production facility, which is located adjacent to the biodiesel factory. The purpose of this project is to design and install a crude glycerol cooling system before the glycerol enters the downstream reactor used in cosmetic product manufacture. The project is required because PMB Biodiesel has identified that reducing the crude glycerol temperature before reactor feed can improve plant operation and is expected to achieve an annual saving of approximately RMB 2.5 million .

The proposed design will cool the crude glycerol stream from 110°C to 55°C at a flowrate of 6000 kg/h, with no phase change expected. The crude glycerol composition is specified as 88.3 wt% glycerol, 11.3 wt% water and 0.4 wt% methanol. Cooling will be achieved by installing a new shell-and-tube heat exchanger using water as the cooling medium. The design will also include the required pumping and piping systems for both the cooling water circuit, from water source tap point to discharge point, and the crude glycerol circuit, from heat exchanger inlet to outlet .

The heat exchanger and associated systems will be designed to comply with TEMA requirements and relevant environmental standards. A control system will be included to maintain a constant crude glycerol supply rate and outlet temperature of 55°C, while considering possible process deviations caused by changes in weather conditions and cooling water source conditions. The new heat exchanger unit will be installed in the allocated area shown in the project layout, with the approximate GPS coordinates 29.800350, 121.564459, and the project is expected to be completed by the end of 2026 .

3.2 Plant Duty

The principal design duty is to cool the crude glycerol stream from 110°C to 55°C at a continuous mass flowrate of 6000 kg/h, with no phase change permitted across the exchanger.

Table 3.1 Key process basis

Parameter	Design basis
Crude glycerol flowrate	6000 kg/h
Crude glycerol inlet temperature	110°C
Crude glycerol outlet temperature	55°C
Composition	88.3 wt% glycerol; 11.3 wt% water; 0.4 wt% methanol
Phase change across exchanger	None
Required crude glycerol output	6000 kg/h at 55°C
Project completion target	End of 2026

3.3 Options

3.3.1 K-T screening method

The K-T method is appropriate at concept stage because it allows options to be ranked on a consistent basis even when only limited cost and vendor data are available. A score of 1 represents poor performance and a score of 5 represents excellent performance. Weighting factors range from 1 (low

importance) to 5 (critical importance). Because this is a concept-stage study, the matrices should be read as relative rankings rather than as final vendor-quoted economics. Detailed side-stream treatment design is discussed later in the process description because, in the current scheme, it is treated as part of the integrated operating concept rather than as a separate top-level concept choice.

3.3.2 Water source selection

Two source-water options were screened for the cooling system: (i) plant/tap water supplied from an existing treated-water connection, and (ii) direct river-water abstraction with front-end screening and pretreatment. The weighted comparison is summarised below.

Table 3.2 Kepner–Tregoe assessment of cooling-water source options

Criterion	Weight	Plant/tap water	River water
Capital cost	4	5	2
Operating cost	3	3	4
Water-quality stability	5	5	2
Operability and maintenance	4	5	2
Environmental and permitting risk	4	4	2
Schedule and constructability	3	5	2
Energy and carbon burden	2	4	2
Weighted total		**113**	**56**

Plant/tap water is therefore selected as the preferred basis. Although direct river abstraction offers lower raw-water cost, it would require intake works, solids control, more extensive pretreatment, and a more complex environmental and permitting case. Because the selected cooling concept is a recirculating loop rather than a once-through system, make-up demand is relatively modest and the tariff penalty of treated water is outweighed by greater source stability and lower project risk.

3.3.3 Cooling system configuration

The principal utility-configuration choice is between a once-through cooling arrangement and a recirculating cooling-water loop incorporating a cooling tower and bypass control. The comparison below was made using the selected plant/tap water source as the make-up basis.

Table 3.3 Kepner–Tregoe assessment of cooling-water circuit configuration options

Criterion	Weight	Once-through	Recirculating loop
Temperature stability and controllability	5	2	5
Fresh-water consumption	5	1	5
Environmental discharge burden	5	1	4
Capital cost	3	5	2
Operating cost	4	2	4
Operability under weather variation	5	2	5
Plot space requirement	2	5	2
Weighted total		**63**	**121**

The recirculating loop is taken forward. Its main advantage is that exchanger inlet temperature can be stabilised by combining cooling-tower duty with a bypass stream, which is important because the brief requires reliable control of the crude glycerol outlet temperature under changing weather conditions. A once-through system appears simpler, but it would couple exchanger performance directly to source-water temperature and discharge constraints while greatly increasing make-up water demand.

The higher capital cost and larger footprint of the recirculating system are accepted because temperature stability and reduced water usage are more important in this project.

3.3.4 Shell-and-tube exchanger configuration

The project brief fixes the exchanger family as shell-and-tube, but the specific TEMA arrangement still requires selection. Three practical configurations were screened: BEM fixed tube-sheet, BEU U-tube, and AES floating head.

Table 3.4 Kepner–Tregoe assessment of shell-and-tube exchanger construction options

Criterion	Weight	BEM fixed tube-sheet	BEU U-tube	AES floating head
Thermal-expansion tolerance	5	2	4	5
Tube-side cleanability	4	5	3	5
Shell-side inspectability	5	2	4	5
Suitability for fouling / viscous service	4	2	3	5
Capital cost	3	5	4	2
Mechanical simplicity / leakage paths	2	5	4	3
Maintenance downtime risk	4	2	3	5
Weighted total		**81**	**96**	**122**

AES floating head is selected. This service combines a substantial temperature difference with the likelihood of scale on the cooling-water side and fouling risk on the crude glycerol side. The AES arrangement best accommodates differential thermal expansion and provides the most flexible basis for inspection, bundle withdrawal, and mechanical cleaning. BEU remains a credible lower-cost alternative, but its U-bend geometry is less attractive where routine tube-side cleaning may be required. BEM was rejected because its fixed tube-sheet arrangement is less forgiving of thermal stress and gives the weakest shell-side maintenance access.

Within the selected AES arrangement, cooling water is assigned to the tube side and crude glycerol to the shell side. Water-side scale is easier to inspect and remove from the inside of straight tubes, while routing the more viscous crude glycerol through the shell avoids unnecessary tube-side pressure drop and provides a mechanically robust arrangement for a potentially fouling process stream.

3.3.5 Cooling-system type

A further concept-stage decision is whether heat should be rejected by an open evaporative cooling system based on a cooling tower or by a closed mechanical cooling system using refrigeration or compressor-based cooling. Both could in principle meet the duty, but they differ materially in energy demand, capital cost, water management burden, and the degree of temperature control they provide.

Table 3.5 Kepner–Tregoe assessment of cooling-system type options

Criterion	Weight	Open evaporative cooling tower	Closed mechanical cooling system
Suitability for moderate-temperature duty	5	5	2
Temperature stability / controllability	4	4	5

Capital cost	4	5	2
Operating power / energy cost	5	5	1
Water consumption / blowdown	3	2	5
Fouling / microbiological control burden	3	2	5
Constructability and maintainability	3	4	2
Weighted total		110	79

An open evaporative cooling system is selected. A closed mechanical system would reduce exposure to airborne contamination and would provide tighter temperature control, but this duty does not require sub-ambient cooling and only needs the crude glycerol to be cooled to 55°C. The cooling-tower route is therefore a better fit for the service because it can provide an adequate exchanger inlet temperature at much lower capital cost and power consumption. The additional chemistry-management burden of the open system is addressed in Section 4 through the selected filtration, dosing, and ion-exchange steps.

3.3.6 Selected concept taken forward

The K-T screening therefore supports the following integrated concept basis for the remainder of this proposal.

Table 3.6 Summary of selected concept basis taken forward for detailed design

Decision item	Selected basis	Principal reason for selection
Water source	Plant/tap water make-up	Highest source stability and lowest project / permitting risk
Cooling configuration	Recirculating loop with cooling tower and bypass	Best control of exchanger inlet temperature with much lower water use
Heat exchanger type	AES shell-and-tube exchanger	Best combination of thermal flexibility and cleanability
Cooling system type	Open evaporative cooling tower system	Adequate for a 55°C product target at much lower power and CAPEX than a closed mechanical system

Accordingly, the process description in Section 4 is based on plant/tap water being mixed with treatment chemicals in T-103 and circulated through E-101 in an open recirculating cooling-water system, with heat rejected through CT-101 and supply temperature stabilised by a bypass line. E-101 is taken as an AES floating-head exchanger with cooling water on the tube side and crude glycerol on the shell side. The recirculating loop includes sand filtration, cation exchange on the recycle stream to reduce scaling potential, and anion exchange on the purge stream before final cooling and discharge.

3.4 Process Description

Crude glycerol from the upstream process is first routed to sealed storage tank T-101. The covered arrangement has been retained because it reduces contamination and rainwater ingress, limits weather-driven temperature fluctuation, and helps to reduce vapour release from the methanol-containing crude glycerol. Once an adequate operating level is established in T-101, the crude glycerol is transferred by centrifugal pump P-101 to the shell side of E-101.

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In E-101 the crude glycerol is cooled from 110°C to 55°C without phase change. The exchanger is currently taken as an AES floating-head shell-and-tube unit. Cooling water is routed through the tubes because water-side scale and deposits are easier to inspect and remove from tube-side surfaces. The crude glycerol is routed on the shell side because the scale-prone cooling water is preferentially assigned to the tube side for easier cleaning. The resulting shell-side crude glycerol flow has a low Reynolds number due to its viscosity, but the calculated shell-side pressure drop remains within the adopted design limit and the AES removable-bundle arrangement provides access for shell-side inspection and cleaning. The cooled crude glycerol is then discharged by centrifugal pump P-102 to buffer tank T-102, from which it is supplied to the downstream process at the required flowrate.

Cooling water is supplied from the plant water source and delivered by pump P-103 to covered mixing tank T-103, where it is mixed with scale inhibitor before use. The treated cooling water is then pumped by P-104 to the tube side of E-101. Entry is from the lower nozzle and exit from the upper nozzle so that air venting is promoted and a stable flooded condition is maintained on the tube side.

Warm cooling water leaving E-101 enters the recirculating cooling-water system. A portion of the flow is sent to cooling tower CT-101 for cooling, while the remainder bypasses the tower. The cooled stream and bypass stream are then remixed so that the exchanger inlet temperature can be held at approximately 33°C. This bypass strategy is important because it allows the plant to cope with seasonal variation in ambient wet-bulb temperature while still maintaining a stable crude glycerol outlet temperature.

The mixed return water is collected in covered buffer tank T-104 and passed through sand bed filter F-101 to remove suspended solids. This step reduces fouling risk in the circulating loop and protects downstream treatment equipment. The current basis is that particles larger than 20 μm are removed in this stage.

Following filtration, a variable purge stream is withdrawn from the circulating loop to control the build-up of dissolved salts and treatment chemicals and to prevent excessive system inventory under off-design conditions. The purge stream is then sent to anion exchanger C-102, where phosphate and molybdate ions are removed to reduce the concentration of regulated discharge species. The treated purge is subsequently routed to buffer tank T-106, where it is cooled prior to discharge so that the receiving-water thermal criterion can be met.

The remaining filtered water is routed to cation exchanger C-101, where calcium and magnesium ions are removed to reduce scaling potential in the cooling-water circuit. The treated water then returns to T-103, where it is blended with fresh make-up water and inhibitor to maintain a stable circulating flowrate and composition.

Each ion-exchange service is provided with a standby unit. When the resin capacity in C-101A or C-102A approaches exhaustion, the duty is transferred to the corresponding standby vessel, allowing regeneration to be carried out without interrupting the main cooling duty. This arrangement has been selected to support the project requirement for minimum shutdown time.

During regeneration, the isolated vessel is connected to the regenerant supply and wastewater discharge systems. The regenerant displaces the accumulated ions from the resin, and the resulting regeneration effluent is routed to a neutralisation and settling tank. In this vessel the wastewater is

neutralised and conditioned to promote solids settling. Settled sludge is removed for solid-waste disposal, while clarified supernatant is discharged only after confirmation of compliance with the relevant discharge requirements.

3.5 Mass Balance

3.5.1 Material Balance Summary

Because the cooling system performs only sensible heat exchange and does not involve chemical reaction or phase change on the crude glycerol side, the total mass flow entering the exchanger is equal to the total mass flow leaving the exchanger. The crude glycerol composition therefore remains unchanged across the exchanger, and the exchanger outlet stream contains the same component mass flowrates as the inlet stream.

Using the design basis of 6000 kg/h total crude glycerol, the component flowrates are 5298 kg/h glycerol, 678 kg/h water and 24 kg/h of methanol. These component flowrates remain unchanged between the exchanger inlet and outlet. The material balance therefore demonstrates direct mass conservation and provides the necessary basis for the equipment datasheet, pump sizing and line sizing calculations.

Detailed material-balance calculations are provided in Appendix A.

Component	Mass fraction	Mass flow (kg/h)	Molar flow (kmol/h)	Mole fraction
Glycerol	0.883	5298	57.59	0.5998
Water	0.113	678	37.67	0.3923
Methanol	0.004	24	0.75	0.0078
Total	1.000	6000	96	1.0000

3.5.2 Energy Balance Summary

The exchanger duty is determined from the sensible heat removed from the crude glycerol as it cools from 110 °C to 55 °C. For the revised basis, the mixture heat capacity is estimated by mass-weighting the specific heat capacities of glycerol, water and methanol. Using specific heat capacities of 2.67 $kJ \cdot kg^{-1} \cdot K^{-1}$ for glycerol, 4.19 $kJ \cdot kg^{-1} \cdot K^{-1}$ for water and specific heat capacities of 6.71 $kJ \cdot kg^{-1} \cdot K^{-1}$ for methanol, the mixed heat capacity is calculated as 2.86 $kJ \cdot kg^{-1} \cdot K^{-1}$.

$$\dot{Q} = \dot{m}_{cg} \times C_{p,\,cg} \times \Delta T_{cg}$$

Substituting the design values gives a heat duty of approximately 943,126 kJ/h, or 261.98 kW. This duty forms the basis for preliminary exchanger sizing and for the cooling water requirement.

If cooling water is assumed to enter at 33 °C and leave at 40 °C, the required cooling water flowrate can be estimated from the same duty using a water heat capacity of 4.19 $kJ \cdot kg^{-1} \cdot K^{-1}$. The corresponding cooling water requirement is approximately 32,295 kg/h, equivalent to about 32.5 m^3/h when a water density of 993 kg/m^3 is used.

Detailed energy-balance calculations are provided in Appendix A.

Calculated item	Value

Mixture heat capacity, Cp,mix	2.86 $kJ \cdot kg^{-1} \cdot K^{-1}$
Crude glycerol temperature drop	55 °C
Heat duty	943,126 kJ/h (261.98 kW)
Cooling water temperature rise (illustrative basis)	7 °C
Cooling water mass flowrate	32,295 kg/h
Cooling water volumetric flowrate	32.5 m^3/h

3.5.3 Remarks on Utilities and Pump Work

At concept design level, the dominant energy term is the exchanger duty required to cool the crude glycerol stream. Pump work is relatively small compared with the exchanger duty and is therefore commonly neglected in the main process energy balance, although it should still be considered in the individual pump datasheets and hydraulic calculations. The final design package should therefore show the main exchanger duty in the report, while pump differential head, NPSH, rated flow and power should be presented in the relevant supporting datasheets.

GCW FrontSheet

3.5.4 Mass Balance table with stream numbers referenced to the PFD

CLIENT:	PMB Biodiesel Co. Ltd.
LOCATION:	PMB Biodiesel site, Ningbo, China
SUBJECT:	M&E Balance for Crude Glycerol Cooling
PROJECT:	Crude Glycerol Cooling System
JOB NO.:	CHEE2050-G3
REV.:	D

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MASS AND ENERGY BALANCE

DOCUMENT NO.: M&EB-001

GROUP NO.: 3

Stream Number		1				2				3				4				5			
Description		Grude glycerol mixture entering tank T-101 from the previous industrial process				Grude glycerol mixture entering the heat exchanger E-101				Grude glycerol mixture entering tank T-102				Grude glycerol mixture to Subsidiary Company				Tap water from water treatment plant			
Component	Mol. Wt.	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol
water	18	678	37.67	39.23%	8296.2	678	37.67	39.23%	8296.2	678	37.67	39.23%	4148.1	678	37.67	39.23%	4148.1	32295.6	1794.20	100.00%	2488.86
glycerol	92	5298	57.59	59.98%	27020.4	5298	57.59	59.98%	27020.4	5298	57.59	59.98%	13510.2	5298	57.59	59.98%	13510.2				
methanol	32	24	0.75	0.78%	23617.9	24	0.75	0.78%	23617.9	24	0.75	0.78%	11792	24	0.75	0.78%	11792				
Total Flow		6000	96.00	100%	58934.5	6000	96.00	100%	58934.5	6000	96.00	100%	29450.3	6000	96.00	100%	29450.3	32295.6	1794.20	100.00%	2488.86
Temperature (°C)		110				110				55				55				33			
Pressure (barg)		1				2				1.5				1.5				2			
Density (kg/m^3)		1200				1200				1200				1200				993			
Flow rate (m^3/h)		5				5				5				5				32.5			
Enthalpy (kJ/h)		1886226.23				1886226.23				943100.40				943100.40				4465512.61			

Stream Number		6				7				8				9				10			
Description		Tap water dosed with scale inhibitor entering the mixing tank T-103				Cooling water entering the heat exchanger E-101				Cooling water entering the cooling tower CT-101				Cooling water entering the storage tank T-104				Cooling water entering the filter F-101			
Component	Mol. Wt.	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol
water	18	32295.6	1794.20	100.00%	2488.86	32295.6	1794.20	100.00%	2488.86	32295.6	1794.20	100.00%	3016.8	31746.21	1763.68	100.00%	2488.86	31746.21	1763.68	100.00%	2488.86
glycerol																					
methanol																					
Total Flow		32295.6	1794.20	100.00%	2488.86	32295.6	1794.20	100.00%	2488.86	32295.6	1794.20	100.00%	3016.8	31746.21	1763.68	100.00%	2488.86	31746.21	1763.68	100%	2488.86
Temperature (°C)		33				33				40				33				33			
Pressure (barg)		0.1				2				1				2				0.2			
Density (kg/m^3)		993				993				993				993				993			
Flow rate (m^3/h)		32.50				32.50				32.50				31.97				31.97			
Enthalpy (kJ/h)		4465512.612				4465512.612				5412742.56				4389548.457				4389548.457			

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Stream Number		11				12				13				14				15			
Description		Cooling water exits the filter, prior to the separation of the purge and circulating water				Circulating water entering anion exchanger column C-101				Circulating water entering tank T-103				Purge water entering the cation exchanger column C-102				Purge water entering the tank T-106			
Component	Mol. Wt.	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol
water	18	31746.21	1763.68	100.00%	2488.86	31619.23	1756.62	100.00%	2488.86	31619.23	1756.62	100.00%	2488.86	126.98	7.05	100.00%	2488.86	126.98	7.05	100.00%	2488.86
glycerol																					
methanol																					
Total Flow		31746.21	1763.68	100%	2488.86	31619.23	1756.62	100%	2488.86	31619.23	1756.62	100%	2488.86	126.98	7.05	100%	2488.86	126.98	7.05	100%	2488.86
Temperature (°C)		33				33				33				33				33			
Pressure (barg)		0.1				0.1				0.1				0.1				0			
Density (kg/m^3)		993				993				993				993				993			
Flow rate (m^3/h)		31.97				31.84				31.84				0.13				0.13			
Enthalpy (kJ/h)		4389548.457				4371990.932				4371990.932				17557.5246				17557.5246			

Stream Number		16			
Description		Waste water leaving the tank T-106			
Component	Mol. Wt.	Mass, kg/h	Mol, kmol/h	Mol, %	h, kJ/kmol
water	18	126.98	7.05	100.00%	2488.86
glycerol					
methanol					
Total Flow		126.98	7.05	100.00%	2488.86
Temperature (°C)		33			
Pressure (barg)		0			
Density (kg/m^3)		993			
Flow rate (m^3/h)		0.13			
Enthalpy (kJ/h)		17557.5246			

Note: Water loss through the cooling tower is approximately 0.527 m³/h. After the initial fill and stable operation, make-up water of 0.527 m³/h shall be added via Stream 5.

REV.	DESCRIPTION	DATE	BY	CHECK	APPROVED	DOCUMENT NO.
	SEE PAGE 1					M&EB-001
						PAGE 2 OF 2

GCW FrontSheet

4 Equipment Specification

4.1 Equipment list, Valve list and Line list

4.1.1 Equipment list & Valve list

REV	REVISION DESCRIPTION	DATE	BY	CHK	APPR
D	Modified for final submition	23/04/2023	Weicheng Qin	Yijie Yao	Chen Ba
C	PID CHECK: MANUAL VALVE TYPES ALIGNED WITH P&ID LEGEND; P-103 AND LINE BASIS CLEANED	21/04/2024	Weicheng Qin	Chen Ba	Chen Ba
B	UPDATED P-101A/B TO P-107A/B PUMP MODELS PER AHJN QUOTATION	18/04/2025	Weicheng Qin	Shubo Ma	Weicheng Qin
A	UPDATED TO MATCH PFD/P&ID + HX/CT CALCS	15/04/2026	Weicheng Qin	Weicheng Qin	Weicheng Qin

CLIENT: PMB Biodiesel Co. Ltd.
LOCATION: PMB Biodiesel Plant
PROJECT: Crude Glycerol Cooling System
JOB NO.: CHEE2050

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EQUIPMENT LIST

SHEET 1 OF 1
DOCUMENT NUMBER EL-001

Tag No.	Equipment Description	Equipment Type	Equipment Duty Capacity	Dimensions H:W:D ***	Weight ** kg	Materials of Construction	Fluid	Connection Type	P&ID NO.	OPERATING PRES. (barg)	OPERATING TEMP. (°C)	DESIGN PRES. (barg)	DESIGN TEMP. (°C)	INSULATION INSUL. CLASS	INSULATION INSUL. MM	COMMENTS	ISSUE/ REV
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
T-101	Hot crude glycerol storage tank	Vertical storage tank	13.7 m³	Φ 2.5 x 2.75 m	900	FRP	Hot crude glycerol	DN40 150# RF	P&ID-001	0	110	0.35	135	H	50	FRP wall thickness 9 mm; main process nozzles DN40.	F
T-102	Cooled crude glycerol storage tank	Vertical storage tank	13.7 m³	Φ 2.5 x 2.75 m	1400	FRP	Cooled crude glycerol	DN40 150# RF	P&ID-001	0	55	0.35	80	PP	25	FRP wall thickness 9 mm; main process nozzles DN40.	F
T-103	Cooling water mixing tank	Mixing tank	5.4 m³ working	2.5 x 2 x 1.8 m	900	FRP	Cooling water / scale inhibitor	DN80 150# RF	P&ID-001	0	40	0.35	65	C	25	10 min hold-up at 32.43 m³/h; working volume 5.4 m³.	F
T-104	Circulating water storage tank	Vertical tank	10.8 m³ working	3.0 x 2.5 x 2.1 m	1200	FRP	Circulating water	DN80 150# RF	P&ID-001	0	40	0.35	65	C	25	20 min hold-up at 32.43 m³/h; working volume 10.8 m³.	F
T-105	Circulating water precipitate settling tank	Settling tank	7.6 m³ working	Φ 2.3 x 1.5 m	1800	FRP+PP	Circulating water / precipitate slurry	DN80 150# RF	P&ID-001	0	40	0.35	65	C	25	HOLD: preliminary settling capacity; 30 min full-flow hold-up at 32.43 m³/h would require approximately 16.2 m³.	F
T-106	Purge water cooling tank	Cooling tank	67.6 m³	5.2 x 5.2 x 2.5 m	44700	RCC	Waste water / purge water	DN32 150# RF	P&ID-001	0	25	0.35	50	-	-	HOLD: purge-water flow/cooling basis to be confirmed; RCC wall thickness 200 mm.	F
CT-101	Circulating water cooling tower	Induced-draft cooling tower	289 kW (82.1 RT)	2.2 x 2.2 x 3.8 m	2800	FRP casing / PVC fill / SS hardware	Circulating cooling water	DN80 150# RF	P&ID-001	0	40	0.35	65	-	-	Selected for normal heat rejection; vendor curve to confirm duty margin.	F
C-101A	Cation exchange column	Ion exchange column	32.4 m³/h	Φ 2 x 2.1 m	1800	SS316 vessel / PP internals	Cooling water	DN80 150# RF	P&ID-001	2	40	2.35	65	C	25	Full-flow cation exchange duty; duty/standby/regeneration per P&ID.	F
C-101B	Cation exchange column	Ion exchange column	32.4 m³/h	Φ 2 x 2.1 m	1800	SS316 vessel / PP internals	Cooling water	DN80 150# RF	P&ID-001	2	40	2.35	65	C	25	Full-flow cation exchange duty; duty/standby/regeneration per P&ID.	F
C-102A	Anion exchange column	Ion exchange column	0.13 m³/h	Φ 0.315 x 2.96 m	1800	PP	Cooling water	DN80 150# RF	P&ID-001	2	40	2.35	65	C	25	Verify duty basis: 0.13 m³/h conflicts with full-flow wording; update either capacity or service basis.	F
C-102B	Anion exchange column	Ion exchange column	0.13 m³/h	Φ 0.315 x 2.96 m	1800	PP	Cooling water	DN80 150# RF	P&ID-001	2	40	2.35	65	C	25	Verify duty basis: 0.13 m³/h conflicts with full-flow wording; update either capacity or service basis.	F
E-101	AES shell-and-tube HX	Shell & tube HX	262.5 kW req.; 315.1 kW rated	4.6 x 1 x 1.5 m	1400	SS316	Crude glycerol / cooling water	DN40/80 RF	P&ID-001	2	110	2.35	135	PP	50	15.25 in shell; 128 tubes; 4-pass; 3.66 m tubes; DN40/DN80 connections.	F
F-101	Circulating water filter	Side-stream cartridge / bag filter	32.4 m³/h	1.2 x 0.7 x 0.7 m	250	316L SS housing	Circulating water	DN80 150# RF	P&ID-001	2	40	2.35	65	C	25	Filter upstream of ion-exchange loop; confirm full-flow vs side-stream basis.	F
P-101A	Crude glycerol feed pump	AHJN JIH stainless steel centrifugal pump & motor	5.0 m³/h normal; 6.3 m³/h @ 20 m selected	Vendor std.	65	SS316L wetted parts	Hot crude glycerol	Pump DN40/25; line DN40 150# RF	P&ID-001	2	110	2.35	135	PP	25	AHJN JIH40-25-125; 6.3 m³/h @ 20 m; DN40/25; final TDH/NPSH by vendor curve.	G
P-101B	Crude glycerol feed pump	AHJN JIH stainless steel centrifugal pump & motor	5.0 m³/h normal; 6.3 m³/h @ 20 m selected	Vendor std.	65	SS316L wetted parts	Hot crude glycerol	Pump DN40/25; line DN40 150# RF	P&ID-001	2	110	2.35	135	PP	25	Standby pump; same model and rating as P-101A.	G
P-102A	Cooled glycerol transfer pump	AHJN JIH stainless steel centrifugal pump & motor	5.0 m³/h normal; 6.3 m³/h @ 20 m selected	Vendor std.	65	SS316L wetted parts	Cooled crude glycerol	Pump DN40/25; line DN40 150# RF	P&ID-001	2	55	2.35	80	PP	25	AHJN JIH40-25-125; 6.3 m³/h @ 20 m; DN40/25; final TDH/NPSH by vendor curve.	G
P-102B	Cooled glycerol transfer pump	AHJN JIH stainless steel centrifugal pump & motor	5.0 m³/h normal; 6.3 m³/h @ 20 m selected	Vendor std.	65	SS316L wetted parts	Cooled crude glycerol	Pump DN40/25; line DN40 150# RF	P&ID-001	2	55	2.35	80	PP	25	Standby pump; same model and rating as P-102A.	G
P-103A	Cooling water make-up / feed pump	AHJN JIH stainless steel centrifugal pump & motor	0.527 m³/h normal; 2.0 m³/h @ 20 m selected	Vendor std.	55	SS316L wetted parts	Tap water / cooling-water make-up	Pump DN25/20; line DN32 with reducers	P&ID-001	2	33	2.35	58	-	-	AHJN JIH25-20-125; smallest selected make-up pump: 2.0 m³/h @ 20 m vs normal 0.527 m³/h; intermittent level-control operation.	G
P-103B	Cooling water make-up / feed pump	AHJN JIH stainless steel centrifugal pump & motor	0.527 m³/h normal; 2.0 m³/h @ 20 m selected	Vendor std.	55	SS316L wetted parts	Tap water / cooling-water make-up	Pump DN25/20; line DN32 with reducers	P&ID-001	2	33	2.35	58	-	-	Standby pump; same model and rating as P-103A.	G
P-104A	Mixed cooling water feed / transfer pump	AHJN JIH stainless steel centrifugal pump & motor	32.43 m³/h normal; 50 m³/h @ 20 m selected	Vendor std.	85	SS316L wetted parts	Mixed cooling water	Pump DN80/65; line DN80 150# RF	P&ID-001	2	40	2.35	65	-	-	AHJN JIH80-65-125; 50 m³/h @ 20 m; oversized vs 32.43 m³/h; check pump/system curve.	G
P-104B	Mixed cooling water feed / transfer pump	AHJN JIH stainless steel centrifugal pump & motor	32.43 m³/h normal; 50 m³/h @ 20 m selected	Vendor std.	85	SS316L wetted parts	Mixed cooling water	Pump DN80/65; line DN80 150# RF	P&ID-001	2	40	2.35	65	-	-	Standby pump; same model and rating as P-104A.	G
P-105A	Circulating water transfer pump	AHJN JIH stainless steel centrifugal pump & motor	32.43 m³/h normal; 50 m³/h @ 20 m selected	Vendor std.	85	SS316L wetted parts	Circulating water	Pump DN80/65; line DN80 150# RF	P&ID-001	2	40	2.35	65	-	-	AHJN JIH80-65-125; 50 m³/h @ 20 m; oversized vs 32.43 m³/h; check pump/system curve.	G
P-105B	Circulating water transfer pump	AHJN JIH stainless steel centrifugal pump & motor	32.43 m³/h normal; 50 m³/h @ 20 m selected	Vendor std.	85	SS316L wetted parts	Circulating water	Pump DN80/65; line DN80 150# RF	P&ID-001	2	40	2.35	65	-	-	Standby pump; same model and rating as P-105A.	G
P-106A	HCl feed pump	AHJN CQB fluoroplastic magnetic centrifugal pump	2.0 m³/h normal assumed; 2.5 m³/h @ 11 m selected	Vendor std.	12	Fluoroplastic wetted parts	HCl	Pump DN25/20; line DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-	AHJN CQB25-20-100F; fluoroplastic wetted parts; 2.5 m³/h @ 11 m; confirm regeneration set point.	G
P-106B	HCl feed pump	AHJN CQB fluoroplastic magnetic centrifugal pump	2.0 m³/h normal assumed; 2.5 m³/h @ 11 m selected	Vendor std.	12	Fluoroplastic wetted parts	HCl	Pump DN25/20; line DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-	Standby pump; same model and rating as P-106A.	G
P-107A	NaOH feed pump	AHJN CQB fluoroplastic magnetic centrifugal pump	2.0 m³/h normal assumed; 2.5 m³/h @ 11 m selected	Vendor std.	12	Fluoroplastic wetted parts	NaOH	Pump DN25/20; line DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-	AHJN CQB25-20-100F; fluoroplastic wetted parts; 2.5 m³/h @ 11 m; confirm regeneration set point.	G
P-107B	NaOH feed pump	AHJN CQB fluoroplastic magnetic centrifugal pump	2.0 m³/h normal assumed; 2.5 m³/h @ 11 m selected	Vendor std.	12	Fluoroplastic wetted parts	NaOH	Pump DN25/20; line DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-	Standby pump; same model and rating as P-107A.	G
FCV101	Hot crude glycerol inlet flow control to T-101	Control valve	5 m³/h	Vendor std.	-	CF8M / SS316 body; F316 trim	Hot crude glycerol	DN25 150# RF	P&ID-001	1	110	1.35	135	-	-	Fail state per P&ID; vendor to confirm Cv, actuator and GA.	F
LCV101	T-101 outlet level control to P-101 suction manifold	Control valve	5 m³/h	Vendor std.	-	CF8M / SS316 body; F316 trim	Hot crude glycerol	DN25 150# RF	P&ID-001	0.1	110	0.45	135	-	-	Fail state per P&ID; vendor to confirm Cv, actuator and GA.	F
HV106	T-101 bottom drain	Manual drain valve	-	Vendor std.	-	316 SS	Hot crude glycerol	DN40 150# RF	P&ID-001	0	110	0.35	135	-	-		F
HV112A	P-101A suction isolation — hot crude glycerol feed	Manual gate valve	-	Vendor std.	-	316 SS	Hot crude glycerol	DN40 150# RF	P&ID-001	0.1	110	0.45	135	-	-		F
HV112B	P-101B suction isolation — hot crude glycerol feed	Manual gate valve	-	Vendor std.	-	316 SS	Hot crude glycerol	DN40 150# RF	P&ID-001	0.1	110	0.45	135	-	-		F
HV113A	P-101A suction drain — hot crude glycerol feed	Manual drain valve	-	Vendor std.	-	316 SS	Hot crude glycerol	DN40 150# RF	P&ID-001	0.1	110	0.45	135	-	-		F
HV113B	P-101B suction drain — hot crude glycerol feed	Manual drain valve	-	Vendor std.	-	316 SS	Hot crude glycerol	DN40 150# RF	P&ID-001	0.1	110	0.45	135	-	-		F

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NRV101A	P-101A discharge non-return — hot crude glycerol feed	Check valve	5 m³/h	Vendor std.	-	316 SS	Hot crude glycerol	DN40 150# RF	P&ID-001	2	110	2.35	135	-	-	F
NRV101B	P-101B discharge non-return — hot crude glycerol feed	Check valve	5 m³/h	Vendor std.	-	316 SS	Hot crude glycerol	DN40 150# RF	P&ID-001	2	110	2.35	135	-	-	F
HV114A	P-101A discharge isolation — hot crude glycerol feed	Manual gate valve	-	Vendor std.	-	316 SS	Hot crude glycerol	DN40 150# RF	P&ID-001	2	110	2.35	135	-	-	F
HV114B	P-101B discharge isolation — hot crude glycerol feed	Manual gate valve	-	Vendor std.	-	316 SS	Hot crude glycerol	DN40 150# RF	P&ID-001	2	110	2.35	135	-	-	F
LCV102	Hot crude glycerol level/flow control upstream of ESDV101 / E-101	Control valve	5 m³/h	Vendor std.	-	CF8M / SS316 body; F316 trim	Hot crude glycerol	DN25 150# RF	P&ID-001	2	110	2.35	135	-	-	F — Fail state per P&ID; vendor to confirm Cv, actuator and GA.
ESDV101	Hot crude glycerol emergency isolation upstream of E-101	Actuated on/off valve	-	Vendor std.	-	CF8M / SS316 body;	Hot crude glycerol	DN40 150# RF	P&ID-001	2	110	2.35	135	-	-	F — Fail closed; vendor to confirm actuator and GA.
HV146	E-101 low-point drain	Manual drain valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	2	110	2.35	135	-	-	F
PSV101	E-101 / hot crude glycerol side overpressure protection	Pressure safety valve	-	Vendor std.	-	316 SS	Mixed cooling water	150# RF / metal line	P&ID-001	2	40	2.35	65	-	-	F — Vendor sizing required from final relief case.
TCV101	Heat exchanger cooling-water temperature control valve	Control valve	32.43 m³/h	Vendor std.	-	CF8M / SS316 body; F316 trim	Mixed cooling water	DN50 150# RF	P&ID-001	2	33	2.35	58	-	-	F — Fail state per P&ID; vendor to confirm Cv, actuator and GA.
FCV102	Cooling water flow control into CT-101	Control valve	32.43 m³/h	Vendor std.	-	CF8M / SS316 body; F316 trim	Circulating water	DN50 150# RF	P&ID-001	1	40	1.35	65	-	-	F — Fail state per P&ID; vendor to confirm Cv, actuator and GA.
FCV103	Cooling water flow control on CT-101 / cooling-water header loop	Control valve	32.43 m³/h	Vendor std.	-	CF8M / SS316 body; F316 trim	Circulating water	DN50 150# RF	P&ID-001	1	40	1.35	65	-	-	F — Fail state per P&ID; vendor to confirm Cv, actuator and GA.
FCV104	Cooling water feed / make-up flow control to T-103	Control valve	0.527 m³/h normal make-up	Vendor std.	-	CF8M / SS316 body; F316 trim	Tap water / cooling water	DN25 150# RF; DN32 line reducers	P&ID-001	2	33	2.35	58	-	-	F — Fail state per P&ID; vendor to confirm Cv, actuator and GA.
HV138A	P-103A suction isolation — cooling water feed	Manual gate valve	-	Vendor std.	-	316 SS	Tap water / cooling-water make-up	DN32 150# RF; reducer to pump DN25	P&ID-001	1	33	1.35	58	-	-	F
HV138B	P-103B suction isolation — cooling water feed	Manual gate valve	-	Vendor std.	-	316 SS	Tap water / cooling-water make-up	DN32 150# RF; reducer to pump DN25	P&ID-001	1	33	1.35	58	-	-	F
HV139A	P-103A suction drain — cooling water feed	Manual drain valve	-	Vendor std.	-	316 SS	Tap water / cooling-water make-up	DN32 150# RF; reducer to pump DN25	P&ID-001	1	33	1.35	58	-	-	F
HV139B	P-103B suction drain — cooling water feed	Manual drain valve	-	Vendor std.	-	316 SS	Tap water / cooling-water make-up	DN32 150# RF; reducer to pump DN25	P&ID-001	1	33	1.35	58	-	-	F
NRV103A	P-103A discharge non-return — cooling water feed	Check valve	0.527 m³/h normal; 2.0 m³/h pump selected	Vendor std.	-	316 SS	Tap water / cooling-water make-up	DN32 150# RF; pump outlet DN20 via expander	P&ID-001	2	33	2.35	58	-	-	F
NRV103B	P-103B discharge non-return — cooling water feed	Check valve	0.527 m³/h normal; 2.0 m³/h pump selected	Vendor std.	-	316 SS	Tap water / cooling-water make-up	DN32 150# RF; pump outlet DN20 via expander	P&ID-001	2	33	2.35	58	-	-	F
HV140A	P-103A discharge isolation — cooling water feed	Manual gate valve	-	Vendor std.	-	316 SS	Tap water / cooling-water make-up	DN32 150# RF; pump outlet DN20 via expander	P&ID-001	2	33	2.35	58	-	-	F
HV140B	P-103B discharge isolation — cooling water feed	Manual gate valve	-	Vendor std.	-	316 SS	Tap water / cooling-water make-up	DN32 150# RF; pump outlet DN20 via expander	P&ID-001	2	33	2.35	58	-	-	F
FCV105	Circulating water / filter flow control around F-101 and T-104	Control valve	-	Vendor std.	-	CF8M / SS316 body; F316 trim	Circulating water	DN50 150# RF	P&ID-001	2	33	2.35	58	-	-	F — Fail state per P&ID; vendor to confirm Cv, actuator and GA.
HV161A	P-104A suction isolation — mixed cooling water transfer	Manual gate valve	-	Vendor std.	-	316 SS	Mixed cooling water	DN80 150# RF/lug	P&ID-001	0.1	33	0.45	58	-	-	F
HV161B	P-104B suction isolation — mixed cooling water transfer	Manual gate valve	-	Vendor std.	-	316 SS	Mixed cooling water	DN80 150# RF/lug	P&ID-001	0.1	33	0.45	58	-	-	F
HV162A	P-104A suction drain — mixed cooling water transfer	Manual drain valve	-	Vendor std.	-	316 SS	Mixed cooling water	DN80 150# RF	P&ID-001	0.1	33	0.45	58	-	-	F
HV162B	P-104B suction drain — mixed cooling water transfer	Manual drain valve	-	Vendor std.	-	316 SS	Mixed cooling water	DN80 150# RF	P&ID-001	0.1	33	0.45	58	-	-	F
NRV105A	P-104A discharge non-return — mixed cooling water transfer	Check valve	32.43 m³/h	Vendor std.	-	316 SS	Mixed cooling water	DN80 150# wafer/lug/RF	P&ID-001	2	33	2.35	58	-	-	F
NRV105B	P-104B discharge non-return — mixed cooling water transfer	Check valve	32.43 m³/h	Vendor std.	-	316 SS	Mixed cooling water	DN80 150# wafer/lug/RF	P&ID-001	2	33	2.35	58	-	-	F
HV163A	P-104A discharge isolation — mixed cooling water transfer	Manual gate valve	-	Vendor std.	-	316 SS	Mixed cooling water	DN80 150# RF/lug	P&ID-001	2	33	2.35	58	-	-	F
HV163B	P-104B discharge isolation — mixed cooling water transfer	Manual gate valve	-	Vendor std.	-	316 SS	Mixed cooling water	DN80 150# RF/lug	P&ID-001	2	33	2.35	58	-	-	F
HV130A	P-105A suction isolation — circulating water transfer	Manual gate valve	-	Vendor std.	-	316 SS	Circulating water	DN80 150# RF/lug	P&ID-001	0.1	33	0.45	58	-	-	F
HV130B	P-105B suction isolation — circulating water transfer	Manual gate valve	-	Vendor std.	-	316 SS	Circulating water	DN80 150# RF/lug	P&ID-001	0.1	33	0.45	58	-	-	F
HV131A	P-105A suction drain — circulating water transfer	Manual drain valve	-	Vendor std.	-	316 SS	Circulating water	DN80 150# RF	P&ID-001	0.1	33	0.45	58	-	-	F
HV131B	P-105B suction drain — circulating water transfer	Manual drain valve	-	Vendor std.	-	316 SS	Circulating water	DN80 150# RF	P&ID-001	0.1	33	0.45	58	-	-	F
NRV102A	P-105A discharge non-return — circulating water transfer	Check valve	32.43 m³/h	Vendor std.	-	316 SS	Circulating water	DN80 150# wafer/lug/RF	P&ID-001	2	33	2.35	58	-	-	F
NRV102B	P-105B discharge non-return — circulating water transfer	Check valve	32.43 m³/h	Vendor std.	-	316 SS	Circulating water	DN80 150# wafer/lug/RF	P&ID-001	2	33	2.35	58	-	-	F
HV132A	P-105A discharge isolation — circulating water transfer	Manual gate valve	-	Vendor std.	-	316 SS	Circulating water	DN80 150# RF/lug	P&ID-001	2	33	2.35	58	-	-	F
HV132B	P-105B discharge isolation — circulating water transfer	Manual gate valve	-	Vendor std.	-	316 SS	Circulating water	DN80 150# RF/lug	P&ID-001	2	33	2.35	58	-	-	F
HV147A	P-102A suction isolation — cooled crude glycerol transfer	Manual gate valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0.3	55	0.65	80	-	-	F
HV147B	P-102B suction isolation — cooled crude glycerol transfer	Manual gate valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0.3	55	0.65	80	-	-	F
HV148A	P-102A suction drain — cooled crude glycerol transfer	Manual drain valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0.3	55	0.65	80	-	-	F
HV148B	P-102B suction drain — cooled crude glycerol transfer	Manual drain valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0.3	55	0.65	80	-	-	F
NRV104A	P-102A discharge non-return — cooled crude glycerol transfer	Check valve	5 m³/h	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	1.5	55	1.85	80	-	-	F
NRV104B	P-102B discharge non-return — cooled crude glycerol transfer	Check valve	5 m³/h	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	1.5	55	1.85	80	-	-	F
HV149A	P-102A discharge isolation — cooled crude glycerol transfer	Manual gate valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	1.5	55	1.85	80	-	-	F
HV149B	P-102B discharge isolation — cooled crude glycerol transfer	Manual gate valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	1.5	55	1.85	80	-	-	F
HV150	T-102 bottom drain	Manual drain valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0	55	0.35	80	-	-	F
HV151	LCV103 upstream isolation on T-102	Manual gate valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0.1	55	0.45	80	-	-	F
LCV103	T-102 cooled glycerol outlet level control	Control valve	5 m³/h	Vendor std.	-	CF8M / SS316 body; F316 trim	Cooled crude glycerol	DN25 150# RF	P&ID-001	0.1	55	0.45	80	-	-	F — Fail state per P&ID; vendor to confirm Cv, actuator and GA.
HV152	LCV103 upstream bypass isolation	Manual gate valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0.1	55	0.45	80	-	-	F
HV153	LCV103 bypass center valve	Manual gate valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0.1	55	0.45	80	-	-	F
HV154	LCV103 downstream bypass isolation	Manual gate valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0.1	55	0.45	80	-	-	F
HV155	LCV103 downstream isolation on T-102 outlet	Manual gate valve	-	Vendor std.	-	316 SS	Cooled crude glycerol	DN40 150# RF	P&ID-001	0.1	55	0.45	80	-	-	F
HV121	Scale inhibitor manual isolation	Manual gate valve	-	Vendor std.	-	316 SS	Scale inhibitor	DN32 150# RF	P&ID-001	0	25	0.35	50	-	-	F
LCV104	T-104 circulating-water tank level control	Control valve	32.43 m³/h	Vendor std.	-	CF8M / SS316 body; F316 trim	Circulating water	DN50 150# RF	P&ID-001	2	33	2.35	58	-	-	F — Fail state per P&ID; vendor to confirm Cv, actuator and GA.
LCV105	T-103 outlet / ion-exchange feed level control valve	Control valve	32.43 m³/h	Vendor std.	-	WCB lined F46/PFA; FEP-	Circulating water / plastic-class service	DN50 150# RF + plastic adaptor	P&ID-001	0.1	33	0.45	58	-	-	F — Fail state per P&ID; vendor to confirm Cv, actuator and GA.
AV101	F-101 outlet switching valve to ion-exchange train (mixed header)	Actuated three-way switching valve	31.78 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Circulating water	DN32 150# RF + plastic adaptor	P&ID-001	0.1	33	0.45	58	-	-	F — 3-way service/backwash switching valve; vendor GA TBC.
AV102	F-101 outlet switching valve to ion-exchange train (mixed header)	Actuated three-way switching valve	0.13 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Purge water	DN32 150# RF + plastic adaptor	P&ID-001	0.1	33	0.45	58	-	-	F — 3-way service/backwash switching valve; vendor GA TBC.
AV103	Top switch valve at C-101A	Actuated three-way switching valve	31.78 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Circulating water / HCl	DN20/DN25 lined valve + reducers/adaptors	P&ID-001	0.1	33	0.45	58	-	-	F — 3-way service/backwash switching valve; vendor GA TBC.
AV104	Top switch valve at C-101B	Actuated three-way switching valve	31.78 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Circulating water / HCl	DN20/DN25 lined valve + reducers/adaptors	P&ID-001	0.1	33	0.45	58	-	-	F — 3-way service/backwash switching valve; vendor GA TBC.
AV105	Top switch valve at C-102A	Actuated three-way switching valve	0.13 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Purge water / NaOH	DN20/DN25 lined valve + reducers/adaptors	P&ID-001	0.1	33	0.45	58	-	-	F — 3-way service/backwash switching valve; vendor GA TBC.
AV106	Top switch valve at C-102B	Actuated three-way switching valve	0.13 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Purge water / NaOH	DN20/DN25 lined valve + reducers/adaptors	P&ID-001	0.1	33	0.45	58	-	-	F — 3-way service/backwash switching valve; vendor GA TBC.

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AV107	Bottom switch valve at C-101A	Actuated three-way switching valve	31.78 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Circulating water / HCl	DN20/DN25 lined valve + reducers/adaptors	P&ID-001	0.1	33	0.45	58	-	-	3-way service/backwash switching valve; vendor GA TBC.	F
AV108	Bottom switch valve at C-101B	Actuated three-way switching valve	31.78 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Circulating water / HCl	DN20/DN25 lined valve + reducers/adaptors	P&ID-001	0.1	33	0.45	58	-	-	3-way service/backwash switching valve; vendor GA TBC.	F
AV109	Bottom switch valve at C-102A	Actuated three-way switching valve	0.13 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Purge water / NaOH	DN20/DN25 lined valve + reducers/adaptors	P&ID-001	0.1	33	0.45	58	-	-	3-way service/backwash switching valve; vendor GA TBC.	F
AV110	Bottom switch valve at C-102B	Actuated three-way switching valve	0.13 m³/h	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Purge water / NaOH	DN20/DN25 lined valve + reducers/adaptors	P&ID-001	0.1	33	0.45	58	-	-	3-way service/backwash switching valve; vendor GA TBC.	F
HV174	Manual isolation near C-101A side branch	Manual gate valve	-	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Circulating water / HCl	DN20 plastic line	P&ID-001	0.1	33	0.45	58	-	-		F
HV175	Manual isolation near C-101B side branch	Manual gate valve	-	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Circulating water / HCl	DN20 plastic line	P&ID-001	0.1	33	0.45	58	-	-		F
HV176	Manual isolation near C-102A side branch	Manual gate valve	-	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Purge water / NaOH	DN20 plastic line	P&ID-001	0.1	33	0.45	58	-	-		F
HV177	Manual isolation near C-102B side branch	Manual gate valve	-	Vendor std.	-	CPVC/PVDF/PFA-lined wetted parts	Purge water / NaOH	DN20 plastic line	P&ID-001	0.1	33	0.45	58	-	-		F
ESDV102	Ion-exchange / side-stream emergency isolation to T-105	Actuated on/off valve	-	Vendor std.	-	PVC/CPVC/PVDF/PFA-lined wetted parts	Waste / side-stream water	Size per connected line; plastic/lined adaptor	P&ID-001	0.1	33	0.45	58	-	-	Fail closed; vendor to confirm actuator and GA.	F
HV178A	P-106A suction isolation — HCl feed	Manual gate valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	HCl	DN20 plastic/RF adaptor	P&ID-001	0.1	25	0.45	50	-	-		F
HV178B	P-106B suction isolation — HCl feed	Manual gate valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	HCl	DN20 plastic/RF adaptor	P&ID-001	0.1	25	0.45	50	-	-		F
HV179A	P-106A suction drain — HCl feed	Manual drain valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	HCl	DN20 plastic/RF adaptor	P&ID-001	0.1	25	0.45	50	-	-		F
HV179B	P-106B suction drain — HCl feed	Manual drain valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	HCl	DN20 plastic/RF adaptor	P&ID-001	0.1	25	0.45	50	-	-		F
NRV106A	P-106A discharge non-return — HCl feed	Check valve	2.0 m³/h assumed	Vendor std.	-	PVC wetted parts	HCl	DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-		F
NRV106B	P-106B discharge non-return — HCl feed	Check valve	2.0 m³/h assumed	Vendor std.	-	PVC wetted parts	HCl	DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-		F
HV180A	P-106A discharge isolation — HCl feed	Manual gate valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	HCl	DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-		F
HV180B	P-106B discharge isolation — HCl feed	Manual gate valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	HCl	DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-		F
HV187A	P-107A suction isolation — NaOH feed	Manual gate valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	NaOH	DN20 plastic/RF adaptor	P&ID-001	0.1	25	0.45	50	-	-		F
HV187B	P-107B suction isolation — NaOH feed	Manual gate valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	NaOH	DN20 plastic/RF adaptor	P&ID-001	0.1	25	0.45	50	-	-		F
HV188A	P-107A suction drain — NaOH feed	Manual drain valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	NaOH	DN20 plastic/RF adaptor	P&ID-001	0.1	25	0.45	50	-	-		F
HV188B	P-107B suction drain — NaOH feed	Manual drain valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	NaOH	DN20 plastic/RF adaptor	P&ID-001	0.1	25	0.45	50	-	-		F
NRV107A	P-107A discharge non-return — NaOH feed	Check valve	2.0 m³/h assumed	Vendor std.	-	PVC wetted parts	NaOH	DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-		F
NRV107B	P-107B discharge non-return — NaOH feed	Check valve	2.0 m³/h assumed	Vendor std.	-	PVC wetted parts	NaOH	DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-		F
HV189A	P-107A discharge isolation — NaOH feed	Manual gate valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	NaOH	DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-		F
HV189B	P-107B discharge isolation — NaOH feed	Manual gate valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	NaOH	DN20 plastic/RF adaptor	P&ID-001	3	25	3.35	50	-	-		F
FCV106	HCl backwash / regeneration flow control	Control valve	2.0 m³/h	Vendor std.	-	WCB lined F46/PFA; FEP-lined trim	HCl	DN20 150# RF + plastic adaptor	P&ID-001	2.5	25	2.85	50	-	-	Fail state per P&ID; vendor to confirm Cv, actuator and GA.	F
ESDV103	HCl emergency isolation after FCV106	Actuated on/off valve	-	Vendor std.	-	PVDF/PFA/PTFE wetted parts	HCl	DN20 plastic/RF adaptor	P&ID-001	2.5	25	2.85	50	-	-	Fail closed; vendor to confirm actuator and GA.	F
HV186	HCl branch isolation / drain near HC-048 to ion-exchange train	Manual drain valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	HCl	DN20 plastic line	P&ID-001	2.5	25	2.85	50	-	-		F
FCV107	NaOH backwash / regeneration flow control	Control valve	-	Vendor std.	-	WCB lined F46/PFA; FEP-lined trim	NaOH	DN20 150# RF + plastic adaptor	P&ID-001	2.5	25	2.85	50	-	-	Fail state per P&ID; vendor to confirm Cv, actuator and GA.	F
ESDV104	NaOH emergency isolation after FCV107	Actuated on/off valve	-	Vendor std.	-	PVDF/PFA/PTFE wetted parts	NaOH	DN20 plastic/RF adaptor	P&ID-001	2.5	25	2.85	50	-	-	Fail closed; vendor to confirm actuator and GA.	F
HV202	T-105 wastewater branch isolation / drain to WW-043	Manual drain valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	Waste / purge water	DN32 plastic line	P&ID-001	2.5	25	2.85	50	-	-		F
HV201	T-105 bottom solids / precipitate discharge	Manual drain valve	-	Vendor std.	-	PVC/CPVC/PVDF lined wetted parts	Solid precipitate	DN32 plastic line	P&ID-001	0	25	0.35	50	-	-		F
ESDV105	Wastewater / purge-water emergency isolation at T-106 inlet	Actuated on/off valve	-	Vendor std.	-	PVC/CPVC or PFA-lined wetted parts	Waste / purge water	DN32 plastic/RF adaptor	P&ID-001	0	25	0.35	50	-	-	Fail closed; vendor to confirm actuator and GA.	F
TCV102	T-106 purge/wastewater cooling tank temperature control valve	Control valve	-	Vendor GA	-	WCB lined F46/PFA; FEP-lined trim	Purge / waste water	DN32 150# RF + plastic adaptor	P&ID-001	0	25	0.35	50	-	-	Fail state per P&ID; vendor to confirm Cv, actuator and GA.	F

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4.1.2 Line list

REV	REVISION DESCRIPTION	DATE	BY	CHK	APPR
C	MODIFIED FOR FINAL SUBMISSION	2026/4/20	Weicheng Qin	Chen Ba	Yijie Yao
B	UPDATED TO MATCH VENDOR SELECTION	2026/4/18	Weicheng Qin	Shubo Ma	Chen Ba
A	UPDATED TO MATCH PFD/P&ID + HX/CT CALCS	2026/4/15	Weicheng Qin	Weicheng Qin	Weicheng Qin

PROJECT: Crude Glycerol Cooling System

JOB NO.: CHEE2050

The University of Nottingham
UNITED KINGDOM · CHINA · MALAYSIA

LINE LIST

DOCUMENT NUMBER
LL-001

	PIPELINE DESCRIPTION							OPERATING		DESIGN		FIELD TEST		FLEX	INSULATION AND PAINT					ISSUE/
LINE No.	NOM. LINE SIZE (DN) (in)	LINE SPEC.	FLUID CODE	FLUID PHASE	FROM	TO	P&ID NO.	PRES. (barg)	TEMP. (°C)	PRES. (barg)	TEMP. (°C)	MEDIUM	PRES. (1.5 X D. PRES) (barg)	TEMP. HI / LO (°C)	TEMP. (°C)	PAINT CLASS	INSUL CLASS	INSUL. THK (in)	SPECIAL REQUIREMENTS	REV
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
001	40	A1PL	HG	L	PMB BL	T-101 in	P&ID-001	1	110	1.35	135	H	2.03		110		H	1.0		E
002	40	A1PL	HG	L	T-101 out	P-101A suct	P&ID-001	0.1	110	0.45	135	H	0.68		110		H	1.0	Pump nozzle DN40/25; line remains DN40.	F
003	40	A1PL	HG	L	T-101 out	P-101B suct	P&ID-001	0.1	110	0.45	135	H	0.68		110		H	1.0	Pump nozzle DN40/25; line remains DN40.	F
004	40	A1PL	HG	L	P-101A disch	HG feed hdr	P&ID-001	2	110	2.35	135	H	3.53		110		H	1.0	Pump nozzle DN40/25; line remains DN40.	F
005	40	A1PL	HG	L	P-101B disch	HG feed hdr	P&ID-001	2	110	2.35	135	H	3.53		110		H	1.0	Pump nozzle DN40/25; line remains DN40.	F
006	40	A1PL	HG	L	HG feed hdr	E-101 sh in	P&ID-001	2	110	2.35	135	H	3.53		110		H	1.0		E
007	40	A1PL	CG	L	E-101 sh out	P-102A suct	P&ID-001	0.3	55	0.65	80	H	0.98		55		H	1.0	Pump nozzle DN40/25; line remains DN40.	F
008	40	A1PL	CG	L	E-101 sh out	P-102B suct	P&ID-001	0.3	55	0.65	80	H	0.98		55		H	1.0	Pump nozzle DN40/25; line remains DN40.	F
009	40	A1PL	CG	L	P-102A disch	CG hdr	P&ID-001	1.5	55	1.85	80	H	2.78		55		H	1.0	Pump nozzle DN40/25; line remains DN40.	F
010	40	A1PL	CG	L	P-102B disch	CG hdr	P&ID-001	1.5	55	1.85	80	H	2.78		55		H	1.0	Pump nozzle DN40/25; line remains DN40.	F
011	40	A1PL	CG	L	CG hdr	T-102 in	P&ID-001	1.5	55	1.85	80	H	2.78		55		H	1.0		E
012	40	A1PL	CG	L	T-102 out	Prod tk BL	P&ID-001	0.1	55	0.45	80	H	0.68		55		H	1.0		E
013	32	A1WL	CW	L	WT plant BL	P-103 hdr	P&ID-001	1	33	1.35	58	H	2.03		33		C	1.0	P-103 make-up: 0.527 m³/h normal; selected JIH25-20-125, 2.0 m³/h @ 20 m; DN32 line with reducers.	F
014	32	A1WL	CW	L	P-103 hdr	P-103B suct	P&ID-001	1	33	1.35	58	H	2.03		33		C	1.0	P-103 make-up: 0.527 m³/h normal; selected JIH25-20-125, 2.0 m³/h @ 20 m; DN32 line with reducers.	F
015	32	A1WL	CW	L	P-103A disch	CW make-up hdr	P&ID-001	2	33	2.35	58	H	3.53		33		C	1.0	P-103 make-up: 0.527 m³/h normal; selected JIH25-20-125, 2.0 m³/h @ 20 m; DN32 line with reducers.	F
016	32	A1WL	CW	L	P-103B disch	CW make-up hdr	P&ID-001	2	33	2.35	58	H	3.53		33		C	1.0	P-103 make-up: 0.527 m³/h normal; selected JIH25-20-125, 2.0 m³/h @ 20 m; DN32 line with reducers.	F
017	32	A1WL	CW	L	CW make-up hdr	T-103 in	P&ID-001	2	33	2.35	58	H	3.53		33		C	1.0	P-103 make-up: 0.527 m³/h normal; selected JIH25-20-125, 2.0 m³/h @ 20 m; DN32 line with reducers.	F
018	32	A1WL	CW	L	Chem dose BL	T-103 in	P&ID-001	1	33	1.35	58	H	2.03		33		C	1.0		E
019	80	A1WL	CW	L	T-103 noz	P-104 suct riser	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0		E
020	80	A1WL	CW	L	T-103 suct riser	P-104 hdr	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0		E
021	80	A1WL	CW	L	P-104 hdr	P-104A suct	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0	Pump nozzle DN80/65; line remains DN80.	F
022	80	A1WL	CW	L	P-104 hdr	P-104B suct	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0	Pump nozzle DN80/65; line remains DN80.	F
023	80	A1WL	CW	L	P-104A disch	Mix CW hdr	P&ID-001	2	33	2.35	58	H	3.53		33		C	1.0	Pump nozzle DN80/65; line remains DN80.	F
024	80	A1WL	CW	L	P-104B disch	Mix CW hdr	P&ID-001	2	33	2.35	58	H	3.53		33		C	1.0	Pump nozzle DN80/65; line remains DN80.	F
025	80	A1WL	CW	L	Mix CW hdr	E-101 tb in	P&ID-001	2	33	2.35	58	H	3.53		33		C	1.0		E
026	80	A1WL	CW	L	E-101 tb out	CT-101 in riser	P&ID-001	1	40	1.35	65	H	2.03		40		C	1.0		E
027	80	A1WL	CW	L	CT-101 in riser	CT-101 spray in	P&ID-001	1	40	1.35	65	H	2.03		40		C	1.0		E
028	80	A1WL	CW	L	CT-101 basin	P-105A suct	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0	Pump nozzle DN80/65; line remains DN80.	F
029	80	A1WL	CW	L	CT-101 basin	P-105B suct	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0	Pump nozzle DN80/65; line remains DN80.	F
030	80	A1WL	CW	L	P-105A disch	Circ CW hdr	P&ID-001	2	33	2.35	58	H	3.53		33		C	1.0	Pump nozzle DN80/65; line remains DN80.	F
031	80	A1WL	CW	L	P-105B disch	Circ CW hdr	P&ID-001	2	33	2.35	58	H	3.53		33		C	1.0	Pump nozzle DN80/65; line remains DN80.	F
032	80	A1WL	CW	L	Circ CW hdr	T-104 in	P&ID-001	2	33	2.35	58	H	3.53		33		C	1.0		E

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033	80	A1WL	CW	L	T-104 out	F-101 in	P&ID-001	0.2	33	0.55	58	H	0.83		33		C	1.0			E
034	80	A1WL	CW	L	F-101 out	IX svc hdr	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0			E
035	32	U0ES	CW	L	F-101 out hdr	IX top hdr	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0			E
036	20	A1WL	CW	L	IX top hdr	C-101 top mfd	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0			E
037	20	A1WL	CW	L	C-101 btm mfd	Svc out hdr	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0			E
038	80	A1WL	CW	L	T-103 / LCV105	IX feed hdr	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0			E
039	20	U0BS	CW	L	Upr hdr	C-101 top swg	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0			E
040	20	U0DS	CW	L	Upr hdr	C-102 top swg	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0			E
041	20	U0DS	CW	L	C-102A top	C-102B top	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0			E
042	20	U0DS	CW	L	C-102A btm	C-102B btm	P&ID-001	0.1	33	0.45	58	H	0.68		33		C	1.0			E
043	32	U0ES	WW	L	T-105 drain	WW hdr	P&ID-001	0	33	0.35	58	H	0.52		33		C	1.0			E
044	20	U0BS	HC	L	HCl BL	P-106A suct	P&ID-001	0.1	25	0.45	50	B	0.68		25		PP	0.5			F
045	20	U0BS	HC	L	HCl BL	P-106B suct	P&ID-001	0.1	25	0.45	50	B	0.68		25		PP	0.5			F
046	20	U0BS	HC	L	P-106A disch	HC hdr	P&ID-001	3	25	3.35	50	B	5.03		25		PP	0.5			F
047	20	U0BS	HC	L	P-106B disch	HC hdr	P&ID-001	3	25	3.35	50	B	5.03		25		PP	0.5			F
048	20	U0BS	HC	L	HC hdr	IX acid regen	P&ID-001	2.5	25	2.85	50	B	4.28		25		PP	0.5			E
049	20	U0DS	SH	L	NaOH BL	P-107A suct	P&ID-001	0.1	25	0.45	50	B	0.68		25		PP	0.5			F
050	20	U0DS	SH	L	NaOH BL	P-107B suct	P&ID-001	0.1	25	0.45	50	B	0.68		25		PP	0.5			F
051	20	U0DS	SH	L	P-107A disch	SH hdr	P&ID-001	3	25	3.35	50	B	5.03		25		PP	0.5			F
052	20	U0DS	SH	L	P-107B disch	SH hdr	P&ID-001	3	25	3.35	50	B	5.03		25		PP	0.5			F
053	20	U0DS	SH	L	SH hdr	IX caust reg	P&ID-001	2.5	25	2.85	50	B	4.28		25		PP	0.5			E
054	32	U0ES	WW	L	IX regen eff	T-106 in	P&ID-001	0	25	0.35	50	H	0.52		25		PP	0.5			E
055	32	U0ES	WW	L	T-106 out	WW hdr / river	P&ID-001	0	25	0.35	50	H	0.52		25		PP	0.5			E

GCW FrontSheet

4.2 Data Sheet

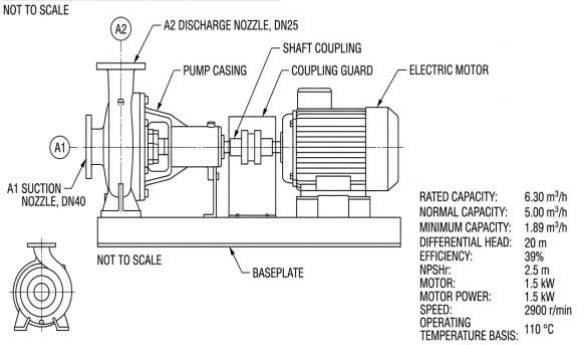
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GCW FrontSheet

1	CLIENT:	PMB Biodiesel Co. Ltd.							
2	LOCATION:	PMB Biodiesel Co. Ltd. site, approx. 29.8003							
3	SERVICE:	Hot crude glycerol feed to E-101							
4	PROJECT:	Crude Glycerol Cooling System					UNIT NO.:	Cooling System	
5	JOB NO.:	CHEE2050					ITEM NO.:	P-101A	NO. REQD.: 2
6									
7	Rev	PUMP TYPE : Horizontal Centrifugal					PUMP DRIVER TYPE :	Electric Motor 2-pole	
8									
9									
10		**FLUID DATA**							
11		Name : Crude glycerol					SP.GR (flowing) :	1.2	
12		Composition : 88.3 wt% glycerol /					Corrosive / Erosive :	No / No due to ean hot liquid; no erosive solids expect	
13		11.3 wt% water / 0.4 wt% methanol					Toxic / Flammable :	Yes / Yes due to anol component; conservative toxic/fla	
14		**OPERATING DATA**							
15		FLOW CASE				NORMAL	RATED	MINIMUM	OTHER
16									
17		Pumping temp. :	Normal	oc		110	110	110	
18			Maximum	oc		110	110	110	
19			Minimum	oc		110	110	110	
20		Vapour pressure @ P.T.(max) :		bara		0.6	0.6	0.6	
21		Viscosity @ P.T. :		cP		8.2	8.2	8.2	
22		Capacity @ P.T. :		m3/h		5	6.3	1.89	
23		NPSH (available) : Note 1		m		5	4.5	5.5	
24		Suction pressure :	Normal	barg		0.25	0.2	0.3	
25			Maximum	barg		0.35	0.3	0.4	
26		Discharge pressure :		barg		2.6	2.55	2.65	
27		Differential head :		m		20	20	20	
28		System design pressure		barg		3.1			
29		**ADDITIONAL PUMP DATA**							
30		Materials of Construction :	Casing -	316 SS		Special Seal Requirements :	Mechanical seal; seal type by vendor		
31			Internals -	316 SS					
32		Efficiency (assumed) :		39% (catalogue)					
33		Pump cooling/heating jacket required :	No						
34		**NOZZLE CONNECTION SUMMARY**							
35		Number	Size DN	Flange Finish	Flange Rating	Service			
36		A1	DN40	RF	150#	SUCTION			
37		A2	DN25	RF	150#	DISCHARGE			
38				–					
39				-					
40		**PROCESS NOTES**				**SKETCH**			
41									
42									
43		1. NPSH is referenced to the top of the pump plinth / bottom of the baseplate.							
44		Selected model: JIH40-25-125 / 1.5 kW-2P, SS316L stainless steel centrifugal							
45		pump.							
46		Nozzles: suction DN40, discharge DN25; provide DN25-to-DN40 reducer / expander							
47		at the line interface as required.							
48		Final seal arrangement, elastomers, minimum continuous flow and NPSHr shall be							
49		confirmed by the vendor.							
50									
51									
52									
53									
54									
55									
56									
57									
58									

HORIZONTAL CENTRIFUGAL PUMP SKICH P-101A

NOT TO SCALE



RATED CAPACITY:	6.30 m³/h
NORMAL CAPACITY:	5.00 m³/h
MINIMUM CAPACITY:	1.89 m³/h
DIFFERENTIAL HEAD:	20 m
EFFICIENCY:	39%
NPSHr:	2.5 m
MOTOR:	1.5 kW
MOTOR POWER:	1.5 kW
SPEED:	2900 r/min
OPERATING TEMPERATURE BASIS:	110 °C

REV.	DESCRIPTION	DATE	BY	CHECK	APPROVED	DOCUMENT NO.
C	Finalised for submission; datasheet aligned with the selec	4.24	Weicheng Qin	Weicheng Qin	Chen Ba	
B	Updated in line with current P&ID and supporting pump c	4.2	Weicheng Qin	Chen Ba	Weicheng Qin	DS-P-101A-3
A	Initial issue for preliminary pump specification.	4.8	Weicheng Qin	Chen Ba	Weicheng Qin	
						SHEET 1 OF 1

GCW FrontSheet

CLIENT	PMB Biodiesel Co. Ltd.
PROJ. NAME	Crude Glycerol Cooling System
LOCATION	PMB Biodiesel Co. Ltd. site, approx. 29.800350, 12
JOB NO.	CHEE2050

CONTROL VALVE DATA SHEET

The University of Nottingham
UK UNITED KINGDOM · CHINA · MALAYSIA

CONTROL VALVE NO. **LCV-101**

PLANT AREA		Cooling System
P & ID NO.		P&ID-001
LINE NO.		HG-002-40-A1PL-H
SIZE & SCHEDULE		DN40 / Sch 40S
VALVE SPEC		CSS-JT / 150# RF / 316L SS
VALVE TYPE	☐ REQUIRED	Jacketed globe/plug
PROCESS FLUID		Hot crude glycerol
DESIGN PRESSURE		1
FLEX HIGH/LOW TEMP		135 / ambient °C
INLET FLUID STATE		Liquid
OPERATION		Level control / pneumatic
SHUTOFF CLASS		FCI 70-2 IV min
FAIL POSITION		FO
HANDWHEEL		No

PROCESS DATA	UNITS	MAX FLOW	NORM FLOW	MIN FLOW	ALT FLOW
MASS FLOWRATE	kg/h	7,560	6,000	3,000	No alt case
OPERATING TEMP	°C	110	110	110	No alt case
P1 (UPSTREAM)	barg	0.4	0.35	0.25	No alt case
P2 (DOWNSTREAM)	barg	0.25	0.25	0.2	No alt case
VISCOSITY (LIQ)	cP	8.2	8.2	8.2	No alt case
CRITICAL PRESS (LIQ)	bara		"CRITICAL" SERVICE ONLY		
UPSTREAM DENSITY	kg/m3	1200	1200	1200	No alt case
MOLECULAR WT (VAP)	g/mol	Liquid only	Liquid only	Liquid only	Liquid only
VAPOR PRESS (LIQ)	bara	0.6	0.6	0.6	No alt case
DP TO SIZE ACTUATOR	barg	0.15	0.1	0.05	No alt case
Cvc	-	21	20	14	No alt case
EST SOUND PRESS	dBA	By Vendor	By Vendor	By Vendor	No alt case
EST VALVE BODY SIZE	DN	DN32	DN32	DN32	No alt case

NOTES & SPECIAL REQUIREMENTS:

Service: LCV-101 on HG-002 between T-101 and P-101A/B suction.

Valve basis: jacketed globe / plug control valve, DN32 body on DN40 line; use DN40–DN32 reducers at the line interface.

Fail action: FO for the T-101 outlet level-control valve.

The jacket is provided for temperature maintenance / viscosity control of hot crude glycerol.

Final trim, actuator, jacket connections and shut-off class shall be confirmed by the vendor.

REV/ ISSUE	0	1	2	3	4	5	6	7	8
DATE	2026.4.8	2026 4.10	2026 4.18	2026 4.24					
BY	Weicheng Qin	Weicheng Qin	Weicheng Qin	Weicheng Qin					
CHECKED	Weicheng Qin	Chen Ba	Weicheng Qin	Weicheng Qin					
APPROVED	Weicheng Qin	Weicheng Qin	Chen Ba	Chen Ba					

4.3 Safety, robustness and turndown

Safety, robustness and turndown have been considered at FEED level for E-101 and the associated utility / control items retained in this equipment-specification package. This section demonstrates that the selected equipment basis has appropriate operating margins, and that the relevant safeguards and robustness checks are referenced to the supporting calculations.

For E-101, robustness is demonstrated by the shell-side heat-transfer coefficient sensitivity in Appendix B. The exchanger continues to satisfy the required process duty when the shell-side coefficient is reduced by 10%, 15% and 20%, and first fails at a 25% reduction. The hydraulic checks also pass: the tube-side pressure drop is 0.36308 bar against the adopted 0.700 bar limit, and the shell-side pressure drop is 0.01241 bar against the adopted 0.300 bar limit. The end-dead-length / tubesheet-allowance sensitivity also retains a positive thermal margin. Vent and drain connections are retained on both shell and tube sides to support commissioning, shutdown, drainage and maintenance, as recorded in Appendix C.

For CT-001, turndown and seasonal operation are addressed through the cooling-tower summer sizing basis, water-balance calculation, winter anti-overcooling philosophy and bypass / fan-control basis in Appendix D. The retained COC = 4 cooling-water basis is supported by the design-basis LSI check in Appendix D.3. The positive LSI cases show that calcium-carbonate scaling is possible under harder-water conditions, so the utility package retains blowdown control, pH/conductivity monitoring through ion analyser, and provision for scale inhibitor dosing. The associated CT-001 line, valve and control interfaces are summarised in Appendix E.

For F-101, robustness is addressed by the FEED-stage pressure sand-bed filter sizing check in Appendix F. The selected 2000 mm vessel gives an actual filtration velocity of approximately 10.3 m/h at the retained water-treatment flow basis, which is within the adopted vendor filtration range. The filter is retained as an associated water-treatment item only; final media grading, clean/dirty pressure drop, backwash sequence, underdrain design and mechanical details remain vendor-confirmation items. The operating rationale for the retained F-101 treatment route, cleaning / backwash provisions and vendor-confirmation items is summarised in Appendix F.8.

For the ion-exchange package, the water-treatment equipment shown on P&ID-001 is retained as an associated rough-estimate utility package rather than as the selected major equipment item. Its safety and operating basis is therefore limited to service isolation, regeneration interface, acid/caustic handling assumptions and final confirmation against make-up / circulating-water chemistry, as recorded in Appendix F. The ion-exchange resin-form basis, duty / standby philosophy, regeneration interface and cleaning / backwash limitations are summarised in Appendix F.8, with final operating-cycle and resin-cleaning requirements remaining subject to resin-supplier confirmation. The LSI check supports the retained cooling-water treatment basis and does not remove or replace the IX package.

For P-101A/B, robustness is provided by the duty/standby pump arrangement, rated-flow margin, NPSH check, minimum-flow basis and motor-power check retained in Appendix G. For LCV-101, turndown and safety basis are addressed through the maximum / normal / minimum flow cases, the selected jacketed globe / plug valve basis, DN40–DN32 line interface, and fail-open action for the T-101 outlet level-control valve, as retained in Appendix G. Final pump curve, minimum continuous

stable flow, valve trim, actuator sizing, shut-off class and vendor mechanical details remain vendor-confirmation items.

5 Plant Layout and Plot Plan

5.1 Layout overview

The layout of the crude glycerol cooling plant can be divided into two main sections. The first is the core module within the 30 m² designated area, which is mainly used for crude glycerol transfer, cooling and discharge. A vertical arrangement is adopted in this area, with the pumps for the crude glycerol transfer and outlet line located on the first floor and the heat exchanger on the second floor. This arrangement was chosen because the heat exchanger is the key equipment in the crude glycerol cooling line. By placing the pumps and the heat exchanger in the same module, the crude glycerol line becomes more compact and easier to manage. An anti-blast wall is provided around this core module to separate it from the surrounding area and reduce the potential impact on nearby equipment if a failure occurs in the high-temperature heat exchanger. Since the crude glycerol enters the exchanger at 110°C and leaves at 55°C, this area is one of the main heat-load zones in the new unit.

The second section includes the storage and cooling-water-treatment facilities arranged around the 30 m² core area. T101 and T102, which act as the crude glycerol storage tanks, are located close to the heat exchanger but outside the 30 m² designated area. And a nearby pump house is provided for pump operation and maintenance. In the centre of the layout, a three-level structure is used to accommodate the main cooling-water-treatment equipment. T105 is located on the first level, C101A/B and C102A/B on the second level, and F101 on the third level. This arrangement makes use of gravity flow, allowing the water to move through the treatment line more smoothly. At the outer edge of the layout, T104, T103, T106, and the cooling tower CT101 are installed, with an anti-blast wall also used to separate the cooling tower from nearby equipment. The layout also includes access routes for inspection and maintenance, together with an outer main road designed to allow heavy vehicle access.

5.2 Equipment area table

Equipment Number	Dimensions (m) L × W × H/ ΦD × H	Footprint area (m²)	Allocated layout area incl. shared spacing (m²)
E101	4.6×1.0×1.5	4.60	30
CT101	2.2 × 2.2×3.8	4.84	16
F101	1.2 × 0.7 × 0.7	0.84	46.24
C101A/B	Φ 2 × 2.1	3.14(each)	11.56(each)
C102A/B	Φ 0.315 × 2.96	0.078(each)	11.56(each)
T101	Φ 2.5 × 2.75	4.91	42.25
T102	Φ 2.5 × 2.75	4.91	42.25
T103	2.5 × 2 × 1.8	5.00	36
T104	3.0 ×2.5 ×2.1	7.50	36
T105	Φ 2.3 × 1.5	4.15	46.24

T106	5.2 × 5.2 × 2.5	27.04	81

The footprint area represents the actual plan area occupied by the equipment itself. The allocated layout area including shared spacing was determined from the layout drawing based on its actual scale. The open space between adjacent equipment was measured and equally apportioned to the neighbouring equipment items. As a result, the final column reflects both the equipment footprint and the shared spacing allowance adopted in the layout. This approach incorporates the safety separation distance between equipment and provides a more realistic estimate of the area required in the proposed arrangement.

For E101, the allocated layout area is large because the exchanger is located within the core module and its surrounding space also needs to accommodate practical spacing, access and separation within that module. For F101 and T105, these values are relatively large because these items are located within a more open part of the layout, so a larger proportion of the surrounding shared space was allocated to them.

5.3 Area limitation and equipment arrangement

Based on this design, the 30 m² designated area should mainly be reserved for the core crude glycerol cooling equipment rather than for all related equipment. This is because the area has already been allocated to the main crude glycerol-side equipment, including the pumps, the heat exchanger, and the anti-blast wall around the core module. In addition, space is still needed for basic access, inspection and maintenance around the equipment. For this reason, the designated area is not suitable for accommodating additional large or non-core equipment.

Heat dissipation is another reason for this arrangement. Since crude glycerol enters the heat exchanger at a relatively high temperature, the exchanger area is already one of the main local heat-load zones in the plant. Therefore, the core area should remain as open as possible to maintain good ventilation and avoid unnecessary thermal interference.

If the cooling tower were placed too close to the heat exchanger, the heat released during operation could raise the local ambient temperature around the exchanger and cause thermal interference. This would not be favourable for the cooling performance of the unit. Large storage tanks are not continuous heat sources, but their large footprint would still reduce the open space around the core module. This could restrict airflow, reduce the effectiveness of local heat dissipation, and make equipment access more difficult.

For these reasons, the cooling tower CT101, the large storage tanks, and the equipment associated with the cooling water line are arranged outside the 30 m² designated area, while the designated area itself is kept mainly for the core crude glycerol cooling equipment. These external units are connected back to the core module through the pipe rack. This gives a clearer separation between the crude glycerol line and the cooling water line, while also improving safety and access for operation and maintenance.

GCW FrontSheet

6 Process Safety Assessment/Evaluation

6.1 Summary and Conclusions

During the safety assessment, the team first identified 9 materials including methanol, glycerol and hydrochloric acid—used in the project that could pose a risk, based on project requirements and the PFD. In accordance with Regulation (EC) No 1272/2008 — Classification, Labelling and Packaging of Substances and Mixtures to classify the risks of each material in accordance with GHS standards, and listed their significant risk statements and key precautionary statements to ensure that personnel could gain a preliminary understanding of potential risks and implement appropriate preventive measures. Subsequently, in the Hazards Inventory, the entire heat exchanger system was categorized into four major systems: the Glycerol Feeding System (System 1), Water Pretreatment System (System 2), Heat Exchange and Cooling Systems (System 3) and Wastewater Treatment and Recycling Systems (System 4). Potential hazard sources (such as Toxic Substances, High-Speed Rotational Equipment, High Pressure, etc.) are examined under the three main categories of Material, Equipment and Process, with notes provided on why the hazard source exists (e.g., 'Toxic Substances' because methanol is toxic; inhalation of its vapors or skin contact may cause blindness or death). Finally, each system has a separate HAZID worksheet, which identifies the risks for that system based on the Hazards Inventory, analyses their trigger mechanisms and potential consequences, and sets out specific actions.

During this process, the team identified that methanol flammability and toxicity, as well as risks associated with high-speed rotating equipment, occurred frequently and posed significant hazards. Consequently, we recommend installing methanol vapor sensors and reinforcing the equipment casings; these recommendations were duly accepted.

However, regarding the risk of explosion, the installation of explosion-proof walls was deemed impractical due to space constraints; this issue therefore remains to be resolved.

After completing the HAZID analysis, the safety team carried out a further HAZOP study based on the actions listed in HAZID worksheet to mitigate potential hazard consequences. In addition, a detailed P&ID was developed based on the PFD, which included additional process parameters, instrumentation and safety devices.

In the HAZOP analysis, safety team has divided the whole process into 12 different nodes: Crude Glycerol Feeding Line (node 1), Cooling Tower (node 2), Shell & Tube Heat Exchanger (node 3), Cooled Glycerol Transmit Line (node 4), Cooling Water Treatment Line (node 5), Cooling Water Feeding Line (node 6), Recycled Water Feeding Line (node 7), Waste Water Discharge Line (node 8), Cation Exchange Line (node 9), Anion Exchange Line (node 10), Anion Exchange Column (node 11) and Ion Exchange Regenerant Treatment Line (node 12). Each node includes equipment and process streams that perform similar functions and share similar fluid properties.

Based on updated P&ID, several potential hazards were identified, including high-temperature process streams, possible equipment damage, health and safety issues and environmental impacts. After considering all possible risks that has been identified, safety team listed several recommendations to improve P&ID drawing to reduce risk level and ensure all the safety status are "closed".

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The installation of explosion-proof walls problem mentioned in HAZID analysis has been carried out in plant layout design part: explosion-proof walls have been installed near the cooling tower CT-101 and heat exchanger E-101. Additional recommendations have been written in the" Outstanding action list" part, which majorly focus on operational and monitoring measures.

6.2 Introduction

6.2.1 Methodology Overview

HAZID is a risk assessment tool used to identify potential significant risk sources in the early stages of a project. It is primarily divided into Materials Assessment, Hazards Inventory and Qualitative Risk Assessment (i.e. the HAZID worksheet). In the Materials Risk Assessment, the team first conducts physical and health hazard assessments for all chemicals involved in the system in accordance with GHS classification standards, identifying the potential risk statements and key precautionary statements for each chemical. Subsequently, in the Hazards Inventory, the equipment shown in the Process Flow Diagram (PFD) is categorized into different functional systems, and potential hazard sources are examined across three main areas: Materials, Equipment and Process. Finally, in the HAZID worksheet, the hazards identified for each system were analyzed in terms of their exposure pathways and consequences. A risk rating was derived by multiplying the 'Likelihood' and 'Severity' factors in the risk matrix. For high-risk items, preliminary mitigation measures were proposed

HAZOP is a more systematic and detailed post-design analysis method that focuses on identifying the consequences of process deviations from the design intent, thereby preventing major issues from occurring in the future and identifying outstanding issues, improvement timelines and methods. HAZOP begins by dividing the entire engineering design into distinct nodes, analyzing the process parameters at each node (such as flow, pressure, temperature and level), and classifying potential deviations according to predefined prompts (such as None, More, Less, Reverse, etc.). Upon identifying potential deviations, the method lists the possible causes and consequences. It then assesses the safety barriers already present in the current design. If existing protective measures are insufficient to reduce the risk to an acceptable level, specific improvement actions are proposed, their status is tracked, and a defined timeline for implementation is provided, followed by an evaluation of their status and outcomes.

In safety assessments, HAZID primarily focuses on external and global hazards, whilst HAZOP focuses on internal systems and operational deviations. The combination of the two ensures comprehensive safety coverage from the early to late stages of design, providing a scientific basis for subsequent design optimization.

6.2.2 Scope of Study

The scope of this safety assessment is explicitly defined as the Crude Glycerol Cooling and Water Treatment plant section. While identified as a specific section, the study maintains functional integrity by encompassing the complete process cycle from feed intake to effluent discharge.

Specifically, the process boundary originates from the PMB Biodiesel Co. Ltd feed stream and terminates at the Cooled Glycerin Product Tank.

The methodology for partitioning systems and nodes is predicated upon the technical logic of the PFD and detailed P&ID drawings (see Appendices L and M). For the HAZID study, the process was categorized into broad functional systems based on primary process lines. In the subsequent HAZOP analysis, a higher level of granularity was applied by dividing the P&ID into 12 discrete nodes, defined by similar equipment functions and fluid properties.

This study adopts a "Full Reporting" format, ensuring a systematic audit of all potential deviations. Beyond primary safety hazards, the analysis rigorously addresses operability and HSE factors, such as shutdown maintenance, equipment downtime, and utility consumption efficiency. Through an iterative collaboration, the design team has integrated the HAZOP mitigation actions into the latest P&ID revision. Consequently, hardware-related risks have been mitigated to the lowest practicable level, shifting the focus of future safety management toward standardized operational procedures.

6.2.3 Assumptions

Under the site conditions, the feed flow rate of crude glycerol is assumed to be 6,000 kg/h, with a feed temperature of 110 °C; The crude glycerol output capacity is 6,000 kg/h, with an outlet temperature of 55 °C; the composition of the crude glycerol feed is 88.3 wt% glycerol, 11.3 wt% water and 0.4 wt% methanol, and no phase changes are expected to occur. It is also assumed that the industrial site meets the necessary requirements.

With regard to standard assumptions, it is assumed that the process water complies with the water quality standards of Mainland China—Zhejiang Province. The team designed the equipment and piping in accordance with TEMA Standard 2020 (Classes R/C/B) and the ASTM A213:316L stainless steel heat exchanger tube material standard. For the discharge of cooling water, reference was made to GB 8978 ‘Comprehensive Standard for Wastewater Discharge’ and GB 18918 ‘Pollutant Discharge Standard for Urban Sewage Treatment Plants’. Regarding the site layout for the equipment, reference was made to GB 50187-2012 ‘Code for General Site Planning of Industrial Enterprises’. For the HAZID analysis, Regulation (EC) No 1272/2008 – Classification, Labelling and Packaging of Substances and Mixtures, were used as standard references. It was assumed that the team would conduct design, assessment and operational activities in full compliance with these standards.

In terms of regulations, the team referred to Regulation (EC) No 1272/2008 – Classification, Labelling and Packaging of Substances and Mixtures, and assumed that the assessment and design were carried out in full compliance with the legal provisions.

The health and safety team assumes that all personnel will adhere to the NOTABLE PRECAUTIONARY STATEMENTS set out in the HAZID worksheet, that designers will implement corrective measures in accordance with the safety measures outlined in the HAZID and HAZOP worksheets, and that the work will be delivered on schedule.

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6.3 Outstanding Action List

The following is a list of outstanding actions identified in the HAZOP study:

Action No.	Description	Priority	Time Scale
1	Establish switching procedures for working pumps and standby pumps (e.g., P-101A and P-101B) to ensure that the standby pump can be smoothly brought online, either manually or automatically, in the event of a main pump failure.	High	Before commissioning
2	Develop a quarterly inspection plan for heat exchanger E-101 to monitor scaling and corrosion risks on shell and tube side.	Medium	30 days before commissioning
3	Validate the DCS control logic for ion exchange column to ensure that the DCS can precisely control three-way valves from AV-101 to AV-110, thereby ensuring ion exchange performance.	High	Before site SAT
4	Install a low-flow/low-level interlock protection (dry-run protection) on the feed pump to ensure that the pump automatically shuts down in the event of no flow or interrupted flow, thereby preventing damage to the mechanical seal.	High	Before commissioning
5	Conduct a specialized review of pipeline 3D routing, valve accessibility, and sensor control loop installation points to ensure that the physical layout complies with industrial safety standards and meets maintenance accessibility requirements.	High	Before commissioning
6	Install a standby filter F-101B in parallel with original filter F-101 and configure inlet and outlet switching valves to ensure maintenance or filter cartridge replacement without shutting down process.	Medium	Before mechanical completion
7	The current design uses manual backwashing; we recommend upgrading to an automatic backwashing system to improve system reliability and ion exchange performance.	High	Detailed design review

Additionally, if the ion exchange column configuration fails to meet expected performance, then transfer the original column to mixed bed ion exchanger.

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7 Functional Design Specification

7.1 Control Philosophy

The heat exchanger system utilises a DCS and PLC as its core control platform, employing multi-level control to achieve automated control of the entire process, from material feeding, buffer storage, heat exchange, filtration, and separation to ion exchange. The field level maintains stable liquid levels in each buffer tank via single-loop feedback control. The flow control loop regulates the feed rate of crude glycerol (6000 kg/h) and the makeup water flow. For components exhibiting significant time delays, such as heat exchanger E-101 and cooling tower CT-101, cascade control is employed, using the heat exchanger outlet temperature ($55℃$) and cooling tower outlet temperature as primary variables, with cooling water flow as the secondary variable. This cascade control rapidly suppresses disturbances to ensure stable outlet temperatures. The control of ion exchangers C-101 and C-102 is performed automatically by the DCS. Based on online concentration analysers (PO_4^{3-} > 0.4 ppm or Ca^{2+} > 20 ppm), the system automatically switches between operational and standby units and initiates the regeneration procedure.

The sand filter F-101 detects potential blockages by monitoring the outlet flow. When the flow rate falls below $3m^3/h$ and remains below this threshold for a sustained period (30s), a filter blockage alarm is sent to the central control room. No automatic valve closure is initiated based solely on the flow signal; the operator is responsible for taking appropriate action.

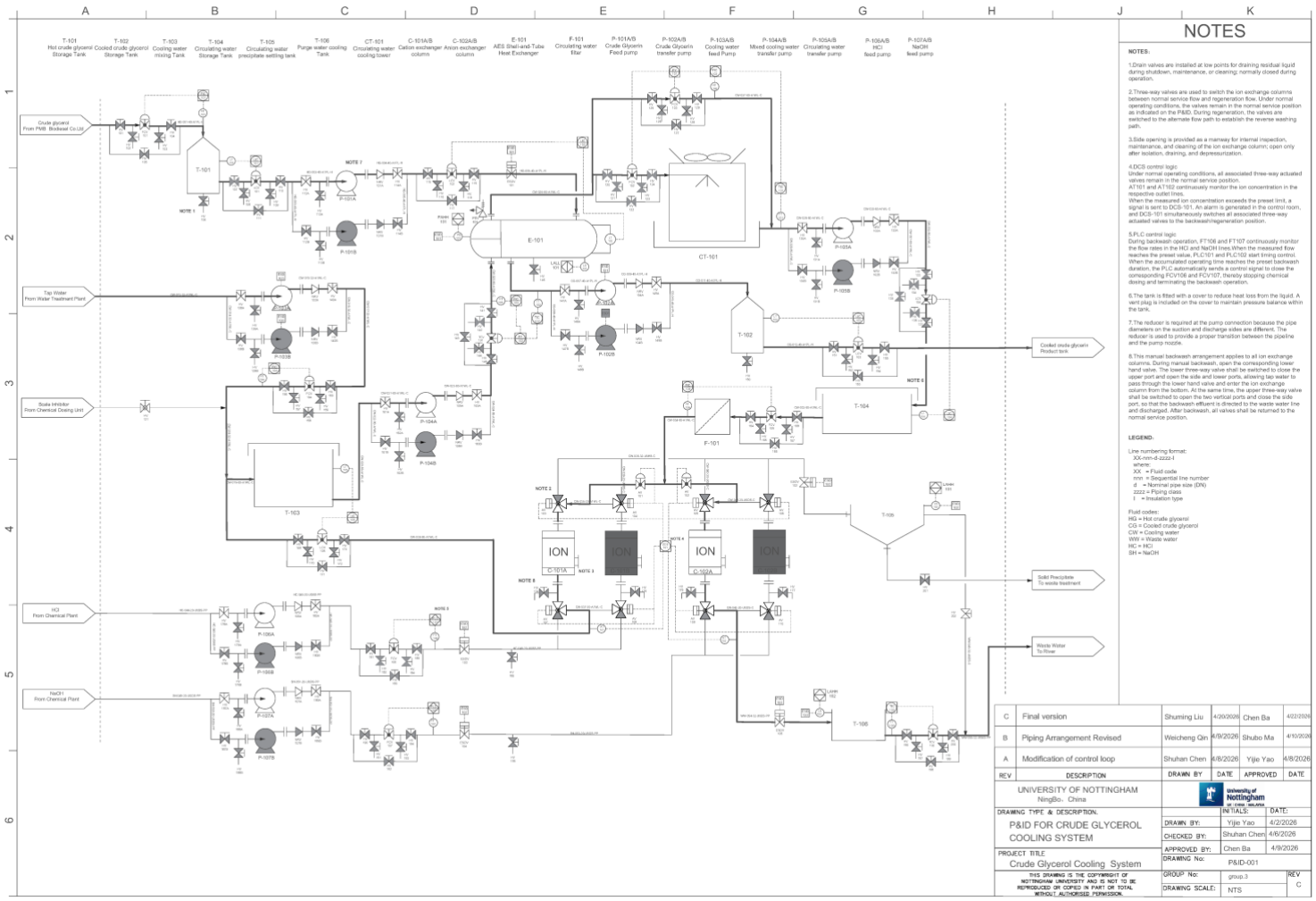
The outlet temperature of T-106 is monitored by TT-103, with TIC-103 regulating TCV-102. TIC-103 keeps TCV-102 closed until the temperature meets the discharge criteria ($\leq 30℃$in summer or $\leq 7℃$ in winter). If the temperature does not meet the criteria, TCV-102 remains closed and the liquid is retained for further cooling.

The safety protection layer operates independently and utilises an ESD for forced shutdown under hazardous conditions. E-101 tube side is protected by PS-101 and PAHH-101 (4.8 bar) which triggers ESD-101 to close ESDV-101 and stop P-101. E-101 shell side LT-102 monitors crude glycerol level; below 10% it closes LCV-102 to prevent gas carry-over. LS-101 provides backup LALL-101 alarm below 5%. T-105 uses LS-102 (90V%) to trigger ESD-102 closing ESDV-102/103/104. T-106 uses LS-103 for high-level overflow protection triggering ESD-103 to close ESDV-105 and stop P-103.

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7.2 Control Loops

Field level:



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F-101:

F-101 is a sand bed filter used to remove suspended solid particles from cooling water. The filter outlet flow is monitored by flow transmitter FT-105. When solid deposition causes a gradual decline in the outlet flow rate, FT-105 sends the signal to controller FIC-105. If the flow rate falls below a preset threshold ($3m^3/h$) and remains below this threshold for a sustained period (30 s) to avoid false trips due to momentary fluctuations, FIC-105 sends a filter blockage alarm to the central control room to alert the operator. No automatic valve closure is initiated based solely on the flow signal; the operator is responsible for taking appropriate action.

T-101:

T-101 is the crude glycerin buffer tank, and the level sensor LT-101 monitors its liquid level. When fluctuations in the upstream flow cause changes in the liquid level of the tank, LT-101 transmits the level signal to the controller LIC-101. LIC-101 converts the signal into control commands and simultaneously sends the signal to the central control room and the actuator of LCV-101, which adjusts the valve opening according to the commands to control the flow of crude glycerol leaving T-101.

T-102:

T-102 is a buffer tank for crude glycerin after cooling, designed to prevent upstream flow reductions from affecting downstream operations. Its liquid level is monitored by the sensor LT-103, which transmits electrical signals to the controller LIC-103. LIC-103 sends the signal simultaneously to the central control room and the actuator of LCV-103, which adjusts the valve opening to control the flow of crude glycerin leaving T-102, thereby stabilizing the liquid level.

T-103:

T-103 is a mixing tank for circulating water, tap water, and inhibitor, used to compensate for evaporative and purge losses and maintain a stable inhibitor concentration. The liquid level in the tank is detected by the float-type sensor LT-105, whose signal is sent to the controller LIC-105, which simultaneously sends the signal to the central control room and to the actuator of the flow control valve LCV-105. By adjusting the valve opening, the inflow rate is controlled to stabilize the liquid level.

T-104:

T-104 is a circulating water buffer tank that provides a safe redundancy space to prevent overflow caused by filter clogging. The liquid level in the tank is detected by the float-type sensor LT-104, which converts the level signal into an electrical signal transmitted to controller LIC-104. The controller simultaneously sends the signal to the central control room and the actuator of LCV-104. The actuator controls the inflow by adjusting the valve opening to prevent overflow.

T-105:

T-105 is a sedimentation tank used to precipitate impurity ions discharged from the ion exchanger during regenerating. Its liquid level is preset and controlled by PLC-101 and PLC-102 to ensure that the regeneration liquid enters T-105 through the ion exchanger at the correct flow rate and velocity.

T-106:

T-106 is a purge cooling tank; the tank level is monitored by level transmitter LT-105, which sends the signal to level controller LIC-105, and LIC-105 regulates the inlet control valve LCV-105 to maintain the liquid level within the normal operating range. The outlet liquid temperature is monitored by temperature transmitter TT-103, which sends the signal to temperature controller TIC-103; TIC-103 keeps the discharge control valve TCV-102 closed until the temperature meets the discharge criteria (≤ 30℃ in summer or ≤ 7℃ in winter), at which point TCV-102 is opened to allow discharge into the river. If the temperature does not meet the criteria, TCV-102 remains closed, and the liquid is retained in T-106 for further cooling. In addition, a separate high-level switch LS-103 is provided as independent overflow protection; when the liquid level reaches the high-high setpoint, LS-103 triggers the LAHH-103 alarm and initiates ESD-103, which closes the emergency shutdown valve ESDV-105 and stops pump P-103, cutting off both makeup water and inlet flow to prevent overflow.

Tap water input:

The flow rate of the supplied tap water is regulated by FT-104, and the flow signal is transmitted to FIC-104. FIC-104 simultaneously sends the data to the central control room and the actuator of FCV-104, which adjusts the valve opening to control the flow.

Crude glycerol input:

FT-101 regulates the crude glycerol flow, and the flow signal is transmitted to FIC-101. FIC-101 simultaneously sends the signal to the central control room and the actuator of FCV-101, which adjusts the valve opening to control the inflow.

Purge water temperature:

The liquid temperature at the T-106 outlet is monitored by TT-103, and the temperature signal is transmitted to TIC-103. TIC-103 simultaneously sends the data to the central control room and the actuator of TCV-102, which controls the retention time of the purge water in the T-106 by adjusting the valve opening, thereby stabilizing the outlet temperature.

Regulatory Control Level

CT-101:

The CT-101 outlet adopts cascade control to address the lag in temperature variations, ensuring that the outlet temperature remains relatively constant during winter and summer. The primary variable is the temperature detected by TT-101, CW-028-80-A1WL-C, and the secondary variable is the flow detected by FT-102, CW-027-80-A1WL-C. Both

signals are sent to the TIC-101 and FIC-102 controllers, which synchronously transmit the data to the central control room and the computing unit FY-101. After calculation, FY-101 allocates commands to the actuators of FCV-102 and FCV-103, stabilizing the outlet temperature of CT-101 by adjusting the flow distribution between the bypass and the cooling tower inlet.

C-101 and C-102:

The operation and regenerating of ion exchangers C-101 and C-102 are controlled by the DCS-101, with multiple distributed computers respectively operating AV-101 to AV-110. During normal operation, AV-103 and AV-107 (or AV-104 and AV-108) connect the cation exchanger C-101 to the pipelines CW-039-20-U0BS-C and CW-038-80-U0BS-C; AV-105 and AV-109 (or AV-106 and AV-110) connect the anion exchanger C-102 to the pipelines CW-041-20-U0DS-C and SH-054-30-U0ES-PP. During regeneration, the DCS automatically switches the outlet sequence of each three-way valve, connecting HC-048-20-U0BS-PP, SH-053-20-U0DS-PP, and CW-035-30-U0ES-C, and discharges the regeneration liquid into the sedimentation tank T-105. Concentration detectors AT-101 and AT-102 send the ion concentration data in the water to DCS-101. DCS-101 compares the measured values with the set thresholds ($PO_4^{3-} > 0.4PPM$ or $Ca^{2+} > 20PPM$), and when the concentration exceeds the limit, it automatically switches the current ion exchanger to the standby unit and starts the regeneration program of the original device.

E-101:

Due to the lag in temperature changes at the heat exchanger outlet, the shell-side outlet temperature of E-101 is jointly monitored by TT-102 and FT-103. This is a cascade control system, where the primary controlled variable is the fluid temperature in CG-007-40-A1PL-H, and the secondary controlled variable is the flow in CW-025-80-A1WL-C. When TT-102 and FT-103 detect temperature and flow data, respectively, they send the data to TIC-102 and FIC-103, which simultaneously transmit the data to the central control room and FY-102. After signal processing, FY-102 sends the output signal to the TCV-101 actuator, which adjusts the valve opening to control the amount of cooling water entering E-101, thereby stabilizing the crude glycerin outlet temperature.

Safety layer:

E-101:

The E-101 tube side is protected by a pressure relief valve PSV-101 for thermal expansion and overpressure relief; a pressure switch PSHH-101 installed on the hot crude glycerol feed line triggers PAHH-101 alarm and initiates ESD-101 to close ESDV-101 and stop pump P-101 when pressure exceeds 4.8 bar, preventing hot crude glycerol from continuing to enter the heat exchanger. The E-101 shell side is equipped with a level sensor LT-102 monitoring the hot crude glycerol level to prevent gas from upstream entering downstream equipment. When the liquid level falls below the set limit (10V%), LT-102 sends a signal to controller FIC-102, which closes LCV-102 to stop the outflow, thereby preventing more gas from being carried downstream. A separate low-low level switch LS-

101 is provided as an independent backup; when the level drops below 5V%, it triggers the LALL-101 alarm to alert the operator.

T-105:

To prevent overflow of strong acid and strong alkali regeneration liquid, a liquid level control switch LS-102 is installed as high-level protection. When the PLC malfunction causes the liquid level to rise and trigger LS-102 (90V% of the vessel), LAHH-102, and ESD-102 are simultaneously activated, and ESD-2 quickly shuts off ESDV-102, ESDV-103, and ESDV-104, cutting off the entry of waste liquid to prevent overflow.

T-106:

T-106 is a purge cooling tank. A separate high-level switch LS-103 is provided as independent overflow protection. When the liquid level reaches the high-high setpoint, LS-103 triggers LAHH-103 alarm and initiates ESD-103, which closes ESDV-105 and stops pump P-103, isolating both makeup water and purge liquid inlet to prevent overflow. Temperature control and discharge are managed by TIC-103 via TCV-102 and are independent of ESD-103.

7.3 Start-up and Shut-down Sequences

7.3.1 Start up

1. Initial Water Make-up:

Open HV-138, HV-140, and P-103, and set FCV-104 to maximum; feed water into T-103 until LIC-105 triggers FCV-105 to close completely.

2. Establish the E-101 Cooling Loop:

Open HV-161, HV-163, and P-104 to allow cooling water to flow from T-103 into E-101. Confirm cooling water flow is established and the tube side is fully flooded.

3. Connect the T-104 storage tank to the piping:

Once the liquid level has risen sufficiently, open HV-130, HV-132, LCV-104 and P-105 to pump cooling tower effluent into T-104. Simultaneously, open AV-103, AV-107 ,AV-105, and AV-109 via the DCS to connect to CW-036-20-A1WL-C, CW-037-20-A1WL-C, CW-041-20-U0DS-C and CW-042-20-U0DS-C, respectively.

4. Start filtration:

Once the liquid level in T-104 reaches LIC-104h, partially open FCV-105 to allow water to enter F-101 without triggering an alarm.

5. Establish the main circulation:

Set FIC-104 to the target value and reduce the system water inflow; the drop in the T-103 liquid level will trigger FCV-109 to open, establishing the cooling water circulation. Wait for the cooling water in all equipment to stabilise.

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6. Introducing crude Glycerol:

Set FCV-101 to the target value to allow hot glycerol to enter T-101. Switch LCV-101 to manual mode; once the liquid level has accumulated, open P-101 to pump the hot raw glycerol to E-101. Open LCV-102 fully.

7. Heat Exchanger Stabilisation:

First, shut down HV-147. Once the tube in E-101 is full of cooling water and the temperature is below 55℃, open HV-147, HV-149, and P-102 to allow glycerol to enter T-102 up to the operating level. Open LCV-103 to allow the cooled glycerol to flow into the next process.

8. System Balancing and Adjustment:

Gradually adjust the FIC-104 setpoint to balance the makeup water, discharge volume, and evaporation rate, ensuring that the liquid levels in each cooling water tank no longer fluctuate significantly. Adjust FCV-101 to the setpoint and wait for the E-101 outlet temperature to stabilise.

Adjust the FIC-103 flow rate and set TIC-102 to 55℃ until the E-101 outlet temperature is less than 55℃. Adjust the LIC-103 setpoint so that the flow rate of cooled glycerol equals the input flow rate of hot glycerol.

9. Finally, activate automatic control:

Once the liquid levels in all equipment have stabilised and the heat exchanger outlet temperature has become constant, switch all indicator controllers to automatic mode; switch the indicator controllers in the cascade control loop to cascade mode.

7.3.2 Shut down

1. Stop the glycerol feed:
Close FCV-101 to cut off the hot glycerol supply. Switch LCV-101 and LCV-102 to manual mode and open them fully.

2. Drain the glycerol side:
Once the T-101 level has dropped to the P-101 limit, close P-101. Wait for the E-101 shell side to drain completely, which can be confirmed by LS-101, triggering the low-level alarm; at this point, close P-102 and reset the alarm.

3. Switch over and drain the ion exchanger piping:
After draining the glycerol side, use the DCS to close the AV-101 side recirculation, fully open the AV-102 side, and set all control valves on this line to manual mode with the valve opening set to maximum.

4. Stop water-side feed and drain:
Close P-103. The system liquid level will drop sequentially in the order: T-103, CT-101, T-104, F-101. Close the corresponding pumps when P-104 and P-105 reach their limit values. Once the water side is largely drained, open all water-side drain valves to discharge residual cooling water, thereby completing the shutdown.

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8 Environmental Impact and Sustainability Assessment

Purpose and basis. This section assesses the environmental impact and sustainability of the proposed crude glycerol cooling package on the basis of 1 tonne of cooled crude glycerol delivered at 55 °C. The package processes 6000 kg/h of crude glycerol and is assumed to operate for 8000 h/y, corresponding to 48,000 t/y of cooled crude glycerol. The quantified boundary includes crude-glycerol storage and transfer, the shell-and-tube heat exchanger, the recirculating cooling-water loop, the cooling tower, filtration, ion-exchange treatment, chemical regeneration, wastewater neutralisation/settling and the T-106 discharge-cooling basin. Upstream biodiesel production and downstream cosmetic processing are excluded from the quantified boundary.

Table 7.1 Summary of quantified environmental metrics for the cooling package.

Metric	Annual value	Per tonne cooled crude glycerol	Comment
Electricity	141,520 kWh/y	2.95 kWh/t	No direct fuel or steam use within the package boundary.
Steam	0	0	No steam used within the cooling package boundary.
Gas/oil fuel	0	0	No fired heater within the cooling package boundary.
Fresh tap water	4,216 m^3/y	0.0878 m^3/t	Cooling-water make-up and treatment related water use.
Wastewater	1,040 m^3/y	0.0217 m^3/t	Cooling-water purge and regeneration wastewater after treatment.
NaOH solution, 5 wt%	3.31452 m^3/y	0.0691 kg solution/t	Active NaOH = 0.00345 kg/t, assuming density = 1000 kg/m^3.
HCl solution, 2 wt%	8.586 m^3/y	0.1789 kg solution/t	Active HCl = 0.00358 kg/t, assuming density = 1000 kg/m^3.
Sodium tripolyphosphate	24 kg/y	0.000500 kg/t	Additional inhibitor
Sodium molybdate	8 kg/y	0.000167 kg/t	Additional inhibitor
Solid precipitate	38.4 kg/y	0.000800 kg/t	Equivalent to 0.80 g/t of cooled crude glycerol.
Electricity-related CO_2	70.4 tCO_2/y	1.47 kgCO_2/t	Calculated from Zhejiang electricity emission factor.

8.1 Raw materials per tonne of product

For this project, the most meaningful raw-material metric is incremental consumption by the cooling package rather than the crude-glycerol process stream itself. On the chosen functional-unit basis, the process stream remains 1.0 t crude glycerol per tonne of cooled crude glycerol, while the additional material consumption arises from make-up water, regenerants and treatment chemicals. Fresh water use is 4216 m^3/y, or 0.0878 m^3/t. The chemical inventory comprises 3.31452 m^3/y of 5 wt% NaOH solution, 8.586 m^3/y of 2 wt% HCl solution, 24 kg/y of sodium tripolyphosphate and 8 kg/y of sodium molybdate. Assuming solution density equal to water, the active NaOH and HCl consumptions are 165.7 kg/y and 171.7 kg/y respectively, equivalent to 0.00345 kg NaOH/t and 0.00358 kg HCl/t.

8.2 Energy – steam, gas/oil, electricity

No steam, gas or oil fuel is consumed within the quantified cooling-package boundary. The environmental energy burden therefore comes almost entirely from purchased electricity for pumps, fans and ancillary controls. The total electricity demand is 141,520 kWh/y, equivalent to 2.95 kWh/t of cooled crude glycerol. This low electrical intensity indicates that the principal sustainability trade-off in the selected design is not combustion-related emissions, but the interaction between recirculating cooling, chemical treatment and wastewater management. In detailed design, further reductions in electricity use could be achieved through pump/fan turndown and weather-responsive control of the cooling-water system.

8.3 Water per tonne of product

The selected design uses a recirculating cooling-water system rather than once-through cooling, so the relevant water metric is make-up water rather than total circulation rate. Fresh make-up water consumption is 4216 m^3/y, equivalent to 0.0878 m^3/t of cooled crude glycerol, while wastewater generation is 1040 m^3/y, or 0.0217 m^3/t, mainly as purge water and treated regeneration effluent. This relatively low water intensity is a key sustainability advantage of the selected configuration, whereas a once-through system would substantially increase freshwater abstraction and direct discharge burden.

8.4 Waste Emissions per Tonne of Product and Compliance with Environmental Discharge Standards

8.4.1 Emissions to air

There is no direct combustion source within the cooling package, so no routine stack emissions are expected. The main air-emission concerns are cooling-tower drift and fugitive methanol releases from crude-glycerol handling, because the crude glycerol contains 0.4 wt% methanol. The design response is therefore to use covered tanks, closed transfer, good pump-seal integrity and avoidance of open drains or unnecessary venting, consistent with the control approach required for fugitive VOC sources (MEE, 2019).

8.4.2 Emissions to the aquatic environment

The wastewater stream is generated from cooling-water purge and ion-exchange regeneration, with a total design quantity of 0.13 m^3/h or 1040 m^3/y. The regeneration effluent is routed to T-105 for neutralisation and solids settling, and the clarified supernatant is discharged only after compliance confirmation. In this assessment, GB 8978-1996 is taken as the national reference framework for direct wastewater discharge, while GB 3838-2002 is used to judge whether the receiving surface-water objective is protected (MEE, 1996; MEE, 2002).

Because detailed wastewater composition data are not yet available, compliance with the applicable industrial discharge standard is treated as a design assumption rather than a demonstrated fact. The final design should therefore specify commissioning tests for pH, suspended solids, COD, methanol, chloride, phosphate, molybdate and any other regulated species before routine discharge is accepted.

8.4.3 Thermal pollution control

Cooling-tower blowdown is estimated to leave the tower at about 33 °C and is then routed through T-106, an outdoor reinforced-concrete open cooling basin, before discharge. The selected basin is 5.2 m

× 5.2 m × 2.5 m with 2.0 m normal operating depth and 0.5 m freeboard. As weekly winter receiving-water temperature data for Ningbo were not available, the preliminary T-106 sizing used the February monthly average background water temperature with a 15% design margin. Final compliance must be judged against the receiving-water thermal criterion in GB 3838-2002, which limits anthropogenic changes in ambient surface-water temperature to a weekly maximum average rise of no more than 1 °C and a weekly maximum average fall of no more than 2 °C (MEE, 2002).

8.4.4 Emissions to land / solid waste

The estimated solid precipitate from T-105 is 38.4 kg/y, which corresponds to 0.00080 kg/t or 0.80 g/t of cooled crude glycerol. This is a very low land-waste intensity. Nevertheless, the precipitate and any spent media should be collected and sent to an approved treatment/disposal route. If future sampling shows hazardous characteristics, the disposal route should be upgraded accordingly.

8.5 Carbon footprint per tonne of product

The operational carbon footprint of the cooling package is calculated from purchased electricity only, because there is no fired heater, boiler or other direct fuel combustion inside the quantified boundary. Using the Zhejiang provincial electricity CO_2 emission factor of 0.4974 $kgCO_2/kWh$, the annual electricity-related carbon footprint is 70.4 tCO_2/y, equivalent to 1.47 $kgCO_2/t$ of cooled crude glycerol. This confirms that the carbon footprint of the package is comparatively modest and that the more critical environmental issues are wastewater verification and thermal discharge control rather than direct greenhouse-gas emissions.

The provincial electricity factor used here is taken from the published 2023 electricity CO_2 emission factors issued by the Ministry of Ecology and Environment and the National Bureau of Statistics (MEE and NBS, 2025).

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10 Project Schedule

The Gantt chart in Figure 9.1 summarises the project programme for the PMB Biodiesel crude glycerol cooling unit from 24 February 2026 to 31 December 2026. It shows the sequential development of the project from design and FEED through procurement, fabrication, construction and installation, testing and commissioning. The schedule includes the major project items required by the brief, namely the shell-and-tube heat exchanger, cooling-water pumping and piping system from the water source tap point to the discharge point, crude glycerol piping from the heat exchanger inlet to outlet, and the control system required to maintain a crude glycerol flowrate of 6000 kg/h and an outlet temperature of 55°C. Key milestones such as FEED design freeze, vendor purchase order award, equipment delivery to site, installation completion, hot commissioning completion and final project completion are identified. The critical path runs through project kick-off, material and energy balance, heat exchanger thermal and mechanical design, FEED review, vendor award, heat exchanger manufacture, equipment delivery, installation, pressure testing, hot commissioning, performance testing and final handover.

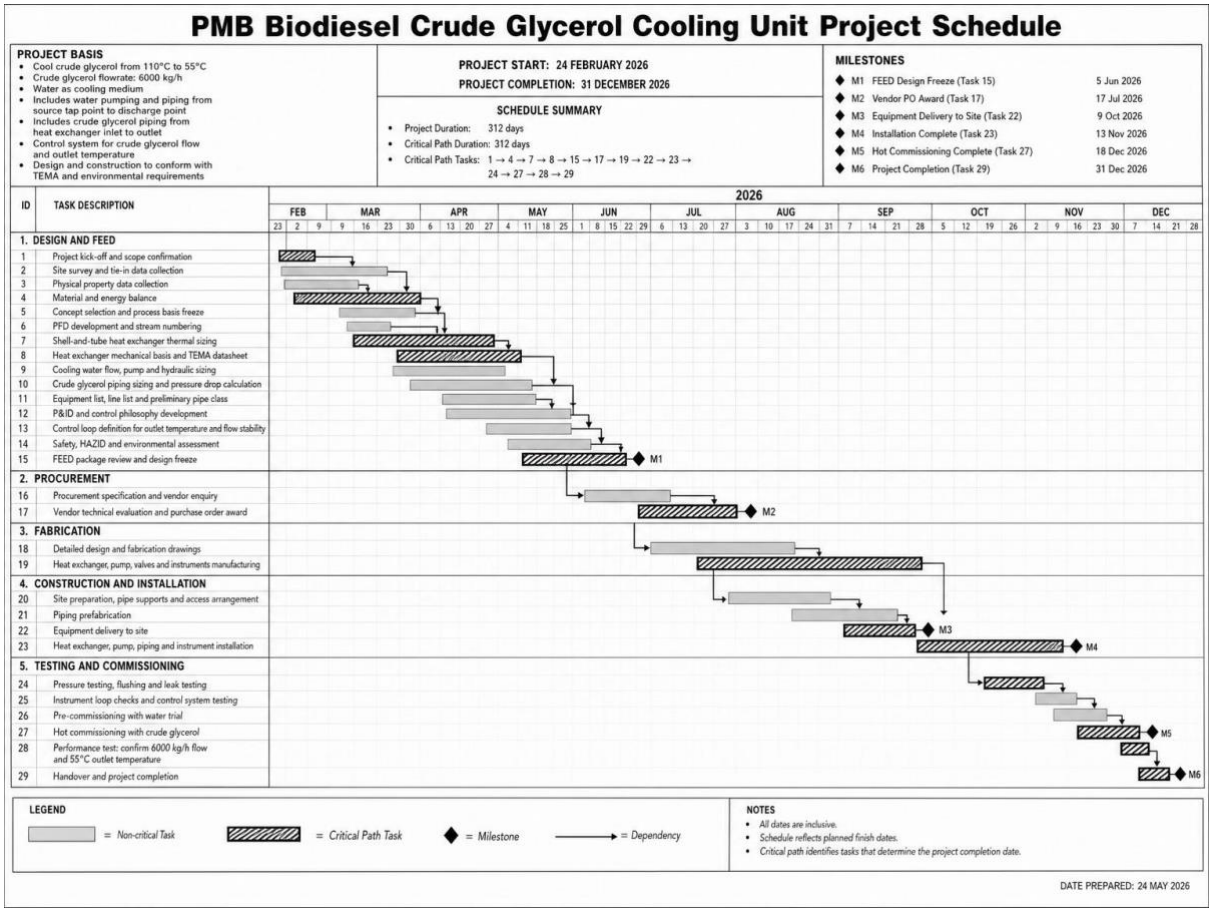


Figure 9.1 Project programme Gantt chart.

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12 Economic Analysis

12.1 Basis for the economic analysis

This economic analysis has been prepared for the Finance Director to demonstrate the value of the proposed crude glycerol cooling project. The project installs a shell-and-tube heat exchanger and supporting water, pumping and treatment equipment to cool crude glycerol before it is used in the downstream cosmetic-product operation.

This analysis treats the project as a plant improvement initiative rather than a new stand-alone product line. In essence, it represents an operational optimisation of an existing facility with an established market position. Accordingly, the annual income is taken as the expected annual operating cost saving, or equivalent revenue benefit, of RMB 2,500,000. This value is therefore used as the annual revenue basis for the discounted cash flow (DCF) assessment.

Table 12.1 Basis and assumptions for the economic analysis

Item	Basis used in this analysis
Plant duty	Cooling of 6,000 kg/h crude glycerol from 110 deg C to 55 deg C; no phase change
Annual operating time	8,000 h/year, based on 350 operating days with allowance for maintenance and shutdown
Project life	20 years
Capital estimate class	Preliminary estimate with an intended accuracy of approximately +/-30%
Costing method	Itemised delivered equipment cost plus percentage factors for installation, piping, instruments, electrical systems, buildings, yard works and service facilities(based on Methods for capital estimates)
Revenue basis	RMB 2,500,000/year project benefit or saving
Discount rate / MARR	12%
Currency	RMB

12.2 Preliminary estimate of capital cost

The total capital investment is estimated at RMB 4,715,536. The estimate is based on an itemised purchased-equipment list with a delivered equipment total of RMB 795,200. Installation and project allowances have then been calculated using delivered-equipment-cost percentage factors for a fluid-processing plant. At this stage, detailed pipe, valve and instrument take-offs are represented by the installed piping and instrumentation allowances; supporting calculations are provided in Appendix N. The specific CAPEX source references used for the equipment and cost basis are listed in Appendix Q.1.

Table 12.2 Summary of capital and operating costs

Cost item	Cost (RMB)	Basis / note
Purchased equipment delivered cost (PEC)	795,200	Itemised major equipment list; included in total direct plant cost
Total direct plant cost	2,862,720	PEC plus installation, piping, instrumentation, electrical, buildings, yard and service facilities

Cost item	Cost (RMB)	Basis / note
Total indirect plant cost	1,145,088	Engineering, construction expenses, legal expenses, contractor fee and contingency
Fixed capital investment	4,007,808	Total direct plant cost + total indirect plant cost
Working capital	707,728	Short-term operating funds estimated at approximately 15% of total capital investment
Total capital investment	4,715,536	Fixed capital investment + working capital
Annual operating cost	358,757	Detailed in Section 9.4 and Appendix B

The purchased-equipment cost is the basis for the direct and indirect plant cost allowances. Total direct plant cost is the largest capital component because it includes the installed plant assets required to make the system operable. Working capital is included to cover the short-term cash requirement during early operation.

12.3 Revenue for each year of the projected plant life

The project does not generate revenue from the sale of an independent product. Instead, the economic benefit is taken as the annual operating saving achieved by cooling the crude glycerol before downstream use. As no escalation or degradation has been assumed, the annual benefit is considered constant over the 20-year project life.

Table 12.3 Projected annual revenue, OPEX and net operating cash flow

Year	Revenue / benefit (RMB/year)	Annual OPEX (RMB/year)	Net operating cash flow (RMB/year)
1-20	2,500,000	358,757	2,141,243

Note: No escalation has been applied; therefore, the annual revenue benefit, OPEX and net operating cash flow are assumed constant throughout the 20-year project life.

12.4 Operating costs for each year of operation

The operating cost estimate includes raw materials and consumables, utilities, maintenance, waste treatment and disposal. The specific OPEX source references used for unit prices and utility costs are listed in Appendix Q.2, and the supporting operating-cost calculations are provided in Appendix O.

Table 12.4 Annual operating cost summary

OPEX category	Annual cost (RMB/year)	Basis / note
Raw materials and consumables	1,194	NaOH, HCl, sodium tripolyphosphate and sodium molybdate
Utilities: power and water	121,633	Electricity for pumps and cooling tower fan, plus tap water make-up
Maintenance	235,776	Estimated as 5% of total capital investment
Waste treatment and disposal	154	Wastewater disposal allowance
Total annual OPEX	358,757	Rounded annual operating expenditure

The annual OPEX is assumed to remain constant over the projected plant life. Detailed raw material, consumable, water and electricity calculations are provided in Appendix O.

12.5 Payback, NPV and DCF IRR

The annual net operating cash flow is calculated as revenue minus annual OPEX. With revenue of RMB 2,500,000/year and annual OPEX of RMB 358,757/year, the annual net cash flow is RMB 2,141,243/year. The following financial equations and detailed calculations are provided in Appendix P.

- Annual net cash flow = revenue - annual OPEX
- Simple payback period = total capital investment / annual net cash flow
- NPV = the total present worth value of all cash flow minus total present worth value of all capital investments
- DCF IRR = discount rate at which NPV equals zero

Table 12.5 Profitability metrics

Metric	Calculated value
Total capital investment (C0)	4,715,536 RMB
Annual revenue / project benefit	2,500,000 RMB/year
Annual OPEX	358,757 RMB/year
Annual net cash flow (Ct)	2,141,243 RMB/year
Simple payback period	2.20 years
ROI	45.41%
NPV at 12% over 20 years	11277259 RMB
DCF IRR	45.37%

The simple payback period is approximately 2.20 years, which is short relative to the 20-year assumed operating life. The NPV is positive at the 12% MARR, and the DCF IRR is well above the MARR. On this basis, the project is economically attractive under the base-case assumptions.

12.6 A summary sensitivity analysis showing the effects of variation in selling price, raw material costs and utility costs.

A summary sensitivity analysis was carried out to assess the effect of key economic and operating uncertainties on the profitability of the crude glycerol cooling project. The variables tested were the annual project saving, total capital investment, utility cost and process turndown.

Table 12.6 Sensitivity of profitability indicators

Case	Annual net cash flow (RMB/year)	NPV at 12% (RMB)	DCF IRR (%)	Payback (years)
Base case	2,141,243	11,278,358	45.37	2.20
Annual saving -20%	1,641,243	7,543,636	34.72	2.87
Annual saving +20%	2,641,243	15,013,080	56.00	1.79
CAPEX -30%	2,141,243	12,693,019	64.87	1.54
CAPEX +30%	2,141,243	9,863,697	34.84	2.86
Utility cost +20%	2,116,916	11,096,652	44.87	2.23
80% process turndown	1,665,839	7,727,356	35.24	2.83
60% process turndown	1,190,435	4,176,354	24.95	3.96

The sensitivity results show that the annual project saving has the largest influence on the economic performance. This is expected because the annual saving is the main source of income for the project. Changes in CAPEX also have a significant effect, particularly because the capital cost

estimate is preliminary and has an intended accuracy of approximately ±30%. Utility cost has a smaller effect because the annual electricity and water costs are low relative to the annual project benefit.

The project remains financially viable for all tested cases. Even under less favourable conditions, including a 20% reduction in annual saving, a 30% increase in CAPEX and operation at 60% process turndown, the NPV remains positive and the DCF IRR remains above the 12% MARR.

12.7 Economic feasibility of the project

Based on a profitability analysis, this project demonstrates a high degree of economic viability. A range of quantitative indicators, including Return on Investment (ROI), Payback Period (PBP), Net Present Value (NPV) and Internal Rate of Return (IRR), collectively indicate that the project will not only recoup the initial investment rapidly but also generate substantial residual economic value over its operational lifespan. The specific analysis is as follows:

Firstly, calculations show that the project's ROI reaches 45.41%. In the chemical industry, an ROI exceeding 15% is generally considered attractive. This project's figure is nearly three times the industry benchmark, meaning that for every RMB 1 invested, approximately RMB 0.45 in net profit is generated annually, fully demonstrating the efficiency of capital utilisation.

The PBP calculation indicates an estimated payback period of 2.2 years, whilst the project has a lifespan of 20 years. This implies that the project will recoup all initial capital expenditure (CAPEX) in just over two and a half years. This extremely short payback period significantly reduces long-term financial risk and ensures that the majority of the project's lifecycle is dedicated to generating net profit.

Meanwhile, the Net Present Value (NPV) has been calculated at RMB 11277259. The fact that the NPV is significantly greater than zero demonstrates that the project is expected to generate genuine incremental economic value. This figure represents the excess return generated for the company after covering all costs (including working capital and invested capital) and accounting for the time value of money at a 12% discount rate. A positive NPV is the most direct criterion indicating that the project should be accepted.

Furthermore, based on discounted cash flow calculations, the project's DCF IRR stands at 45.37%, which is nearly four times the market average rate of return (MARR, typically 8%–16% for such projects). This indicates that the project retains exceptional profitability resilience even in the face of market fluctuations, such as future declines in revenue or increases in operating costs.

Although the project's annual operating costs amount to RMB 358,757, this represents only approximately 14.4% of annual revenue (RMB 2,500,000). This substantial margin between revenue and operating costs ensures the project possesses robust cash flow to withstand market risks.

In summary, the short payback period (2.2 years), high return on investment of 45.41%, and significant positive net present value (RMB 11,277259) mutually reinforce one another, providing strong evidence that the project is economically sound and viable, and possesses substantial financial investment value.

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12.8 Decision-making advice

As the design team for this project, we strictly adhere to the NSPE Code of Professional Conduct and hereby strongly recommend that your company approve and commence this project. We are committed to prioritising public safety and health, and will resolutely reject any requests to sacrifice equipment safety margins in order to reduce costs; should such a situation arise, we will fulfil our duty to report it in accordance with professional ethics; Furthermore, we shall provide technical services within the scope of our competence, strictly uphold our duty of fiduciary loyalty, proactively disclose any potential conflicts of interest, and reject all improper benefits. In addition, we commit to implementing sustainable development principles, incorporating provisions for future environmental upgrades into the design, and strictly fulfilling our confidentiality obligations to ensure the absolute security of your company's process data. In summary, we recommend that your company proceed with the project on the basis of acknowledging the aforementioned professional ethics; the design team will adhere to these standards to provide your company with compliant and safe engineering deliverables (National Society of Professional Engineers, 2019).

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Appendix A Supporting Calculations for Material and Energy Balance

A.1 Assumptions

The calculations in this appendix assume steady-state operation at 6000 kg/h crude glycerol, with crude glycerol cooled from 110 °C to 55 °C and no phase change. A representative crude glycerol density of 1200 kg/m³ is used for preliminary volumetric flow calculations, while a representative cooling water density of 993 kg/m³ is used for the cooling water volumetric flow estimate.

A.2 Density Basis

Since the mass fraction of methanol in the mixture is very small, when calculating the density of the crude glycerol mixture, the calculation method for the glycerol-water mixture can be adopted.

Average crude glycerol temperature:

$$T_{cg} = (110 + 55) / 2 = 82.5\,℃$$

Density of pure water component in crude glycerol:

$$\rho_0(82.5) = 1000\left(\left|1 - \frac{T - 3.98}{615}\right|^{1.71}\right) = 970.4\ kg/m^3$$

Density of pure glycerol component in crude glycerol:

$$\rho_{g@82.5} = 1200\ kg/m^3$$

Volume contraction coefficient of crude glycerol:

$$\kappa(82.5{,}88.3\%) = 1 + A\sin\left({w_g}^{1.31}\pi\right)^{0.81} = 1.0059$$

Density of crude glycerol mixture:

$$\rho_s(82.5{,}88.3\%) = \kappa(82.5{,}88.3\%)\left[\rho_0(82.5) + \frac{\rho_g(82.5) - \rho_0(82.5)}{1 + \frac{\rho_g(82.5)}{\rho_0(82.5)}\left(\frac{1}{w_g} - 1\right)}\right]$$

$$= 1208\ kg/m^3$$

$$\approx 1200\ kg/m^3$$

The density of cooling water:

$$\rho_0(36.5) = 1000\left(\left|1 - \frac{T - 3.98}{615}\right|^{1.71}\right)$$

$$= 993.4\ kg/m^3$$

$$\approx 993\ kg/m^3$$

Average crude glycerol temperature=82.5 °C

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Crude glycerol density = 1200 kg/m³

Average cooling water temperature =36.5 °C

Cooling water density = 993 kg/m³

A.3 Component Mass Flowrates

Mass flowrate of glycerol:

$$\dot{m}_{gg} = \dot{m}_{cg} \times w_g = 5298\ kg/h$$

Mass flowrate of water:

$$\dot{m}_{gw} = \dot{m}_{cg} \times w_w = 678\ kg/h$$

Mass flowrate of methanol:

$$\dot{m}_{gm} = \dot{m}_{cg} \times w_m = 24\ kg/h$$

Where,

$\dot{m}_{cg}$ - mass flowrate of crude glycerol

$w_g/w_w/w_m$ – mass fraction of glycerol/water/methanol in crude glycerol

Since there is no reaction and no phase change, these component mass flowrates are the same at both the exchanger inlet and outlet.

Take the mass balance of the crude glycerol side-inlet and side-outlet heat exchanger as an example (stream 2 & stream 3):

$$\dot{m}_{cg,2} = \dot{m}_{gg,2} + \dot{m}_{gw,2} + \dot{m}_{gm,2} = 5298 + 678 + 24 = 6000\ kg/h$$

$$\dot{m}_{cg,3} = \dot{m}_{gg,3} + \dot{m}_{gw,3} + \dot{m}_{gm,3} = 5298 + 678 + 24 = 6000\ kg/h$$

$$\dot{m}_{cg,2} = \dot{m}_{cg,3}$$

This proves that the mass of crude glycerol entering and leaving the heat exchanger is conserved.

A.4 Molar Flowrates

Molar flow rate of glycerol:

$$\dot{n}_{gg} = \frac{\dot{m}_{gg}}{MW_g} = 57.59\ kmol/h$$

Molar flow rate of water:

$$\dot{n}_{gw} = \frac{\dot{m}_{gw}}{MW_w} = 37.67\ kmol/h$$

Molar flow rate of methanol:

$$\dot{n}_{gm} = \frac{\dot{m}_{gm}}{MW_m} = 0.75\ kmol/h$$

Total molar flow rate of crude glycerol:

$$\dot{n}_g = \dot{n}_{gw} + \dot{n}_{gg} + \dot{n}_{gm} = 96.00\ kmol/h$$

Mole fraction of glycerol in crude glycerol:

$$Mol\%_{gg} = \frac{\dot{n}_{gg}}{\dot{n}_g} \times 100\% = 59.98\% = 0.5998$$

Mole fraction of water in crude glycerol:

$$Mol\%_{gw} = \frac{\dot{n}_{gw}}{\dot{n}_g} \times 100\% = 39.23\% = 0.3923$$

Mole fraction of methanol in crude glycerol:

$$Mol\%_{gm} = \frac{\dot{n}_{gm}}{\dot{n}_g} \times 100\% = 0.75\% = 0.0075$$

The mole fractions of crude glycerol:

$$Mol\%_{gw} + Mol\%_{gg} + Mol\%_{gm} = 100\% = 1$$

A.5 Volumetric Flowrate of Crude Glycerol

$$\dot{V}_{cg} = \frac{\dot{m}_{cg}}{\rho} = \frac{6000}{1200} = 5\ m^3/h$$

Approximately 1.6% of the cooling tower inlet water is lost (mainly due to evaporation and drift). Therefore, the outlet water flow rate is obtained by subtracting this loss from the inlet flow rate.

The calculation of the water loss flow rate is as follows:

$$\dot{V}_{CT_loss} = \dot{V}_{CT_in} \times 1.6\% = 0.527\ m^3/h$$

Where,

$\dot{V}_{CT_in}$ – the volume flow rate of the cooling water entering the cooling tower,

The flow rate of water at the outlet of the cooling tower:

$$\dot{V}_{CT_out} = \dot{V}_{CT_in} - \dot{V}_{CT_{loss}} = 31.97\ m^3/h$$

A.6 Mixture Heat Capacity

Heat capacity of crude glycerol mixture:

$$C_{p,cg} = \sum_{i=1}^{N} \left(C_{p,i} \times w_i\right) = 2.86\ kJ/(kg \cdot ℃)$$

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A.7 Heat Duty

Heat duty:

$$\dot{Q} = \dot{m}_{cg} \times C_{p,\,cg} \times \Delta T_{cg} = 943{,}126\ kJ/h = 261.98\ kW$$

Where,

ΔT_{cg} – temperature difference of crude glycerol, which is 110℃-55℃

A.8 Cooling Water Requirement

Mass flow rate of cooling water:

$$\dot{m}_{cw} = \frac{\dot{Q}}{C_{p,\,cw} \times \Delta T_w} = 32295.6\ kg/h$$

Volume flow rate of cooling water:

$$\dot{V}_{cw} = \frac{\dot{m}_{cw}}{\rho_w} = 32.5\ m^3/h$$

Wthere,

$C_{p,\,cw}$ – Heat capacity of cooling water

ΔT_w - temperature difference of cooling water, which is 33℃-40℃

ρ_w – density of cooling water

A.9 Note on Pump Work

Pump work is small relative to the main exchanger heat duty and is therefore omitted from the main process energy balance. However, pump head, rated flow, efficiency assumptions and power requirements should still be calculated separately in the pump datasheets and in the hydraulic line calculations.

Take P-101 as an example. The design power (P) is 1.5kW and the efficiency (η) is 0.39. Then the heat generated by the pump can be calculated.

Heat created by P-101:

$$Heat = P \times (1 - \eta) \times 3600 = 3294\ kJ/h$$

P-101 is used to transport the crude glycerin feeds, which is the crude glycerol at 110℃ (stream 2 in PFD). The proportion of the heat generated by the pump in this stream can be calculated using the following formula:

$$\frac{Heat}{h_{g@110}} \times 100\% = \frac{3294}{1889632.8} \times 100\% = 0.17\%$$

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Appendix B. Supporting calculations for E-101

B.1 Design Basis and Traceability

B.1.1 Process Conditions

- Hot-side fluid: crude glycerol (shell side)
- Cold-side fluid: cooling water (tube side)
- Hot-side inlet temperature $T_{h,in} = 110\,°C$
- Hot-side outlet temperature $T_{h,out} = 55\,°C$
- Cold-side inlet temperature $T_{c,in} = 33\,°C$
- Cold-side outlet temperature $T_{c,out} = 40\,°C$
- Hot-side mass flowrate $m_s = 1.667\,kg/s$
- Cold-side mass flowrate $m_t = 8.971\,kg/s$

B.1.2 Physical Properties and Fouling Resistances

- shell-side fluid density $\rho_s = 1200\,kg/m^3$
- shell-side fluid viscosity $\mu_s = 0.00817\,Pa \cdot s$
- shell-side fluid thermal conductivity $k_s = 0.3056\,W/(m \cdot K)$
- shell-side fluid heat capacity $C_{p,s} = 2862.9\,J/(kg \cdot K)$
- tube-side fluid density $\rho_t = 993.45\,kg/m^3$
- tube-side fluid viscosity $\mu_t = 0.00070075\,Pa \cdot s$
- tube-side fluid thermal conductivity $k_t = 0.6164584\,W/(m \cdot K)$
- tube-side fluid heat capacity $C_{p,t} = 4180\,J/(kg \cdot K)$
- shell-side fouling resistance $R_{f,o} = 0.00035\,m^2 \cdot K/W$
- tube-side fouling resistance $R_{f,i} = 0.000146\,m^2 \cdot K/W$
- tube-wall thermal conductivity $k_{wall} = 16\,W/(m \cdot K)$

Property basis and references. Tube-side water properties were taken at the mean bulk water temperature of 36.5 °C. Shell-side crude-glycerol properties were estimated at the mean bulk process temperature of 82.5 °C.

Tube-side water density, viscosity, thermal conductivity and heat capacity were taken from the NIST Chemistry WebBook and checked against IAPWS water-property formulations (National Institute of Standards and Technology, n.d.; International Association for the Properties of Water and Steam, 1997). Shell-side density was estimated from the aqueous-glycerol density model of Volk and Kahler (2018). Shell-side viscosity was estimated from the glycerol-water correlation of Cheng (2008), cross-checked against glycerine handbook data. Shell-side thermal conductivity was taken from Physical Properties of Glycerine and Its Solutions (Glycerine Producers' Association, 1963, Table 24), using the 90 wt% glycerol value at 80 °C as the nearest available basis. Shell-side heat capacity was estimated by ideal mass-weighted mixing using glycerol Cp(T) literature data together with water and methanol pure-component data.

The shell-side stream was treated as 88.3 wt% glycerol, 11.3 wt% water and 0.4 wt% methanol. Because methanol is only a trace component, its effect was handled as a secondary engineering correction rather than a dedicated ternary-property fit. Fouling resistances were selected from standard exchanger design practice (TEMA / Kern). The adopted tube-wall thermal conductivity, k_wall = 16 W/m-K, is consistent with 316/316L stainless steel datasheet values (Alleima, 2025)

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B.1.3 Selected Geometry

- Selected shell diameter $D_s = 15.25\ \text{in}$
- Tube outside diameter $D_o = 0.01905\ \text{m}$
- Tube inside diameter $D_i = 0.0158\ \text{m}$
- Tube pitch $P_t = 0.0254\ \text{m}$
- Number of tubes $N_t = 128$
- Number of tube passes $N_p = 4$
- Tube length $L = 3.66\ \text{m}$
- Baffle spacing $B = 9.7\ \text{in}$

B.2 Heat Duty Calculation

The required duty is established from the heat balance and is consistent with the inlet and outlet conditions specified in the project brief.

B.2.1 Hot-side Required Duty

$$Q_{hot} = m_s C_{p,s}(T_{h,in} - T_{h,out})$$

$$Q_{hot} = 1.667 \times 2862.9 \times (110 - 55)/1000 = 262.485\ \text{kW}$$

B.2.2 Cold-side Calculated Duty

$$Q_{cold} = m_t C_{p,t}(T_{c,out} - T_{c,in})$$

$$Q_{cold} = 8.971 \times 4180 \times (40 - 33)/1000 = 262.491\ \text{kW}$$

B.2.3 Adopted Design Duty

$$Q_{req} = \frac{Q_{hot} + Q_{cold}}{2}$$

$$Q_{req} = \frac{262.485 + 262.491}{2} = 262.488\ \text{kW}$$

B.3 LMTD and Correction Factor

Use of the LMTD correction factor is consistent with standard shell-and-tube heat-exchanger design practice. This section follows the locked workbook FEED basis and the Kern-type design method (Kern, 1950).

B.3.1 Terminal Temperature Differences

$$\Delta T_1 = T_{h,in} - T_{c,out} = 110 - 40 = 70$$

$$\Delta T_2 = T_{h,out} - T_{c,in} = 55 - 33 = 22$$

B.3.2 Counter-current LMTD

$$LMTD = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)}$$

$$LMTD = \frac{70 - 22}{\ln(70/22)} = 41.470$$

B.3.3 R,P and F

$$R = \frac{T_{h,in} - T_{h,out}}{T_{c,out} - T_{c,in}} = \frac{110 - 55}{40 - 33} = 7.85714$$

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$$P = \frac{T_{c,out} - T_{c,in}}{T_{h,in} - T_{c,in}} = \frac{40 - 33}{110 - 33} = 0.090909$$

$$F = \frac{\sqrt{R^2 + 1}}{R - 1} \times \frac{\ln\left(\frac{1 - P}{1 - PR}\right)}{\ln\left(\frac{2/P - 1 - R + \sqrt{R^2 + 1}}{2/P - 1 - R - \sqrt{R^2 + 1}}\right)} = 0.958665$$

B.3.4 Corrected Temperature Difference

$$\Delta T_{corr} = F \times LMTD$$

$$\Delta T_{corr} = 0.958665 \times 41.470 = 39.756$$

B.4 Geometry Conversion

In line with normal FEED analysis, imperial inputs are first converted to SI units and then used to calculate heat-transfer area and equivalent diameter (Kern, 1950).

$$D_s = 15.25 \times 0.0254 = 0.38735 \text{ m}$$

$$B = 9.7 \times 0.0254 = 0.24638 \text{ m}$$

$$C = P_t - D_o = 0.0254 - 0.01905 = 0.00635 \text{ m}$$

$$A_o = N_t \pi D_o L = 128 \times \pi \times 0.01905 \times 3.66 = 28.0373 \text{ m}^2$$

$$D_e = \frac{4(P_t^2 - \pi D_o^2/4)}{\pi D_o} = 0.0240704 \text{ m}$$

$$A_{t,pass} = \left(\frac{N_t}{N_p}\right)\frac{\pi D_i^2}{4} = 0.00627414 \text{ m}^2$$

$$\frac{L}{D_s} = \frac{3.66}{0.38735} = 9.449$$

$$\frac{B}{D_s} = \frac{0.24638}{0.38735} = 0.6361$$

B.5 Tube-side Heat-transfer Calculation

Tube-side flow is turbulent under selected basis. The internal heat-transfer coefficient is therefore estimated using the Dittus–Boelter correlation for turbulent flow in smooth tubes (Dittus and Boelter, 1930).

$$v_t = \frac{m_t}{\rho_t A_{t,pass}} = \frac{8.971}{993.45 \times 0.00627414} = 1.43927 \text{ m/s}$$

$$Re_t = \frac{\rho_t v_t D_i}{\mu_t} = \frac{993.45 \times 1.43927 \times 0.0158}{0.00070075} = 32238.94$$

$$Pr_t = \frac{C_{p,t}\mu_t}{k_t} = \frac{4180 \times 0.00070075}{0.6164584} = 4.75155$$

$$Nu_t = 0.023 Re_t^{0.8} Pr_t^{0.4}$$

$$Nu_t = 0.023 \times 32238.94^{0.8} \times 4.75155^{0.4} = 173.447$$

$$h_i = \frac{Nu_t k_t}{D_i} = \frac{173.447 \times 0.6164584}{0.0158} = 6767.27 \text{ W/m}^2 \cdot \text{K}$$

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B.6 Shell-side Heat-transfer Calculation

This section adopts the simplified FEED shell-side rating which is consistent with a Kern-type shell-side calculation (Kern, 1950).

$$A_s = \frac{D_s C B}{P_t} = \frac{0.38735 \times 0.00635 \times 0.24638}{0.0254} = 0.0238588 \text{ m}^2$$

$$G_s = \frac{m_s}{A_s} = \frac{1.667}{0.0238588} = 69.8693 \text{ kg/m}^2 \cdot \text{s}$$

$$Re_s = \frac{G_s D_e}{\mu_s} = \frac{69.8693 \times 0.0240704}{0.00817} = 205.848$$

$$Pr_s = \frac{C_{p,s}\mu_s}{k_s} = \frac{2862.9 \times 0.00817}{0.3056} = 76.5376$$

$$Nu_s = 0.36 Re_s^{0.55} Pr_s^{1/3}$$

$$Nu_s = 0.36 \times 205.848^{0.55} \times 76.5376^{1/3} = 28.6228$$

$$h_o = \frac{Nu_s k_s}{D_e} = \frac{28.6228 \times 0.3056}{0.0240704} = 363.398 \text{ W/m}^2 \cdot \text{K}$$

B.7 Overall Thermal Resistance, U and Gross Predicted Duty

Overall resistance is evaluated using the usual resistance-in-series approach for shell-and-tube exchangers, with fouling included on both sides (Kern, 1950).

$$R_{clean,shell} = \frac{1}{h_o} = \frac{1}{363.398} = 0.00275181 \text{ m}^2 \cdot \text{K/W}$$

$$R_{wall} = \frac{D_o \ln(D_o/D_i)}{2k_{wall}} = 0.000111357 \text{ m}^2 \cdot \text{K/W}$$

$$R_{clean,tube} = \frac{D_o/D_i}{h_i} = \frac{0.01905/0.0158}{6767.27} = 0.000178166 \text{ m}^2 \cdot \text{K/W}$$

$$R_{total} = R_{clean,shell} + R_{wall} + R_{clean,tube} + R_{f,o} + R_{f,i}$$

$$R_{total} = 0.00275181 + 0.000111357 + 0.000178166 + 0.00035 + 0.000146$$
$$= 0.00353733 \text{ m}^2 \cdot \text{K/W}$$

$$U_{dirty} = \frac{1}{R_{total}} = \frac{1}{0.00353733} = 282.699 \, W/m^2 \cdot K$$

$$Q_{\text{pred}} = \frac{U_{\text{dirty}} \text{AoΔTcorr}}{1000} = \frac{282.699 \times 28.0373 \times 39.756}{1000} = 315.112 \text{ kW}$$

$$\text{Overdesign} = \frac{Q_{\text{pred}} - Q_{req}}{Q_{req}} \times 100\% = \frac{315.112 - 262.488}{262.488} \times 100\% = 20.05\%$$

B.8 Robustness, Pressure Drop and Nozzle Basis

This section summarises the robustness check, tube-side and shell-side pressure drops, and nozzle basis. The pressure-drop assessment follows the FEED method, with the engineering treatment of tube-side flow and heat transfer traceable to classic internal-flow correlations and early heat-exchanger design literature (Sieder and Tate, 1936; Kern, 1950).

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B.8.1 Shell-side ho Sensitivity Check

Base: $h_o = 363.398\ \mathrm{W/(m^2 \cdot K)}$, U_{dirty}=282.699 W/m²·K, $Q = 315.112$ kW, PASS
$h_o - 10\%$: U_{dirty}=260.208 W/m²·K, $Q = 290.042$ kW, PASS
$h_o - 15\%$: U_{dirty}=248.574 W/m²·K, $Q = 277.075$ kW, PASS
$h_o - 20\%$: U_{dirty}=236.671 W/m²·K, $Q = 263.806$ kW, PASS
$h_o - 25\%$: U_{dirty}=224.487 W/m²·K, $Q = 250.226$ kW, FAIL

Therefore, the FEED robustness boundary for the slelected geometry is that the shell-side h_o may be reduced by 20% while the required duty is still met, the design starts to fail at a 25% reduction.

B.8.2 Tube-side pressure drop

The nozzle assumption is: $D_{n,t} = 0.07793$ m (3 in Sch 40 equivalent ID)

$$s_t = \frac{\rho_t}{1000} = \frac{993.45}{1000} = 0.99345$$

$$G_t = \frac{m_t}{A_{t,pass}} = \frac{8.971}{0.00627414} = 1429.84\ \text{kg/m}^2 \cdot \text{s}$$

$$f_t = 0.4137 Re_t^{-0.2585} = 0.0282663$$

$$\Delta P_{f,t} = \frac{f_t N_p L G_t^2}{2000 D_i s_t \phi_t} = 26949.5\ \text{Pa}$$

$$a_r = 2N_p - 1.5 = 6.5$$

$$\Delta P_{r,t} = 5 \times 10^{-4} a_r \frac{G_t^2}{s_t} = 6688.23\ \text{Pa}$$

$$G_{n,t} = \frac{m_t}{\pi D_{n,t}^2/4} = 1880.80\ \text{kg/m}^2 \cdot \text{s}$$

$$\Delta P_{n,t} = 0.00075 \frac{G_{n,t}^2}{s_t} = 2670.54\ \text{Pa}$$

$$\Delta P_{tube,total} = \Delta P_{f,t} + \Delta P_{r,t} + \Delta P_{n,t}$$

$$\Delta P_{tube,total} = 26949.5 + 6688.23 + 2670.54 = 36308.26\ \text{Pa}$$

$\Delta P_{\text{tube,total}} = 0.36308$ bar

B.8.3 Shell-side pressure drop

The nozzle assumption is: $D_{n,s} = 0.04089$ m (1.5 in Sch 40 equivalent ID).

$$s_s = \frac{\rho_s}{1000} = \frac{1200}{1000} = 1.2$$

$$f_1 = 0.0076 + 0.000166 \left(\frac{D_s}{0.0254}\right) Re_s^{-0.125} = 0.00520571$$

$$f_2 = 0.0016 + 0.000058 \left(\frac{D_s}{0.0254}\right) Re_s^{-0.157} = 0.00107650$$

$$f_s = 144 f_1 - 1.25 \left(1 - \frac{B}{D_s}\right)(f_1 - f_2) = 0.479125$$

$$n_b + 1 = \text{round}(L/B, 0) = 15$$

$$\Delta P_{f,s} = \frac{f_s G_s^2 D_s (n_b + 1)}{2000 D_e s_s \phi_s} = 235.246 \text{ Pa}$$

$$G_{n,s} = \frac{m_s}{\pi D_{n,s}^2 / 4} = 1268.82 \text{ kg/m}^2 \cdot \text{s}$$

$$\Delta P_{n,s} = 0.00075 \frac{G_{n,s}^2}{s_s} = 1006.19 \text{ Pa}$$

$$\Delta P_{shell,total} = \Delta P_{f,s} + \Delta P_{n,s} = 235.246 + 1006.19 = 1241.43 \text{ Pa}$$

$Shell - side\ total\ \Delta P = 0.01241$ bar

B.8.4 Nozzle Selection Basis

Tube-side inlet / outlet nozzle:3 in NB; tube-side nozzle velocity $v_{n,t} = 1.893\ \text{m/s}$

Shell-side inlet / outlet nozzle:1.5 in NB; shell-side nozzle velocity $v_{n,s} = 1.058\ \text{m/s}$

The selected nozzle sizes are retained as the FEED-stage connection basis for E-101. The calculated nozzle velocities are acceptable for the adopted tube-side and shell-side flowrates, and the corresponding nozzle sizes, design pressure, design temperature, flange rating and materials of construction are recorded in the completed exchanger datasheet and associated line-list basis. Detailed mechanical items such as nozzle reinforcement, final nozzle orientation, pressure-part thickness, welding details and fabrication checks remain subject to detailed mechanical/vendor verification in accordance with TEMA mechanical design practice and the applicable piping-code requirements (Tubular Exchanger Manufacturers Association, Inc., 2024; American Society of Mechanical Engineers, 2024).

B.9 Supplementary Check: End Dead Length / Tubesheet Allowance

This section checks whether the selected thermal design would still meet duty if part of the effective tube length is occupied by tubesheets at both ends.

$$A_{gross} = N_t \pi D_o L = 28.0373 \text{ m}^2$$

Taking $t_{TS} = 32\ \text{mm/side}$ as a provisional assumption gives:

$$L_{eff} = L - 2t_{TS} = 3.66 - 2 \times 0.032 = 3.596 \text{ m}$$

$$A_{eff} = N_t \pi D_o L_{eff} = 27.5470 \text{ m}^2$$

$$\frac{A_{eff}}{A_{gross}} = \frac{27.5470}{28.0373} = 0.98251$$

$$Q_{eff} = Q_{pred}(\frac{A_{eff}}{A_{gross}}) = 315.112 \times 0.98251 = 309.602\ kW$$

$$Margin_{eff} = \frac{Q_{eff} - Q_{req}}{Q_{req}} \times 100\% = \frac{309.602 - 262.488}{262.488} \times 100\% = 17.95\%$$

The 25-40 mm/side sensitivity results are:
25 mm/side: $L_{eff} = 3.610$ m, $A_{eff} = 27.6543\ \text{m}^2$, Q_{eff}=310.808 kW, Margin = 18.41%
32 mm/side: $L_{eff} = 3.596$ m, $A_{eff} = 27.5470\ \text{m}^2$, Q_{eff}=309.602 kW, Margin = 17.95%
40 mm/side: $L_{eff} = 3.580$ m, $A_{eff} = 27.4244\ \text{m}^2$, Q_{eff}=308.225 kW, Margin = 17.42%

Therefore, within the assumption range, the dead-length allowance does not change the designed exchanger from thermally adequate to thermally inadequate.

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Appendix C. Structural and connection basis for E-101

Appendix C records the structural interpretation and connection basis used to support the E-101 datasheet.

Table C.1 Connection and nozzle schedule

Nozzle	Service	Side	Size	Connection basis	Remark
N1	Crude glycerol inlet	Shell	1.5 in NB	ASME 150 RF	To process feed line
N2	Crude glycerol outlet	Shell	1.5 in NB	ASME 150 RF	To cooled-product transfer line
N3	Cooling-water inlet	Tube	3 in NB	ASME 150 RF	To utility supply line
N4	Cooling-water outlet	Tube	3 in NB	ASME 150 RF	To return / discharge line
N5	Shell vent	Shell	1 in NPS	Threaded coupling	High-point vent for commissioning and maintenance
N6	Shell drain	Shell	1 in NPS	Threaded coupling	Low-point drain / isolation before opening
N7	Tube vent	Tube	1 in NPS	Threaded coupling	High-point vent
N8	Tube drain	Tube	1 in NPS	Threaded coupling	Low-point drain

Notes: Vent and drain points are included on both shell and tube sides to support commissioning, isolation, draining and maintenance.

Appendix D. Supplementary associated utility equipment design for CT-001 cooling tower

This supplement adds the cooling-tower package that supports E-101. CT-001 is not presented as the selected major equipment item for the coursework requirement; rather, it records the FEED utility basis, water balance, seasonal control philosophy, and connection basis so the overall cooling-water system can be adopted.

D.1 Process design basis and selected summer sizing point

Cooling-tower performance is commonly expressed in terms of range and approach, and thermal selection is normally anchored to the summer entering-air wet-bulb condition and the required leaving cold-water temperature. In this framework, the process fixes the required range, while the tower selection fixes the achievable approach for the chosen ambient condition (ASHRAE, 2024; Perry, Green and Southard, 2008; SPX Cooling Technologies, 2009; SPX Cooling Technologies, 2019a). For CT-001, the exchanger workbook fixes the summer cooling-water programme at 33.0/40.0 °C, and the cooling-tower workbook converts that requirement into a practical utility design basis.

Where dry-bulb and relative-humidity pairs are converted to wet-bulb conditions for seasonal checks, standard psychrometric relationships are assumed (ASHRAE, 2024; ASHRAE, 2025; SPX Cooling Technologies, 2009)

Table D.1 CT-001 design basis and selected summer sizing point

Item	Value	Unit	Comment

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Tower type	Induced-draft counterflow package cooling tower	-	
Required duty from E-101	262.488	kW	From E-101 basis
Vendor selection duty	288.737	kW	10% design margin
Vendor selection capacity	82.10	RT	For vendor enquiry
Circulating-water mass flow	8.971	kg/s	Calculated from $Q = mC_p\Delta T$
Circulating-water flow	32.425	m³/h	Summer hydraulic basis
Summer dry-bulb / wet-bulb	34.5 / 29.5	°C	Retained from (Ningbo Municipal Government, 2019)
Summer relative humidity	69.4	%	Retained from (Ningbo Municipal Government, 2019)
Cold-water supply to E-101	33.0	°C	Tower outlet in summer
Hot-water return to tower	40.0	°C	Tower inlet in summer
Range	7.0	°C	40.0 - 33.0
Approach	3.5	°C	33.0 - 29.5
Cycles of concentration	4.0	-	operating assumption
Drift-rate fraction	0.005	% of circulation	packaged-tower assumption

D.2 CT-001 water balance and utility demand

The method treats year-round process heat duty as fixed at the exchanger requirement, so the tower water balance is derived directly from the exchanger duty using Q=mCpΔT on the circulating-water side and then from standard open-cooling-tower evaporation, drift, blowdown and cycles-of-concentration relationships (Perry, Green and Southard, 2008; SPX Cooling Technologies, 2019a; SPX Cooling Technologies, 2019b).

Table D.2 CT-001 water balance

Metric	Value	Unit	Comment
Evaporation loss	0.3953	m³/h	Duty-based estimate
Drift loss	0.00162	m³/h	At 0.005% of circulation
Blowdown loss	0.1302	m³/h	From COC assumption of 4
Total make-up requirement	0.5271	m³/h	Equivalent to 12.65 m³/day

D.3 Design-basis LSI check for CT-001 COC = 4 operation

A design-basis Langelier Saturation Index (LSI) check was completed for CT-001 to assess whether the retained cycles of concentration, COC = 4, is reasonable for the cooling-tower circulating water. The designed CT-001 basis is a cooling-water programme of 33.0/40.0 °C, a circulating-water flow of

32.425 m³/h, evaporation loss of 0.3953 m³/h, blowdown of 0.1302 m³/h, and make-up requirement of 0.5271 m³/h. The hot-water return temperature of 40.0 °C was used for the LSI check because it is the highest normal water temperature in the cooling-water circuit.

The LSI check is used only to support the retained COC = 4 operating basis. It estimates the calcium-carbonate scaling tendency of the circulating water after concentration in the cooling tower. It does not replace the water-treatment package and does not define a final vendor water-treatment programme.

D.3.1 LSI equation

$$\mathrm{LSI} \approx \mathrm{pH} + \log_{10}\left(\frac{CaH}{2000}\right) + \log_{10}\left(\frac{MAlk}{1000}\right) - (2.65 - 0.0167T)$$

where pH is the pH of the circulating water, CaH is the calcium hardness of the circulating water in meq/L, MAlk is the total alkalinity of the circulating water in meq/L, and T is the circulating-water temperature in °C. A positive LSI indicates a calcium-carbonate deposition tendency, while a negative LSI indicates a calcium-carbonate dissolving tendency. (Langelier, 1936)

Table D.3 Symbols used in the LSI calculation

Symbol	Meaning	Unit
pH	pH of the circulating water	dimensionless
CaH	Calcium hardness of the circulating water	meq/L
MAlk	Total alkalinity of the circulating water	meq/L
T	Circulating-water temperature	°C

D.3.2 Water-quality basis and assumptions

The available raw-water data include calcium, magnesium, sodium, chloride, nitrate, sulphate and feed-water pH. The feed-water pH is 7.5. Measured M-alkalinity was not available, so M-alkalinity was estimated by ionic charge balance from the available raw-water composition.

The LSI check is therefore a design-basis estimated LSI rather than a measured operating LSI. The only estimated water-chemistry input is M-alkalinity. Calcium concentration, pH, temperature and COC are taken from the supported water-quality and CT-001 design basis.

Table D.4 Design-basis assumptions for the LSI check

Parameter	Value
COC	4
Temperature for LSI check	40.0 °C
Feed-water pH used as base circulating-water pH	7.5
M-alkalinity basis	Estimated by ionic charge balance
Logarithm basis	Base 10

The pH was not multiplied by COC. The base-case calculation uses pH = 7.5, and a separate pH sensitivity is included because the actual circulating-water pH may differ from the feed-water pH during cooling-tower operation.

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D.3.3 Calculation method

$$\mathrm{CaH_{raw}} = \frac{Ca}{20.04}$$

$$\mathrm{CaH_{circ}} = \mathrm{CaH_{raw}} \times \mathrm{COC}$$

$$\mathrm{COC} = 4$$

Because measured M-alkalinity was not available, raw-water M-alkalinity was estimated by ionic charge balance. The calculation assumes that the unmeasured charge balance is represented by alkalinity species, mainly bicarbonate alkalinity.

$$\mathrm{MAlk_{raw}} = \frac{Ca}{20.04} + \frac{Mg}{12.15} + \frac{Na}{22.99} - \frac{Cl}{35.45} - \frac{\mathrm{NO_3}}{62.00} - \frac{\mathrm{SO_4}}{48.03}$$

All concentrations in the M-alkalinity estimate are in mg/L, and the result is in meq/L.

$$\mathrm{MAlk_{circ}} = \mathrm{MAlk_{raw}} \times \mathrm{COC}$$

D.3.4 Worked calculation for average water case

Table D.5 Average raw-water values used for the LSI worked calculation

Parameter	Raw water Min	Raw water Ave	Raw water Max	Circulating Min×4	Circulating Ave×4	Circulating Max×4
Calcium (mg Ca/L)	8	20	35	32	80	140
Chloride (mg Cl/L)	5	15	30	20	60	120
Conductivity (µS/cm)	50	150	300	200	600	1200
Total hardness (mg/L as $CaCO_3$)	30	60	87	120	240	348
Total hardness (mg Ca/L)	12	24	34.8	48	96	139.2
Magnesium (mg/L)	2	5	8	8	20	32
Nitrate (mg/L)	0.5	1.0	2.0	2.0	4.0	8.0
TOC (mg/L)	0.5	1.2	1.8	2.0	4.8	7.2
Sodium (mg/L)	8	15	25	32	60	100
Sulphate (mg/L)	8	20	40	32	80	160
Reactive silica (mg SiO_2/L)	3	8	12	12	32	48
Total silica (mg SiO_2/L)	4	10	15	16	40	60

Parameter	Average raw-water value
Calcium, Ca	20 mg/L
Magnesium, Mg	5 mg/L

Sodium, Na	15 mg/L
Chloride, Cl	15 mg/L
Nitrate, NO_3	1 mg/L
Sulphate, SO_4	20 mg/L
Feed-water pH	7.5

$$\mathrm{CaH_{raw}} = \frac{20}{20.04} = 0.998\ \mathrm{meq/L}$$

$$\mathrm{CaH_{circ}} = 0.998 \times 4 = 3.992\ \mathrm{meq/L}$$

$$\mathrm{MAlk_{raw}} = \frac{20}{20.04} + \frac{5}{12.15} + \frac{15}{22.99} - \frac{15}{35.45} - \frac{1}{62.00} - \frac{20}{48.03} = 1.206\ \mathrm{meq/L}$$

$$\mathrm{MAlk_{circ}} = 1.206 \times 4 = 4.825\ \mathrm{meq/L}$$

$$\mathrm{LSI} = 7.5 + \log_{10}(\frac{3.992}{2000}) + \log_{10}(\frac{4.825}{1000}) - (2.65 - 0.0167 \times 40) = +0.50$$

The average water case therefore gives an estimated LSI of approximately +0.50 at COC = 4, pH = 7.5 and 40 °C.

D.3.5 LSI results at COC = 4

Table D.6 Design-basis LSI estimate at COC = 4, pH = 7.5 and T = 40 °C

Case	Circulating CaH / meq L^{-1}	Estimated circulating MAlk / meq L^{-1}	Estimated LSI
Minimum water case	1.60	2.38	−0.20
Average water case	3.99	4.83	+0.50
Maximum water case	6.99	7.12	+0.91

At the supported feed-water pH of 7.5, the estimated LSI range at COC = 4 is −0.20 to +0.91. The average water case gives LSI ≈ +0.50, while the maximum water case gives LSI ≈ +0.91, indicating a mild positive calcium-carbonate scaling tendency under harder-water conditions.

D.3.6 pH sensitivity

The base-case calculation uses the supported feed-water pH of 7.5. Since circulating-water pH may differ from feed-water pH during cooling-tower operation, a pH sensitivity check was included. This sensitivity is important because pH enters the LSI equation directly; therefore, increasing pH by 0.5 increases LSI by approximately 0.5.

Table D.7 LSI sensitivity to circulating-water pH at COC = 4 and T = 40 °C

Case	LSI at pH 7.5	LSI at pH 8.0	LSI at pH 8.3
Minimum water case	−0.20	+0.30	+0.60
Average water case	+0.50	+1.00	+1.30
Maximum water case	+0.91	+1.41	+1.71

The pH sensitivity shows that the maximum estimated LSI remains below approximately +1.71 for pH values up to 8.3. This supports retaining COC = 4 as the CT-001 design basis, provided that conductivity monitoring, blowdown control, and chemical treatment provisions are retained.

D.3.7 Interpretation for CT-001 COC = 4 operation

The design-basis LSI check supports the retained COC = 4 operating basis for CT-001. At pH = 7.5 and 40 °C, the estimated LSI values are −0.20, +0.50 and +0.91 for the minimum, average and maximum water-quality cases respectively. The average case is at the upper edge of the normal neutral-control

band, while the maximum case indicates a mild positive calcium-carbonate scaling tendency. Therefore, the CT-001 supporting utility package retains blowdown control, pH/conductivity monitoring through ion analyser, and provision for scale inhibitor dosing. The ion-exchange package shown on P&ID-001 is retained as an associated rough-estimate utility item pending confirmed make-up and circulating-water chemistry.

D.4 Winter control philosophy and anti-overcooling basis

The winter case does not change tower size; it changes the operating philosophy. Because the winter wet-bulb is far lower than the summer design value, uncontrolled operation would overcool the utility header. Accordingly, winter operation is based on fan turndown or VFD control together with hot-water bypass mixing, while final crude-glycerol outlet control remains on the exchanger itself (ASHRAE, 2024; SPX Cooling Technologies, 2016a; SPX Cooling Technologies, 2016b).

Table D.8 Winter control and anti-overcooling basis

Item	Value	Unit	Comment
Winter dry-bulb	4.7	°C	Adopted Ningbo winter basis
Winter relative humidity	79	%	Adopted Ningbo winter basis
Calculated winter wet-bulb	3.3	°C	Adopted Ningbo winter basis
Minimum practical winter approach	4.0	°C	
Tower-only winter cold-water outlet	7.3	°C	If fully exposed to winter air
Selected winter utility-header target	30.0	°C	Retained provisional setpoint
Required hot-water return from E-101	37.0	°C	For winter header balance
Hot-water bypass fraction	76.43	%	Mixing fraction required
Tower-treated fraction	23.57	%	Complement of bypass
Minimum fan speed	20	% of rated speed	assumption
First fan cut-back trigger	CW header < 32	°C	control trigger
Bypass valve action	Open when CW header < 30	°C	control trigger

Table D.9 28/30/32 °C winter header sensitivity check (control context: ASHRAE, 2024; SPX Cooling Technologies, 2016a; SPX Cooling Technologies, 2016b)

Winter header setpoint	Full-open product outlet	Controlled CW flow	% of summer flow	Verdict
28 °C	36.9 °C	10.4 m³/h	32.1%	Thermally feasible
30 °C	40.3 °C	11.9 m³/h	36.8%	Thermally feasible
32 °C	43.7 °C	14.0 m³/h	43.2%	Thermally feasible

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All three winter header targets remain thermally feasible when the exchanger retains local cooling-water flow control. A 30 °C winter utility-header target is therefore retained as a practical mid-range FEED setpoint: it avoids the strongest overcooling seen at 28 °C while remaining clearly distinct from the 33 °C summer basis. This 28/30/32 °C comparison is an internal operating sensitivity built on the exchanger and tower rather than a vendor performance guarantee.

D.5 FEED structural basis, dimensions status and connections

The cooling-tower workbook is a utility-sizing and operating-philosophy model rather than a vendor mechanical datasheet. As a result, the process duty, water balance and control basis are well defined, but final packaged-unit dimensions, fan motor data, sound data and detailed component selections remain vendor-confirmation items. The table below therefore states the retained basis and makes clear which items are already fixed and which remain to be confirmed during quotation. Final procurement should be taken forward against vendor published data and, where available, CTI-certified thermal ratings rather than this workbook alone (Cooling Technology Institute, 2026).

Table D.10 CT-001 structural basis and connection status (Cooling Technology Institute, 2026)

Item	Provisional basis	Status / note
Configuration	Induced-draft counterflow package cooling tower	
Duty class	Approximately 82 RT (288.7 kW vendor duty)	Sized from E-101 summer basis
Packaged arrangement	Single packaged unit basis	Vendor confirmation required
Construction	Vendor standard corrosion-resistant cooling-tower construction, typically FRP / PVC / galvanised or equivalent	FRP as final selection
Fan and control	VFD and/or fan cycling with winter hot-water bypass	Required by operating philosophy
Hot-water connection	Main circulating-water return from E-101, provisional DN80 / 3 in NB	Consistent with P&ID-001
Cold-water connection	Main circulating-water supply to E-101 header, provisional DN80 / 3 in NB	Consistent with P&ID-001
Make-up connection	Treated-water make-up to basin or circulating-water tank, provisional DN20	Provisional size
Blowdown connection	Continuous / intermittent bleed to wastewater system, provisional DN20	Provisional size
Drain and overflow	Low-point drain and overflow to safe disposal, provisional DN30	Provisional size
Additional items	Drift eliminator, fill pack, basin level control, access ladder / platform, upstream strainer, chemical dosing points as required	To be selected at vendor selection
Overall dimensions	Vendor confirmation required for the selected 82 RT class package tower	By vendor

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Appendix E. Associated line, valve and control summary for CT-001

This appendix records the principal cooling-water and control connections associated with the CT-001 supplement.

Table E.1 Principal associated lines for CT-001

Service	Function	Provisional nominal size	Note
Cooling-water return to tower	Hot water from E-101 to CT-001	DN80 / 3 in NB	Summer design 40.0 °C
Cooling-water supply from tower	Cold water from CT-001 to E-101	DN80 / 3 in NB	Summer design 33.0 °C
Make-up water	Water-balance make-up to basin / CW tank	DN20	About 0.527 m³/h
Blowdown	Cycles-of-concentration control bleed	DN20	About 0.130 m³/h
Drain / overflow	Maintenance and upset disposal	DN30	Route to safe wastewater handling
Optional winter bypass	Hot-water mixing around tower	TBC by hydraulic design	Required for anti-overcooling control

Table E.2 Associated valves and control intent for CT-001

Item	Function
Fan-speed control or fan cycling	Prevent winter overcooling and reduce power demand
Hot-water bypass valve	Winter mixing control
Basin make-up valve	Level control
Local TIC / FCV	Final 55 °C product outlet control

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Appendix F. Supplementary FEED-stage water-treatment equipment basis for F-101 and the P&ID-001 ion-exchange trains

F.1 Design basis and interpretation of the water-treatment package

P&ID-001, Crude Glycerol Cooling System, shows four ion-exchange columns, HCl and NaOH regeneration pumps, outlet ion monitoring and three-way valve arrangements for switching between normal service and regeneration/backwash paths. For rough sizing, the left-hand C-101A/B group is treated as the circulating-water ion-exchange train and the right-hand C-102A/B group is treated as the discharge-water ion-exchange train, consistent with the design.

The estimate was developed as a hydraulic rough-sizing exercise rather than a full stoichiometric ion-loading design. Separate-bed ion-exchange operating limits and practical service-loading checks were interpreted using vendor guidance for cation and anion exchange resin operation, including recommended bed-volume service-flow ranges, linear-velocity limits, bed-depth guidance and backwash freeboard requirements (DuPont, 2019; DuPont, 2020; Purolite, n.d.a; Purolite, n.d.b).

F.2 Selected rough-sizing basis

The circulating-water train was given a 10% FEED flow margin above the stated normal flowrate of 32.42 m³/h, giving a design basis of 35.66 m³/h. The discharge-water train was conservatively rounded up from "less than 1 m³/h" to a vendor-aligned design flow of 1.5 m³/h. Each A/B pair was treated as 1 × 100% duty plus 1 × 100% standby/regeneration, so each individual vessel was sized for the full design flow of its respective service. (DuPont, 2019; DuPont, 2020; Purolite, n.d.a; Purolite, n.d.b)

Table F.1 Hydraulic design basis for the ion-exchange trains

P&ID group	Service	Normal flow (m³/h)	Design flow (m³/h)	Selected linear velocity (m/h)	Operating philosophy
C-101A/B	Circulating water	32.42	35.66	20	1 × 100% duty + 1 × 100% standby/regeneration
C-102A/B	Discharge water	<1.00	1.50	19.3	1 × 100% duty + 1 × 100% standby/regeneration

F.3 Rough sizing of the circulating-water train (C-101A/B)

Design flow for the circulating-water train:

Q_{design}=1.10×32.42=35.66 m³/h.

Using a selected service linear velocity of 20 m/h, which sits comfortably inside the cited 10–50 m/h vendor range, the required vessel cross-sectional area is:

$A = Q/v = 35.66/20 = 1.783\ \text{m}^2$.

The corresponding vessel internal diameter is:

$D = (4A/\pi)^{0.5} = 1.51\ \text{m}$.

A nominal 1.5 m internal diameter vessel was therefore adopted for rough FEED sizing.

With a provisional resin bed depth of 1.5 m, the resin volume per vessel becomes:

V_{resin}=πD²H/4 = 2.65 m³.

The resulting service loading is:

Service rate=Q/V_{resin}=35.66/2.65=13.45 BV/h.

This remains within the cited preferred 8–40 BV/h range. Applying the recommended minimum freeboard of 75% of resin bed depth gives 1.13 m freeboard, so 1.2 m was adopted for layout purposes. Allowing a small additional height for distributors and nozzle clearance, an indicative straight-side height of about 3.0 m per vessel was adopted for the rough-estimate plot-space basis (DuPont, 2020; Purolite, n.d.a; Purolite, n.d.b).

F.4 Rough sizing of the discharge-water train (C-102A/B)

The Ø315 mm / 1.5 m³/h basis is retained from the project-supplied vendor table; the Taicang Friendship product page is used as the supplier source for the PP/PVC ion-exchange-column equipment family (Taicang Friendship Antiseptic Equipment Factory, n.d.).

Using the same first-pass course rule with $C = 6\ meq/L$, the theoretical diameter is:

$$D = \sqrt{(Q \times C \times 0.01)} = \sqrt{(1.50 \times 6 \times 0.01)} = 0.300\ m.$$

The selected vendor standard is therefore Ø315 mm, i.e. the theoretical 0.300 m value rounded up to the next available standard vessel diameter. The selected vendor vessel cross-sectional area is:

$$A = \pi D^2/4 = \pi(0.315)^2/4 = 0.0779\ m^2.$$

The corresponding superficial service velocity at the selected design flow is:

$$v = Q/A = 1.50/0.0779 = 19.3\ m/h.$$

This gives:

$$10\ m/h < 19.3\ m/h < 50\ m/h.$$

This lies within the cited preferred service velocity range. The vendor table does not state resin bed depth; therefore, the report records the vendor-aligned body height and overall height rather than imposing an unsupported bed-depth value.

For the right-hand discharge-water train, the adopted report dimensions are therefore Ø315 mm ID, 2500 mm body height and 2960 mm overall height. The associated vendor nozzles are DN20 inlet/outlet/vent, DN25 wash connection and DN30 resin inlet/outlet connection. The Ø315 mm / 1.5 m³/h basis is retained from the project-supplied vendor table, while the Taicang Friendship product page is used as the supplier source for the PP/PVC ion-exchange-column equipment family (Taicang Friendship Antiseptic Equipment Factory, n.d.).

Table F.2 Rough-estimate vessel dimensions and vendor-aligned discharge-water selection

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P&ID group	Adopted ID / standard	Resin bed depth	Resin volume	Service loading	Freeboard	Height basis
C-101A/B	1.50	1.50	2.65	13.45	1.20	3.0
C-102A/B	0.315 m (Ø315 vendor standard)	by vendor	by vendor	Vendor-rated at 1.5 m³/h	by vendor	2.50 m body; 2.96 m overall

F.5 Equipment design basis for F-101 pressure sand-bed filter

F-101 is selected as a FEED-stage vertical pressure sand-bed filter for suspended-solids and turbidity removal ahead of downstream water-treatment duty. The retained 2000 mm vessel basis is supported by Jiangsu YLD supplier information for pressure sand filters, including the YLD-MF-2000 class used as the vendor-aligned reference for rough equipment estimation (Jiangsu YLD Water Processing Equipment Co., Ltd., n.d.).

F.6 Sizing check for F-101 pressure sand-bed filter

The design flowrate adopted for F-101 is $32.43\ m^3/h$. Using the lower-bound vendor filtration rate of $10\ m/h$, the nominal required filtration area is:

$$A_{required} = Q / v = 32.43 / 10 = 3.24\ m^2$$

The selected vendor type uses a nominal vessel internal diameter of $2000\ mm$, which gives an actual cross-sectional area of:

$$A_{actual} = \pi D^2 / 4 = \pi(2.0)^2 / 4 = 3.14\ m^2$$

The corresponding actual filtration velocity is therefore:

$$v_{actual} = Q / A_{actual} = 32.43 / 3.14 = 10.3\ m/h$$

This value remains within the vendor-reported operating range of 10-14 m/h, therefore the selected 2000 mm pressure sand filter is considered acceptable for rough estimation and FEED-stage specification (Jiangsu YLD Water Processing Equipment Co., Ltd., n.d.).

The calculated superficial filtration velocity of 10.3 m/h is also consistent with standard FEED design ranges for pressure filters, for which typical approach velocities are approximately 5-15 m/h with dirty headloss of the order of 2-10 m. Therefore, the selected 2000 mm pressure sand-bed filter is acceptable for FEED-stage sizing. The filter is retained as a FEED-stage water-treatment item only; final media grading, clean / dirty pressure drop, backwash sequence, underdrain design and vendor mechanical details shall be confirmed by the supplier.

F.7 Equipment specification for F-101 pressure sand-bed filter

F-101 is retained as a water-treatment item for suspended-solids / turbidity removal ahead of downstream treatment. The basis below is intended as a rough equipment specification only and is not a final vendor datasheet or procurement package.

Item: F-101 Vertical Pressure Sand-Bed Filter

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Service: Removal of suspended solids / turbidity from water stream prior to downstream treatment

Type: Vertical pressure multimedia sand filter

Selected vendor basis: Jiangsu YLD Water Processing Equipment Co., Ltd., YLD-MF-2000 class / 2000 mm pressure sand filter reference (Jiangsu YLD Water Processing Equipment Co., Ltd., n.d.)

Design flowrate: $32.43\ m^3/h$

Selected vessel ID: $2000\ mm$

Actual filtration area: $3.14\ m^2$

Operating filtration velocity: $10.3\ m/h$

Vendor nominal capacity: $\sim 30\ m^3/h$

Vendor total vessel height: $\sim 3100\ mm$

Installed OAH allowance: $\sim 3800\ mm$

Nozzle size: DN100 inlet / DN100 outlet

Working pressure basis: $0.6\ MPa$

Vendor pressure capability: $0.06 - 1.0\ MPa$

Backwash: Manual or automatic

Filter media depth: $1.0 - 1.2\ m$

Backwash intensity: $16 - 18\ L/m^2 \cdot s$

F.8 Integrated operability and design-rationale basis for F-101 and the ion-exchange package

This section records the FEED-stage operability rationale for the filtration and ion-exchange items retained as associated water-treatment equipment in Section 4.3. It is not a vendor mechanical design and does not replace the F-101 sizing and specification basis in Sections F.5-F.7.

F.8.1 Source-water quality and suspended-solids loading basis

The selected make-up water source is plant / tap water rather than direct untreated river-water abstraction. Therefore, the suspended-solids loading to F-101 is expected to be lower and more stable than for untreated river water. External solids ingress is mainly associated with the open cooling-tower / mixing interfaces and normal cooling-water circulation. F-101 is retained to reduce suspended-solids carry-over, protect the circulating-water loop and reduce fouling risk to downstream water-treatment equipment.

F.8.2 Normal treatment-route basis

Based on the P&ID-001 basis used for this equipment specification, the streams routed to F-101 and the ion-exchange vessels are treated through the retained water-treatment route rather than intentionally bypassed during normal operation. This supports E-101 protection, cooling-tower cleanliness and discharge-water control. Isolation and maintenance provisions are still retained for cleaning, regeneration and vendor-confirmed service requirements; therefore, this basis does not remove maintenance isolation or temporary operating procedures.

F.8.3 F-101 cleaning / backwash basis

Routine automatic online backwashing is not fixed as a normal operating function at this FEED stage. However, because deep-bed pressure filters are normally designed with cleaning / backwash capability, the final F-101 backwash or cleaning connections, underdrain arrangement, dirty-water routing, clean / dirty pressure-drop criteria and media grading remain vendor-confirmation and detailed-design items.

F.8.4 Ion-exchange resin-form basis

The cation-exchange vessel is retained on an H-form resin basis and the anion-exchange vessel is retained on an OH-form resin basis for the FEED-stage demineralisation assumption. This basis is used only for rough equipment specification and water-treatment interface definition. Final resin grade, resin form, operating cycle, regeneration requirement and discharge-water pH control shall be confirmed against the make-up / circulating-water chemistry and supplier recommendations.

F.8.5 Ion-exchange duty / standby, regeneration and cleaning philosophy

The ion-exchange vessels are retained on a duty / standby operating basis. When the duty vessel approaches resin exhaustion, flow is transferred to the standby vessel and the isolated vessel is regenerated through the acid / caustic interface. A separate online hydraulic backwash step is not fixed at this FEED stage; the need for resin bed backwash or cleaning remains a resin-supplier confirmation item. Regeneration effluent is routed to the neutralisation / settling system.

F.8.6 FEED limitations and detailed-design actions

The following items remain open for detailed design and vendor confirmation: final F-101 media grading; clean and dirty-bed pressure drop; backwash / cleaning connection requirements; underdrain arrangement; dirty-water routing; resin-supplier operating cycle; resin regeneration, cleaning and any hydraulic backwash requirements; and final confirmation against make-up-water, circulating-water and purge-water chemistry.

Appendix G. P-101A pump and LCV-101 control-valve supporting calculations

G.1 Design basis and input data

Parameter	Value used	Basis
Normal crude glycerol mass flow	6000 kg/h	Project brief
Crude glycerol temperature to P-101A / LCV-101	110 °C	Project brief
Crude glycerol composition	88.3 wt% glycerol / 11.3 wt% water / 0.4 wt% methanol	Project brief
Phase change	None	Project brief
Density for hydraulic sizing	1200 kg/m³	Design basis
Specific gravity	1.20	Calculated from density
Viscosity	8.2 cP	
Selected pump model	JIH40-25-125/1.5 kW-2P	AHJN product specification manual / selected project model
Selected pump catalogue duty	6.30 m³/h at 20 m	AHJN product specification manual / selected project model
Selected pump material	SS316L stainless steel centrifugal pump	Selected project model / AHJN JIH material range
Selected pump temperature capability	-50 to 260 °C	JIH series equipment capability; not process duty
AHJN catalogue density basis	1000 kg/m³	AHJN product specification manual
Selected pump efficiency	39%	AHJN product specification manual

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Parameter	Value used	Basis
Selected pump NPSH required	2.5 m	AHJN product specification manual
Selected pump weight	65 kg	AHJN product specification manual

$$SG = density \div 1000 = 1200 \div 1000 = 1.20$$

At the normal operating point, the crude glycerol component mass flows are:

Component	Mass fraction	Mass flow at 6000 kg/h
Glycerol	0.883	5298 kg/h
Water	0.113	678 kg/h
Methanol	0.004	24 kg/h
Total	1.000	6000 kg/h

G.2 P&ID and datasheet interface

Item	Meaning used in calculation	Reference / datasheet field
P-101A / P-101B	Crude glycerol feed pump duty / standby	P&ID-001 and pump datasheet
LCV-101	T-101 outlet level-control valve on pump suction line	P&ID-001 and control valve datasheet
HG-002-40-A1PL-H	T-101 to P-101A/B suction line	LCV-101 line basis
HG-004-40-A1PL-H	P-101A/B discharge line to E-101	Pump discharge line basis
A1PL-H	150# RF / 1.5 mm CA / process / 316L SS / heat conservation	Piping class interpretation

The selected P-101A pump discharge nozzle is DN25 while the P&ID-001 discharge line basis is DN40. The datasheet therefore records DN25 discharge nozzle and notes that a DN25-to-DN40 reducer / expander is required at the pump discharge connection shown or noted on P&ID-001.

G.3 P-101A pump supporting calculations

G.3.1 Normal volumetric capacity

The normal volumetric capacity is calculated from the project mass flowrate and design density.

$$Q_{normal} = \frac{\dot{m}_{normal}}{\rho} = 6000 \,/\, 1200 = 5.00 \text{ m}^3\text{/h}$$

G.3.2 Rated capacity using selected pump

The selected pump catalogue flow is 6.30 m³/h. This is used as the rated pump capacity because it is the selected supplier model. The exact JIH40-25-125 performance row is retained from the project-selected pump data, while the AHJN supplier information is used as the source for the JIH stainless-steel centrifugal-pump series basis (Anhui Jiangnan Pump Valve Group Co., Ltd., n.d.).

$$\text{Rated margin} = \frac{Q_{rated}}{Q_{normal}} = 6.30 \,/\, 5.00 = 1.26 = 126\%$$

The selected rated flow is 26% above normal and therefore exceeds the minimum 10% rated-flow margin required for throttled pump service.

G.3.3 Minimum pump capacity

Until the manufacturer confirms the minimum continuous stable flow, the datasheet minimum flow is set as 30% of the selected rated capacity.

$$Q_{min} = 0.30 \times Q_{rated} = 0.30 \times 6.30 = 1.89 \text{ m}^3/\text{h}$$

If actual operation can fall below the vendor-confirmed minimum flow, a minimum-flow spillback / recycle to T-101 should be considered.

G.3.4 Temperature basis

The P-101A process pumping temperature remains 110 °C because this is the designed hot crude glycerol process temperature. The selected JIH series temperature capability of -50 to 260 °C is recorded as vendor equipment capability only and does not change the LCV-101 process temperature or the P&ID-001 hydraulic duty (Anhui Jiangnan Pump Valve Group Co., Ltd., n.d.).

$$T_{design} = T_{max,operating} + 25 = 110 + 25 = 135\,°C$$

If the actual crude glycerol maximum operating temperature is later revised above 110 °C, the process datasheets must be revised using the updated maximum operating temperature plus the design-temperature margin, and the vapour pressure / NPSH basis must be recalculated.

G.3.5 Differential head, discharge pressure and design pressure

The selected JIH40-25-125 performance row is retained from the project-supplied pump data, while the AHJN website is used as the supplier source for the JIH stainless-steel centrifugal-pump series (Anhui Jiangnan Pump Valve Group Co., Ltd., n.d.). The selected pump catalogue head is 20 m. The equivalent crude-glycerol pressure rise is calculated with the project specific gravity of 1.20, rather than the AHJN table density basis of 1000 kg/m³:

$$\Delta P_{pump} = \frac{H \times SG}{10.197} = 20 \times 1.20 / 10.197 = 2.35 \text{ bar}$$

Flow case	Suction pressure	Discharge pressure
Normal	0.25 barg	0.25 + 2.35 = 2.60 barg
Rated	0.20 barg	0.20 + 2.35 = 2.55 barg
Minimum	0.30 barg	0.30 + 2.35 = 2.65 barg

$$P_{design} = MNOP + 0.35 = (0.40 + 2.35) + 0.35 = 3.10 \text{ barg}$$

G.3.6 NPSH and power check

The selected JIH40-25-125 performance row provides the retained NPSHr and duty-point basis used in this FEED-stage check; the AHJN supplier information supports the retained JIH stainless-steel centrifugal-pump series selection (Anhui Jiangnan Pump Valve Group Co., Ltd., n.d.). The datasheet rated-case NPSH available is 4.5 m, giving the required 2.0 m NPSH margin.

$$\text{NPSH margin} = NPSH_A - NPSH_r = 4.5 - 2.5 = 2.0\,m$$

The rated-point brake power check is calculated using the selected rated capacity, selected head, project design density and selected AHJN efficiency. This avoids using the catalogue water-density basis directly for the crude glycerol motor check (Anhui Jiangnan Pump Valve Group Co., Ltd., n.d.).

$$P_{hydraulic} = \frac{\rho \times g \times Q_{rated} \times H}{1000} = 1200 \times 9.81 \times (6.30 / 3600) \times 20 / 1000 = 0.412 \text{ kW}$$

$$P_{brake} = \frac{P_{hydraulic}}{\eta} = 0.412 / 0.39 = 1.06 \text{ kW}$$

The selected 1.5 kW motor is higher than the recalculated brake power of 1.06 kW and is acceptable for the FEED datasheet, subject to final vendor confirmation against the pump curve and liquid

properties. The AHJN manual weight of 65 kg has been retained in the pump datasheet notes for mechanical/layout reference (Anhui Jiangnan Pump Valve Group Co., Ltd., n.d.).

G.4 LCV-101 control-valve supporting calculations

G.4.1 Flow cases

The normal mass flow is the crude glycerol capacity. The maximum case is matched to the selected P-101A rated capacity. The minimum case is set as 50% of normal flow for level-control valve rangeability.

$$\dot{m}_{max} = 6.30 \times 1200 = 7560 \text{ kg/h}$$

$$\dot{m}_{min} = 0.50 \times 6000 = 3000 \text{ kg/h}$$

Case	Mass flow	Volumetric flow at 1200 kg/m³
Maximum	7560 kg/h	6.30 m³/h
Normal	6000 kg/h	5.00 m³/h
Minimum	3000 kg/h	2.50 m³/h

G.4.2 Valve type and body size

LCV-101 is a modulating control valve for T-101 outlet level control. The selected valve is the NEWAY CSS-JT Series jacket control valve, which is a jacketed globe-and-plug straight-travel control valve. It is used instead of a gate valve because LCV-101 is a modulating control valve, not a manual isolation valve. The jacketed body is justified by the hot viscous crude glycerol service and reduces cool-down, crystallisation and high-viscosity risk during abnormal or intermittent operation (NEWAY Valve, 2016; NEWAY Valve, n.d.).

DN40 line to DN32 CSS-JT jacketed globe/plug valve body

Reducers / expanders are required around the DN32 jacketed globe/plug valve body. Final trim, actuator sizing, rangeability, jacket connection details, shut-off class and noise shall be confirmed by the valve vendor.

The NEWAY CSS-JT catalogue basis is retained for LCV-101 because the CSS-JT jacketed control-valve series is identified for low-leakage service and services inclined to crystallisation, with pneumatic or electric actuator options. The broader NEWAY control-valve catalogue also supports the IEC 60534 design-standard and FCI 70-2 leakage-class basis used in the datasheet (NEWAY Valve, 2016; NEWAY Valve, n.d.).

G.4.3 Liquid valve sizing

Liquid valve capacity is calculated using the standard liquid sizing equation shown below. The conversion factors used are 1 m³/h = 4.4029 US gpm and 1 bar = 14.5038 psi.

$$C_v = Q_{gpm} \times \sqrt{\frac{SG}{\Delta P_{psi}}}$$

Case	Q	ΔP	Calculated C_v	Datasheet C_{vc}
Maximum	6.30 m³/h = 27.7 gpm	0.15 bar = 2.18 psi	20.5	21
Normal	5.00 m³/h = 22.0 gpm	0.10 bar = 1.45 psi	20.0	20
Minimum	2.50 m³/h = 11.0 gpm	0.05 bar = 0.73 psi	14.2	14

The calculated values match the LCV-101 datasheet values of C_{vc} = 21 / 20 / 14 for maximum, normal and minimum cases.

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G.4.4 LCV design pressure and fail action

LCV-101 is on the low-pressure suction-side line. The maximum upstream operating pressure basis is 0.40 barg; using the low-pressure design rule gives 0.75 barg, which is rounded to 1.0 barg on the datasheet.

$$P_{design,LCV} = 0.40 + 0.35 = 0.75 \text{ barg}$$

The fail position is FO. This is consistent with the T-101 outlet level-control philosophy because opening on air / signal failure avoids isolating the pump suction line and helps prevent T-101 overfill. The full safeguarding action remains subject to the final control philosophy and hazard review.

G.5 Input and output parameters

Datasheet	Input parameter	Output parameter
P-101A	$\dot{m}_{normal}$ = 6000 kg/h, ρ = 1200 kg/m³, selected Q_{rated} = 6.30 m³/h, H = 20 m, η = 39%, $NPSH_r$ = 2.5 m	Q_{normal} = 5.00 m³/h, Q_{rated} = 6.30 m³/h, Q_{min} = 1.89 m³/h, ΔP_{pump} = 2.35 bar, P_{design} = 3.10 barg, P_{brake} = 1.06 kW
LCV-101	Line = HG-002-40-A1PL-H, ρ = 1200 kg/m³, SG = 1.20, flow cases, P1/P2 cases, line size DN40	Valve type = Jacketed globe/plug, body size = DN32, C_{vc} = 21 / 20 / 14, design pressure = 1.0 barg, fail position = FO

G.6 Design conditions and materials

Item	Design condition / material basis
P-101A operating temperature	110 °C process duty. AHJN JIH series -50 to 260 °C is pump equipment capability only.
P-101A design temperature	135 °C = 110 + 25 °C.
P-101A design pressure	3.10 barg based on maximum pump discharge operating pressure plus 0.35 bar.
P-101A material	SS316L stainless steel centrifugal pump; final wetted-material certificate, mechanical seal type and elastomers to be confirmed by vendor.
LCV-101 operating temperature	110 °C process duty. Not changed by pump equipment capability.
LCV-101 design pressure	1.0 barg rounded from 0.40 + 0.35 barg.
LCV-101 material / line class	A1PL-H: 150# RF / 1.5 mm CA / process / 316L SS / heat conservation.

G.7 Safety, robustness and turndown

Topic	Design statement
Leakage and flammability	Crude glycerol contains 0.4 wt% methanol, so toxic / flammable classification is conservatively marked on the pump datasheet. The selected JIH pump is a mechanically sealed centrifugal pump; final seal arrangement and leakage-control details shall be confirmed by the vendor for hot crude glycerol service.
Particulate risk	The earlier no-particulate restriction has been removed because P-101A has been changed to a JIH mechanically sealed centrifugal pump. Clean liquid service remains assumed; install or confirm suitable suction screening if suspended solids may be present.
Redundancy	P-101A / P-101B provides duty / standby pumping, improving availability for the crude glycerol cooling duty.

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Topic	Design statement
Flow margin	P-101A selected rated capacity is 6.30 m^3/h, which is 126% of normal. This is higher than the normal 10-20% guidance margin but is justified because it is the AHJN standard catalogue duty point for the selected JIH model.
Minimum flow	P-101A minimum flow is 30% of rated capacity. If operation can fall below vendor minimum continuous stable flow, spillback to T-101 should be considered.
LCV controllability	LCV-101 is a jacketed globe/plug control valve for modulating service. Minimum valve case is 50% normal flow; final trim and rangeability are by vendor.
P&ID consistency	DN25 pump discharge to DN40 line and DN32 LCV jacketed globe/plug body to DN40 line require reducers / expanders shown or noted on P&ID-001.

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Appendix H. HAZOP guidewords used

Table H.1 HAZOP guidewords used in the study

Guideword	Meaning	Comments
No/None	The complete negation of the intention	No part of the design intention is achieved
More/Higher	Quantitative increase in a parameter	More of any relevant physical property than there should be (e.g., flow, temperature, pressure, etc.)
Less/Lower	Quantitative decrease in a parameter	Less of any relevant physical property than there should be (e.g., flow, temperature, pressure, etc.)
Reverse	The logical opposite of a parameter	Reverse flow or pressure
Elsewhere	Applicable for flows, transfers, sources and destinations	Applies to process locations, or locations in operating process
Part of	Only part of the design intent is achieved	Only some of intentions are achieved, some are not
As well as	An additional activity occurs	All design and operating INTENTIONS are achieved together with some addition (e.g. extra composition or phase)
Other than	A different activity occurs	No part of original intention achieved, something quite different occurs

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Appendix I. HAZOP parameters used

The following are examples of possible parameters for process operations

- Flow
- Pressure
- Temperature
- Level
- Composition
- Phase
- Distribution
- Heat Transfer/Duty
- Filtration Quality
- Ion Exchange Performance
- Precipitation/Neutralization Performance

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Appendix J. Example combinations of parameters and guidewords

Table J.1 Example combinations of HAZOP parameters and guidewords

Parameter	**Guideword**
Flow	None More Less Reverse As well as Elsewhere
Pressure	Higher Lower Reverse
Temperature	Higher Lower
Level	Higher Lower None
Composition	Part of As well as Other than
Phase	As well as
Distribution	No Less More
Heat Transfer/Duty	No More Less
Filtration Quality	No More Less
Ion Exchange Performance	No Less More
Precipitation/Neutralization Performance	No Less More

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Appendix K. Node list

Project Title:		PMB Biodiesel Crude Glycerol Cooling System		
P&ID Title/Rev/Description		Rev C		
Group No.:		G3	Study Leader/Date	HAZOP Team/ 2026/4/9
Node No	**Description**	**To**	**From**	**Comments**
1	Crude Glycerol Feeding Line	Heat Exchanger E-101	PMB Biodiesel Co.Ltd	
2	Cooling Tower CT-101	Circulating water precipitate settling tank T-105 and filter F-101	Heat Exchanger E-101	The bypass line is dedicated for winter operation only
3	Heat Exchanger E-101	Cooling Tower CT-101/Cooled Crude Glycerol Storage Tank T-102	Crude Glycerol Feeding Line/Cooling Water Feeding Line	
4	Cooled Crude Glycerol Transmit Line	Subsidiary Company	Heat Exchanger E-101	
5	Cooling Water Treatment Line	Ion Exchange Column C-101 and C-102	Cooling Tower CT-101	
6	Cooling Water Feeding Line	Heat Exchanger E-101	Water Treatment Plant/Chemical Dosing Unit	
7	Recycled Water Feeding Line	Cooling Water Mixing Tank T-103	Cation Exchange Column	
8	Waste Water Discharge Line	Waste Treatment Facility	Ion Exchange Column C-101 and C-102	
9	Cation Exchange Line	Cation Exchange Column C-101	Chemical Plant	
10	Anion Exchange Line	Anion Exchange Column C-102	Chemical Plant	
11	Anion Exchange Column	Waste Water Discharge Line	Anion Exchange Line	
12	Ion Exchange Regenerant Treatment Line	River Discharge	Anion Exchange Column C-102	

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Appendix L. PFDs and marked-up PFDs

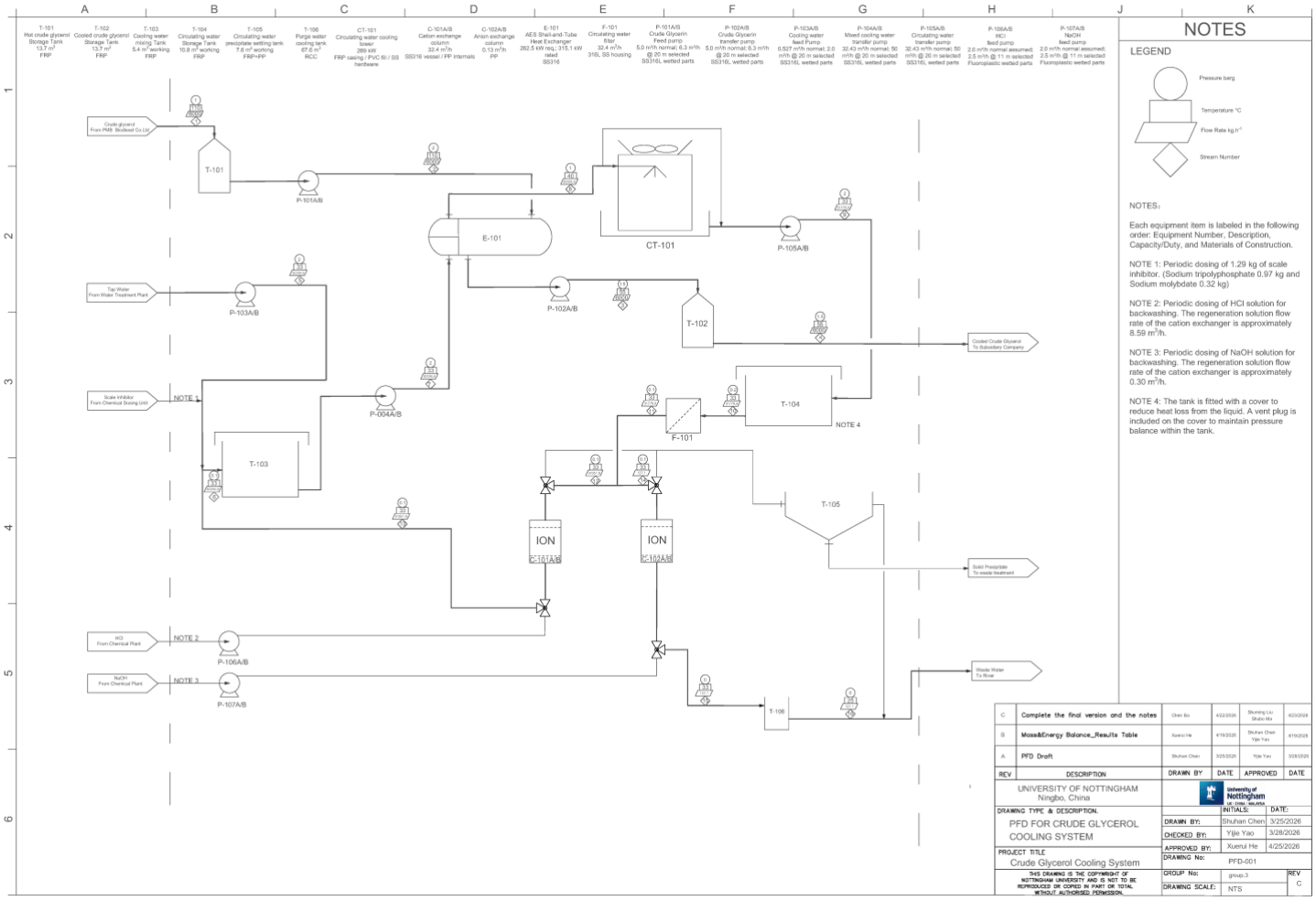


Figure L.1 PFD-001 (original).

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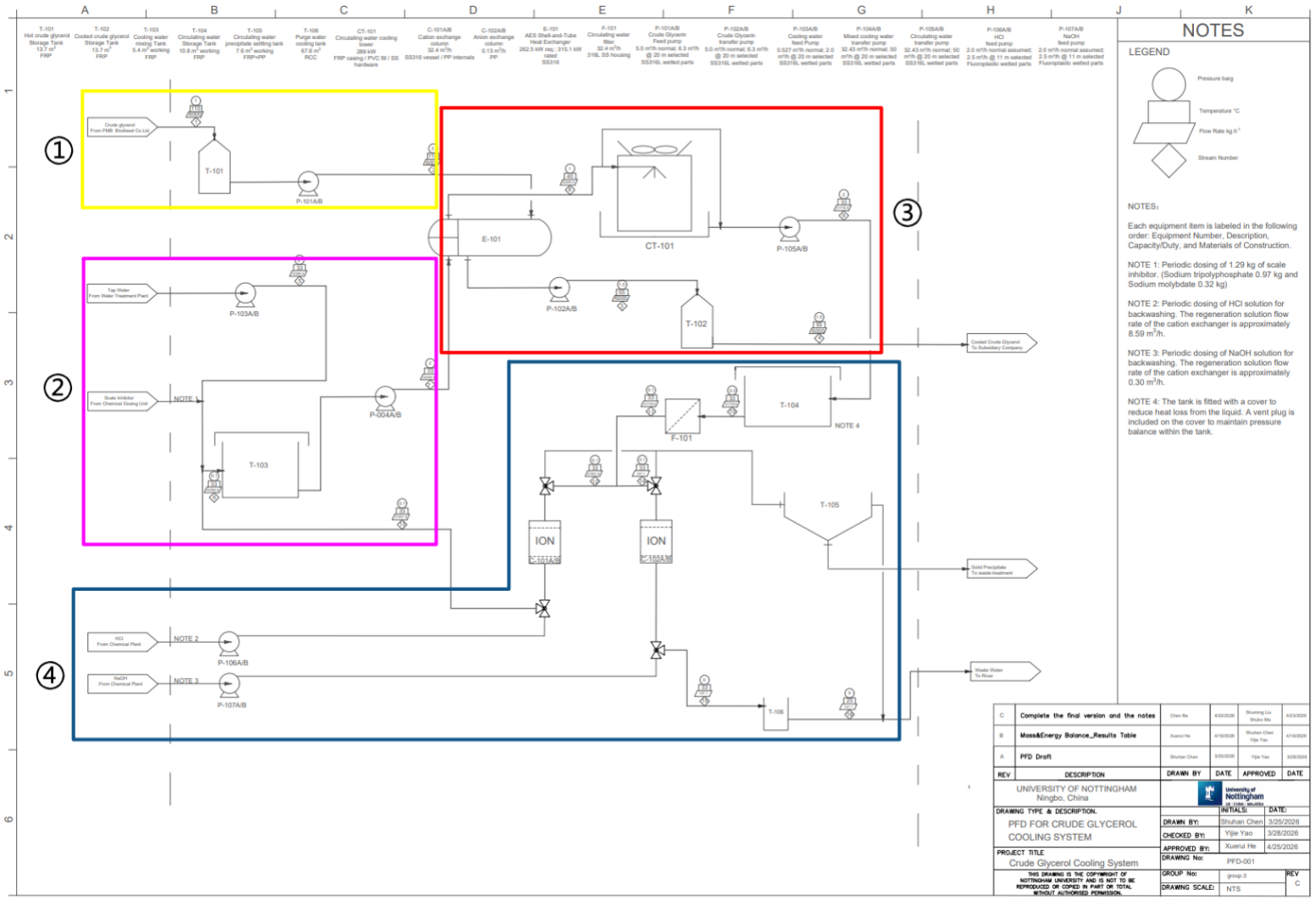


Figure L.2 PFD-001 (marked up).

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Appendix M. P&IDs and marked-up P&IDs

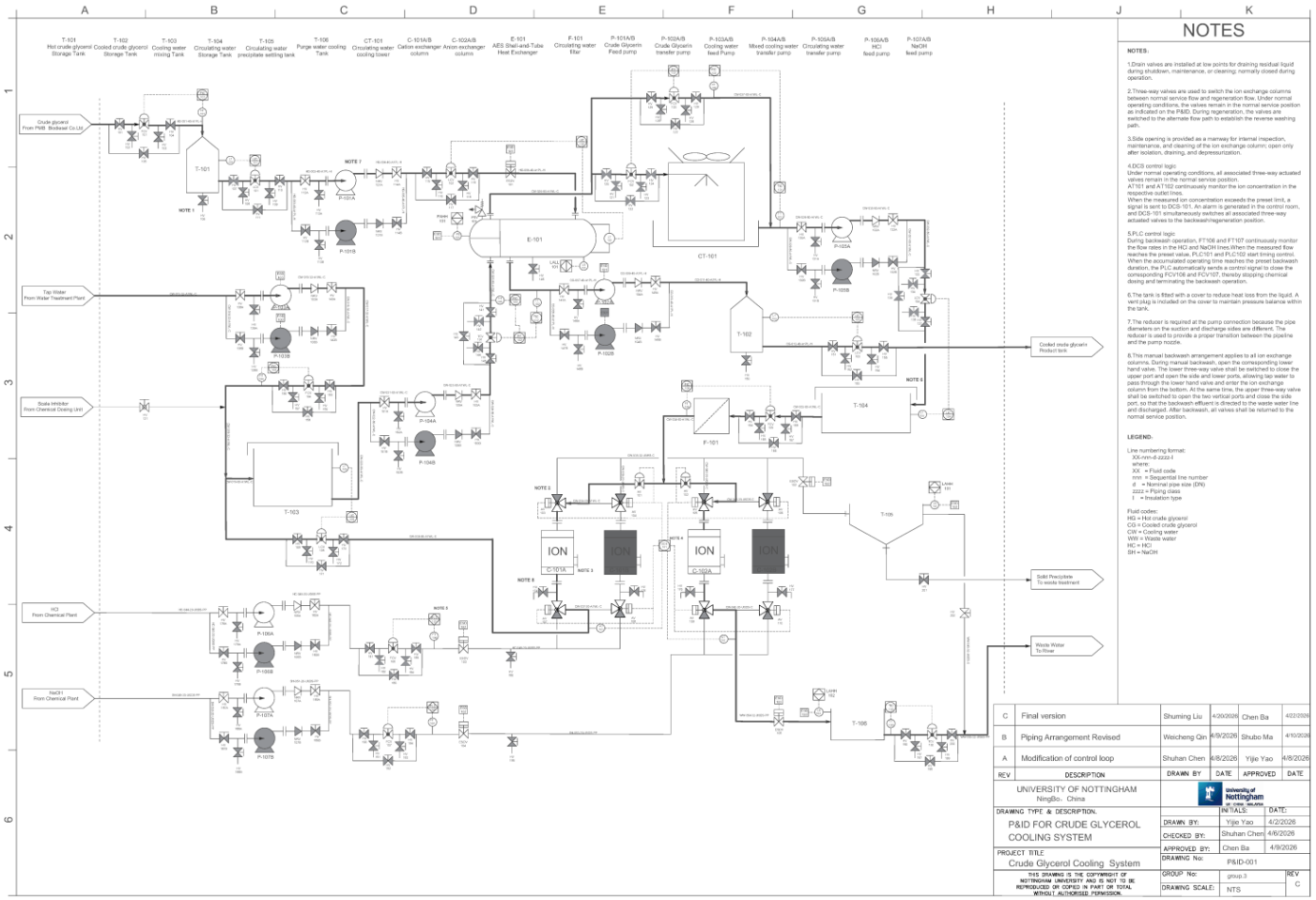


Figure M.1 P&ID-001 (original).

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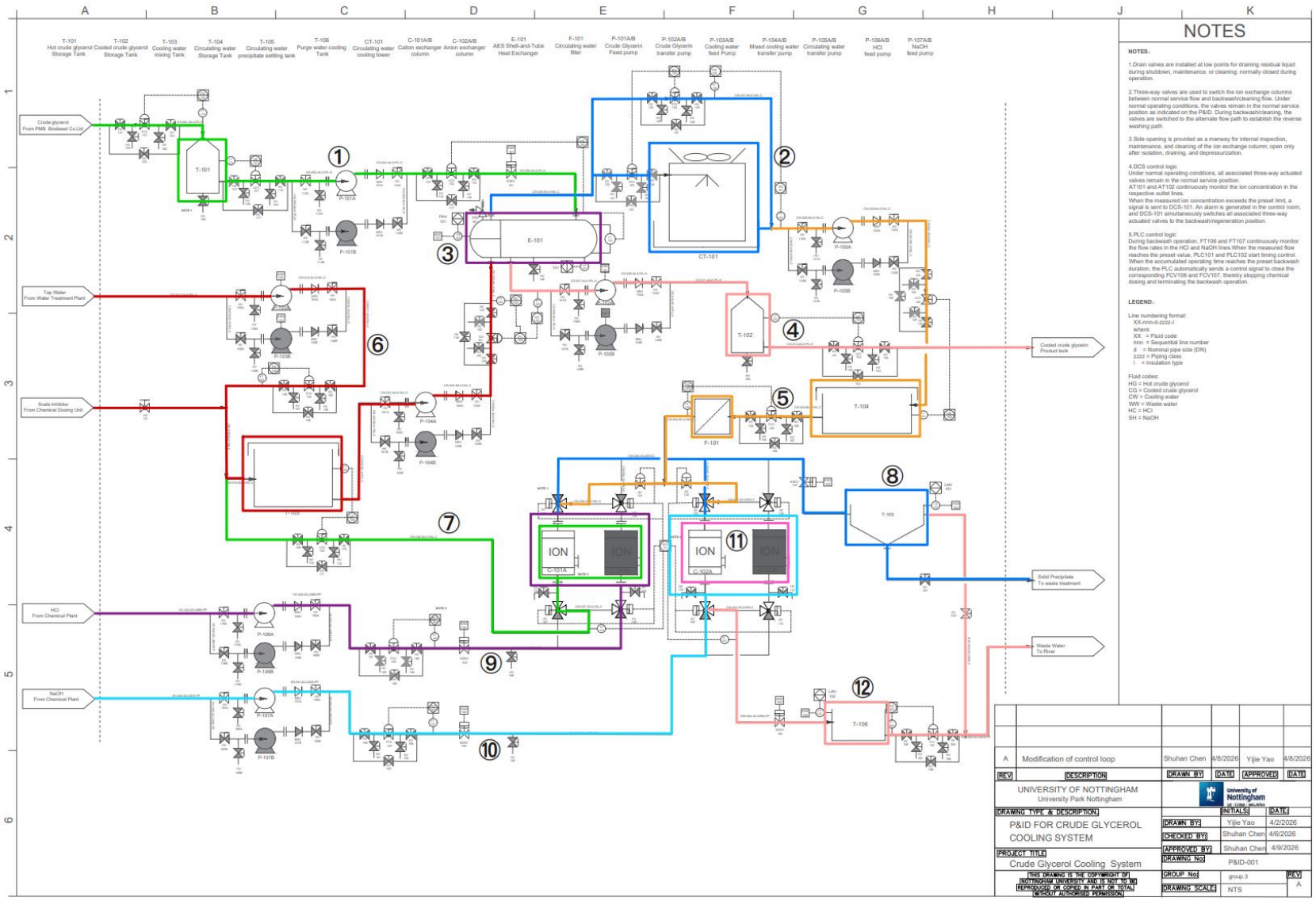


Figure M.2 P&ID-001 (marked up).

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Appendix N Supporting capital cost calculations

N.1 Purchased equipment cost

The delivered purchased-equipment cost (PEC) is RMB 795,200. The itemised equipment basis is shown below.

Table N.1 Purchased equipment delivered cost

Area	Equipment tag / description	Cost (RMB)	Share of PEC (%)
Process side	T-101 Hot crude glycerol storage tank	51,500	6.5
	T-102 Cooled crude glycerol storage tank	51,500	6.5
	E-101 AES shell-and-tube heat exchanger	250,000	31.4
	P-101A/B Crude glycerol feed pump set	9,000	1.1
	P-102A/B Cooled glycerol transfer pump set	9,000	1.1
Cooling water / treatment side	T-103 Cooling water mixing tank	6,700	0.8
	T-104 Circulating water storage tank	8,300	1.0
	T-105 Circulating water precipitate settling tank	42,000	5.3
	T-106 Purge water cooling tank	60,000	7.5
	CT-101 Cooling tower	33,400	4.2
	P-103A/B Cooling water feed pump set	8,600	1.1
	P-104A/B Mixed cooling water transfer pump set	13,700	1.7
	P-105A/B Circulating water transfer pump set	13,700	1.7
	F-101 Vertical pressure sand-bed filter	45,000	5.7
	C-101A/B Cation exchange column pair	142,000	17.9
	C-102A/B Anion exchange column pair	44,000	5.5
	P-106A/B HCl dosing pump set	3,400	0.4
	P-107A/B NaOH dosing pump set	3,400	0.4
Total	Purchased equipment delivered total	795,200	100.0

N.2 Total direct plant cost

Direct plant costs are calculated by multiplying PEC by the selected percentage factors. The total direct plant cost is RMB 2,862,720.

Table N.2 Total direct plant cost calculation

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Direct cost item	Factor / formula	Cost (RMB)	Share of total direct (%)
Purchased equipment delivered	1.00 x PEC	795,200	27.8
Purchased-equipment installation	0.47 x PEC	373,744	13.1
Instrumentation and controls (installed)	0.36 x PEC	286,272	10.0
Piping (installed)	0.68 x PEC	540,736	18.9
Electrical systems (installed)	0.11 x PEC	87,472	3.1
Buildings (including services)	0.18 x PEC	143,136	5.0
Yard improvements	0.10 x PEC	79,520	2.8
Service facilities (installed)	0.70 x PEC	556,640	19.4
Total direct plant cost	3.60 x PEC	2,862,720	100.0

N.3 Total indirect plant cost and working capital

Indirect plant costs are estimated using percentage factors based on PEC. The total indirect plant cost is RMB 1,145,088. Working capital is included at RMB 707,728, resulting in a total capital investment of RMB 4,715,536.

Table N.3 Total indirect plant cost calculation

Indirect / capital summary item	Factor / formula	Cost (RMB)	Comment
Engineering and supervision	0.33 x PEC	262,416	Included in indirect plant cost
Construction expenses	0.41 x PEC	326,032	Included in indirect plant cost
Legal expenses	0.04 x PEC	31,808	Included in indirect plant cost
Contractor's fee	0.22 x PEC	174,944	Included in indirect plant cost
Contingency	0.44 x PEC	349,888	Included in indirect plant cost
Total indirect plant cost	1.44 x PEC	1,145,088	Subtotal of indirect allowances

Table N.4 PERCENTAGE OF DELIVERED-EQUIPMENT COST (Loh, Lyons and White, 2002)

	Percent of delivered-equipment cost for
	Fluid processing plant‡
Direct costs	
Purchased equipment delivered (including fabricated equipment, process machinery, pumps, and compressors)	100
Purchased-equipment installation	47
Instrumentation and controls (installed)	36
Piping (installed)	68
Electrical systems (installed)	11
Buildings (including services)	18
Yard improvements	10

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	Percent of delivered-equipment cost for
	Fluid processing plant‡
Service facilities (installed)	70
Total direct plant cost	360
Indirect costs	
Engineering and supervision	33
Construction expenses	41
Legal expenses	4
Contractor's fee	22
Contingency	44
Total indirect plant cost	144
Fixed-capital investment	504
Working capital (15% of total capital investment)	89
Total capital investment	593

Table N.5 Total capital investment calculation

Capital item	Cost (RMB)
Total direct plant cost	2,862,720
Total indirect plant cost	1,145,088
Fixed capital investment	4,007,808
Working capital	707,728
Total capital investment	4,715,536

Appendix O Supporting operating cost calculations

O.1 Raw material, consumable and water costs

Table O.1 Raw material, consumable and water cost calculation

Item	Annual consumption	Unit price	Annual cost (RMB/year)
5% NaOH solution	3.3 m3/year	75 RMB/m3	249
2% HCl solution	8.5 m3/year	95.9 RMB/m3	823
Sodium tripolyphosphate	24 kg/year	2.4 RMB/kg	58
Sodium molybdate	8 kg/year	8 RMB/kg	64
Tap water	4,216 m3/year	4.88 RMB/m3	20,574
Total raw material, consumable and water cost			21,767

O.2 Electricity costs

The annual electricity cost is calculated using 8,000 operating hours per year and an electricity price of RMB 0.7141/kWh.

Table O.2 Annual electricity cost calculation

Equipment	Power (kW)	Operating time (h/year)	Tariff (RMB/kWh)	Annual cost (RMB/year)
P-101A/B Crude glycerol feed pump set	1.50	8,000	0.7141	8,569
P-102A/B Cooled glycerol transfer pump set	1.50	8,000	0.7141	8,569
P-103A/B Cooling water feed pump set	0.75	8,000	0.7141	4,285
P-104A/B Mixed cooling water transfer pump set	5.50	8,000	0.7141	31,420
P-105A/B Circulating water transfer pump set	5.50	8,000	0.7141	31,420
P-106A/B HCl dosing pump set	0.37	8,000	0.7141	2,114
P-107A/B NaOH dosing pump set	0.37	8,000	0.7141	2,114
CT-101 Cooling tower fan	2.20	8,000	0.7141	12,568
Total annual electricity cost				101,059

O.3 Other operating cost allowances

Table O.3 Other OPEX allowances

Item	Calculation basis	Annual cost (RMB/year)
Maintenance	5% x total capital investment = 0.05 x 4,715,536	235,776
Waste treatment and disposal	38.4 m3/year x 4 RMB/m3	154
Product packaging	Not applicable to bulk transfer/storage	0
Environmental protection / CO2 tax	No separate cost specified in preliminary estimate	0
Total annual OPEX	Raw materials/consumables + utilities + maintenance + waste	358,757

Appendix P Financial calculation basis

P.1 Minimum Attractive Rate of Return （MARR）

The Minimum Attractive Rate of Return is the lowest rate of return an investor is willing to accept; the return on any investment in a project must be at least equal to that of the safest alternative investment. The classification of investment types, along with their corresponding risk levels and minimum required rates of return, is shown in the figure

Investment description	Level of risk	Minimum acceptable return m_{ar} (after income taxes), percent/year
Basis: Safe corporate investment opportunities or cost of capital	Safe	4–8

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New capacity with established corporate market position	Low	8–16
New product entering into established market, or new process technology	Medium	16–24
New product or process in a new application	High	24–32
Everything new, high R&D and marketing effort	Very high	32–48+

The heat exchanger system designed in this project represents an optimisation of the existing process flow. The crude glycerine produced by the system is used in the subsidiary's cosmetics manufacturing. As this essentially constitutes an operational optimisation for a factory that has already established its market position, the team has classified the risk level of this investment as low and estimates the minimum acceptable rate of return to be 12% (median).

P.2 ROI calculation

ROI is calculated as the average annual net profit divided by the total capital investment, expressed as a percentage.

Formula:

$$ROI = \frac{Average\ annual\ net\ profit}{Total\ capital\ investment} \times 100\%$$

Determine the Average Annual Net Profit: This is the project's net profit each year after deducting all operating expenses (OPEX) and non-cash expenses such as depreciation from its revenue. estimate it using the following formula:

$$Net\ Profit = Revenue - OPEX - Depreciation$$

$$Net\ Profit = 2500000\ RMB - 358757\ RMB = 2141243 MB$$

Total Capital Investment = 4715536 RMB

$$\text{ROI} = \frac{2141243\ \text{RMB}}{4715536\ \text{RMB}} \times 100\% = 45.41\%$$

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P.3 Payback Period (PBP)

The Payback Period (PBP) is the amount of time required for the project to recover its initial investment. It is calculated by dividing the fixed capital investment by the annual net profit. The formula is:

$$\text{PBP} = \frac{\textit{Fixed capital investment}}{\textit{Average net annual earning}}$$

Where:

- Fixed Capital Investment = 4715536 RMB
- Average net annual earning = 2141243 RMB

The PBP is calculated as:

$$\text{PBP} = \frac{4715536}{2141243} = 2.2 \text{ years}$$

The relationship between cumulative cash flow and the number of years in operation is shown in the table below:

Table P.1 Cumulative cash flow by operating year

Year	Income（RMB）	Cumulative （RMB）
0	-4716634	-4716634
1	2141243	-2575391
2	2141243	-434148
3	2141243	1707095
4	2141243	3848338
5	2141243	5989581
6	2141243	8130824
7	2141243	10272067
8	2141243	12413310
9	2141243	14554553
10	2141243	16695796
11	2141243	18837039
12	2141243	20978282
13	2141243	23119525
14	2141243	25260768
15	2141243	27402011
16	2141243	29543254

17	2141243	31684497
18	2141243	33825740
19	2141243	35966983
20	2141243	38108226

The line chart is shown below:

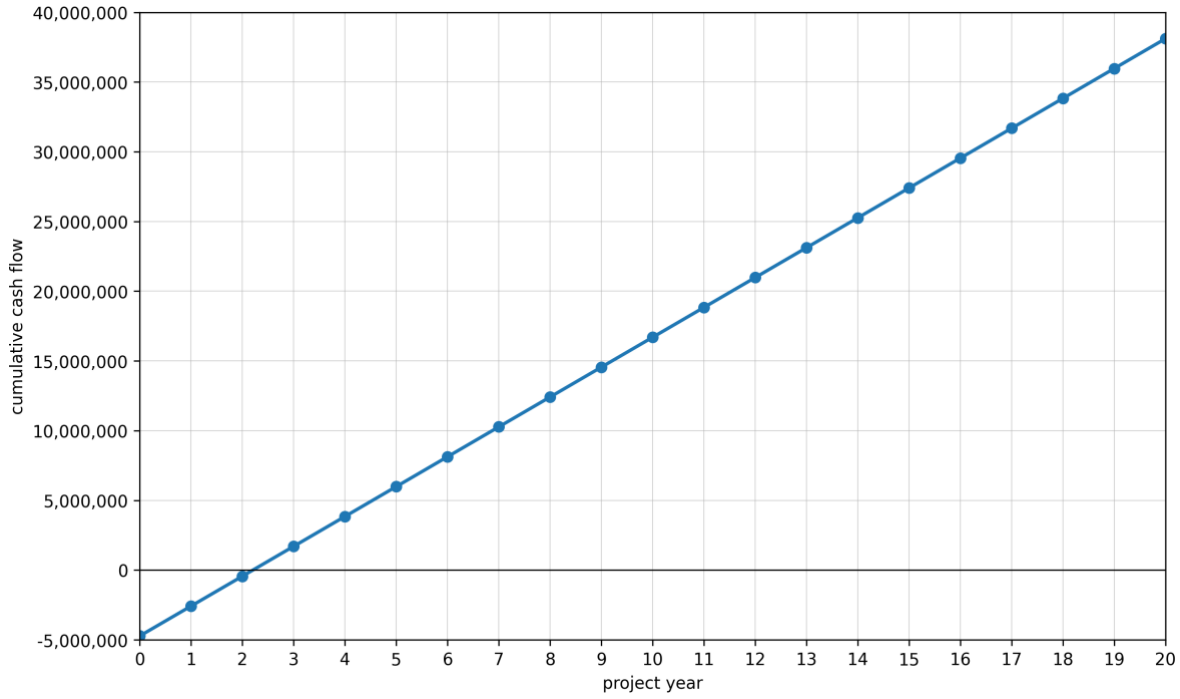


Figure P.1 Cumulative cash flow by operating year.

P.4 Net Present Value (NPV) and Discounted Cash Flow Internal Rate of Return (DCF IRR)

The Net Present Value (NPV) is calculated by discounting the expected future cash flows to the present value, using a specified discount rate. The formula for NPV is:

$$NPV = \sum_{t=1}^{T} \frac{C_t}{(1+r)^t} - C_0$$

Where:

- C_t = Annual Cash Flow
- r = Discount rate
- C_0 = Initial Capital Investment
- T = Project lifetime

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Calculation:
The NPV is calculated by summing the discounted cash flows for each year and subtracting the initial investment. Using a financial calculator or Excel, we get:

$$NPV = 11277259 \text{ RMB}$$

This positive NPV indicates that the project is expected to create value and generate sufficient returns over its lifetime, making it a financially viable investment.

The Discounted Cash Flow Internal Rate of Return (DCFIRR) is the rate at which the NPV equals zero. It is used to assess the rate of return that the project generates, factoring in the time value of money. The formula is:

$$NPV = 0 = \sum_{t=1}^{T} \frac{C_t}{(1+\mathrm{r})^t} - C_0$$

Where:

- C_t = Annual Cash Flow = 2141243 RMB
- C_0 = Initial Capital Investment = 4715536 RMB
- T = Project lifetime = 20 years

Using Excel's IRR function or a financial calculator, we obtain the following result:

DCF IRR= 45.37 %

This high DCF IRR indicates that the project generates a significantly high return on investment, well above the typical MARR (Minimum Attractive Rate of Return) of 12%.

The calculation of NPV, DCF, and IRR in Excel is shown in the table below:

Table P.2 NPV and DCF IRR calculation in Excel

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NPV & DCFIRR calculation (with negative cash flow in first few years)

MARR=	12	%
	0.12	

Year	Cash flow (RMB)	Present worth value (RMB)	PV for DCFIRR (RMB)
0	-4716634	-4716634.00	-4716634.00
1	2141243	1911824.11	1472939.02
2	2141243	1706985.81	1013219.60
3	2141243	1524094.47	696983.33
4	2141243	1360798.64	479447.66
5	2141243	1214998.78	329807.11
6	2141243	1084820.34	226870.92
7	2141243	968589.59	156062.17
8	2141243	864812.13	107353.56
9	2141243	772153.69	73847.41
10	2141243	689422.94	50798.87
11	2141243	615556.20	34944.02
12	2141243	549603.75	24037.63
13	2141243	490717.63	16535.24
14	2141243	438140.74	11374.42
15	2141243	391197.09	7824.35
16	2141243	349283.12	5382.29
17	2141243	311859.93	3702.42
18	2141243	278446.36	2546.86
19	2141243	248612.82	1751.96
20	2141243	221975.74	1205.15

NPV calc=	11277259.87	
"NPV" in Excel=	¥11,277,259.87	
DCFIRR from "Goal Seek"=	0.45	
	45.37	%
NPV for DCFIRR=	¥-0.00	
DCFIRR from "IRR"=	45.37%	

Appendix Q CAPEX and OPEX reference sources

Q.1 CAPEX reference sources

Table Q.1 CAPEX reference sources from capex.xlsx

Estimate area	Item(s) covered	Reference role	Reference link
FRP storage and settling tanks	T-101, T-102, T-103, T-104 and T-105	Supplier quotation/listing used as the reference for FRP tank purchased costs.	Alibaba FRP/GRP tank listing

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Estimate area	Item(s) covered	Reference role	Reference link
Reinforced concrete cooling tank	T-106	Construction/cost reference for the reinforced concrete purge-water cooling tank.	Concrete tank reference
Cooling tower	CT-101	Cooling tower technical model reference and 80 RT/80 T budget cost basis recorded in the CAPEX workbook.	Liang Chi cooling tower model
Shell-and-tube heat exchanger	E-101	Cost-factor reference used for the USD/m² heat-exchanger estimate.	Heat exchanger cost factor
Shell-and-tube heat exchanger	E-101	TEMA AES/BES shell-and-tube exchanger product/type reference.	Xylem TEMA AES/BES reference
Process and cooling-water pumps	P-101A/B to P-105A/B	Pump model and budget purchased-cost reference for JIH centrifugal pumps.	JIH pump listing
Chemical dosing pumps	P-106A/B and P-107A/B	Pump model and budget purchased-cost reference for CQB fluoroplastic magnetic pumps.	CQB dosing-pump listing
Pressure sand-bed filter	F-101	Industrial 2000 mm pressure sand filter listing used as purchased-cost reference.	Pressure sand filter listing
Cation exchange columns	C-101A/B	Cation exchange equipment marketplace quotation/reference used for the budget equipment estimate.	Cation exchange equipment reference
Anion exchange columns	C-102A/B	Anion exchange column product/cost reference used for the budget equipment estimate.	Anion exchange column reference
Ion-exchange resin	Associated resin cost basis	Anion exchange resin price reference retained from the CAPEX workbook.	Anion exchange resin listing
Ion-exchange resin	Associated resin cost basis	Cation exchange resin price reference retained from the CAPEX workbook.	Cation exchange resin listing

Q.2 OPEX reference sources

Table Q.2 OPEX reference sources from OPEX.xlsx

Estimate area	Item(s) covered	Reference role	Reference link
NaOH solution	5% NaOH solution unit cost	Caustic soda liquid supplier price reference used for the NaOH solution cost.	Caustic soda liquid reference

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Estimate area	Item(s) covered	Reference role	Reference link
HCl solution	2% HCl solution unit cost	CEIC/NDRC 36-city average hydrochloric-acid transaction price used for the HCl solution cost.	Hydrochloric acid price reference
Tap water	Industrial tap-water unit cost	CEIC/NDRC 36-city average industrial tap-water usage price used for make-up water cost.	Industrial tap-water price reference
Annual operating hours	8,000 h/year assumption	Operating-hour assumption source recorded in the OPEX workbook; used for utility consumption calculations.	Operating-hour assumption source
Sodium tripolyphosphate	Consumable supplement	Marketplace price reference used for sodium tripolyphosphate consumable cost.	Sodium tripolyphosphate listing
Sodium molybdate	Consumable supplement	Marketplace price reference used for sodium molybdate consumable cost.	Sodium molybdate listing
Electricity	RMB/kWh tariff	Zhejiang general industrial/commercial electricity tariff reference used for pump and cooling-tower fan electricity cost.	Zhejiang electricity tariff PDF
Wastewater treatment and disposal	RMB/m^3 disposal fee	Municipal/non-residential wastewater treatment fee reference used for waste-treatment cost.	Wastewater treatment fee reference

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